

中国科技大学研究生教材系列



研究生学术英语

读写教程

总主编：周厚生 编：李华 孙文卿

外语教学与研究出版社

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总序

习近平总书记在十九大报告中指出：“创新是引领发展的第一动力，是建设现代化经济体系的战略支撑。”因此，要“加强国家创新体系建设，强化战略科技力量”。在人才培养上，“培养造就一大批具有国际水平的战略科技人才、科技领军人才、青年科技人才和高水平创新团队”。

学术英语改革服务于国家宏观的人才培养战略，正在成为最近几年高校英语教学改革热点之一。学术英语课程的设置、开发、构建、实施和推广已成为高校外语课程改革的重要选择。

作为“中国科学院大学研究生教材系列”之一的“果壳学术英语系列教程”着力于学术英语课程的体系化建设，明确学术英语课程之间的层次感，区分基础学术英语能力和更高层次学术英语能力的培养，细化不同类型的学术英语能力，并注重不同类型学术英语能力之间的相互促进。学术英语能力本身是一个十分复杂的体系：从语言使用的角度看，包括学术英语的听、说、读、写等能力；从语言情景的角度看，包括参与学术讨论的能力、撰写学术申请的能力等；从学术思维的角度看，包括批判性思维能力、提出并解决问题的能力、创新思考能力等。学术英语教学应该是系统的、持续的、战略性的工程，唯有系统性地开展学术英语教学，才能更为有效地培养学生的学术英语能力。

“果壳学术英语系列教程”着力于培养有情怀的科学家。学术英语教学不应仅仅止于各项语言技能的传授，还应在提高学术素养、激发学术兴趣、明晰科学伦理意识、注重人文关怀、引发哲学思辨等方面培养具有一定知识广度和思想深度的科技人才。知识广度有助于激发创新性思维，思想深度有助于产生创新性成果。语言教育在充分展示工具性特征的同时，还需兼具开拓崭新视域、融合多维思考、审视固有模式的作用，从而鼓励创造性地提出问题和创造性地解决问题。新时代的新使命呼唤高水平科技人才和科技领军人才，科技人才需要具有国际视野，通晓人类共同关心的普遍问题，展现出不拘于专业领域的知识广度和思想深度，从而具备带领中国科技引领未来世界的能力。

我们认为，学术英语教学应确立人才培养的长期目标，在最初阶段为学生播下学术英语的种子，在接下来的培养过程中持续灌溉，最终助力学生成长成根基扎实的大树，即为国家新时代的新使命做出自己贡献的创新型科技领军人才。

总主编 高原

前言

在建设世界科技强国的背景下，研究生英语教学迎来了新的机遇与挑战。这也赋予了研究生英语教学新的使命：培养学生的学术意识与素养，加强学生的思辨能力，提高学生用英语进行学术研究和学术交流的能力，从而提升我国科研人才的核心竞争力。《研究生学术英语读写教程》在这样一个时代背景下应运而生。

本教程为“果壳学术英语系列教程”中的一册，遵循“以读促写、读写并重”的教学理念，旨在培养学术英语阅读和学术英语写作能力，以期提高学生学术表达的规范性，强化学术素养和学术意识，为研究生阶段的专业学习和学术研究打好语言基础。作为研究生的学术英语读写教材，本教程具有独特的整体性、系统性、严谨性和创新性，适合硕士、博士研究生以及高年级的本科生进行较为系统的学术阅读技能和学术写作技能的学习。

本教程以学科为基础，共包括十个单元，每个单元主题围绕一个学科或专业内容，包括心理学、地学、物理、数学、计算机科学、生物、材料学、化学、管理学和医学十大与理工科研究生专业密切相关的学科内容。每单元课文选材和练习设计既体现了本单元的学科内容，又直接与本单元的学术阅读技能或学术写作技能相关。这两项技能的讲解与练习贯穿每个单元，并且在十个单元的整体编排上循序渐进、环环相扣、前后呼应。

本教程单元结构如下：

一、学术阅读板块 (Academic Reading)

1.1 导入问题 (Lead-in Questions)

1.2 精选文章 (Text A)

1.3 词汇表达 (New Words and Expressions)

1.4 文后练习 (Building Your Vocabulary & Understanding the Text)

1.5 学术阅读技能讲解与练习 (Academic Reading Skills)

二、学术写作板块 (Academic Writing)

2.1 学术论文 (Text B)

2.2 学术写作技能讲解与练习 (Academic Writing Skills)

在学术阅读板块，文前的导入问题 (Lead-in Questions) 有助于学生熟悉单元话题和背景知识。精选文章 (Text A) 为一篇通用学术文章，其中包含的理性思考、人文色彩和社会影响跨越时空，历久弥新，重在培养学生的语言理解能力、文本赏析能力、思

辨能力和作为未来科学家的人文情怀。文中的注释采用脚注的形式，简要介绍文章中提到的人物、概念、术语等。文后的练习部分包括两大类：一是词汇练习（Building Your Vocabulary），帮助学生积累并掌握常用的学术词汇；二是阅读理解练习（Understanding the Text），帮助学生巩固并检验对课文的理解。学术阅读技能（Academic Reading Skills）由讲解和练习两个部分构成：讲解部分语言简明扼要，重点突出；练习部分针对技能讲解设题，循序渐进。在学术阅读板块中，学术阅读技能与本单元 Text A 的课文呼应，在教学中可以鼓励学生运用所学的阅读技能更好地理解 Text A。

在学术写作板块，学术论文（Text B）节选自近期发表的、影响力较大的学科期刊论文，在学术内容和学术语言上体现了本单元的学术写作技能。课文注释亦采用脚注的形式，简要介绍文章中提到的概念、术语等。学术写作技能（Academic Writing Skills）由讲解和练习两个部分构成：讲解部分梳理写作思路，剖析写作步骤，总结写作技巧，与前一板块中的学术阅读技能相衔接，遵循以读促写的教学理念；练习由易及难，紧扣学术写作技能的核心知识点，以便在讲解后可以有针对性地进行练习。

根据每单元的学术能力培养目标，本教程可分成四个模块。1-2 单元为“探究问题”模块，重点培养学生在阅读文献的过程中按照学术规范的要求高效记笔记的能力；3-5 单元为“提出问题”模块，旨在培养学生的批判性思维能力，提高判断信息、分析信息和综述信息的能力，从而使学生能够创造性地发现问题、提出问题。提出有意义的研究问题之后，接下来的研究任务是如何解决问题，即在科学研究中如何进行研究设计和数据收集。由 6-7 单元组成的第三模块便为“解决问题”模块，重点培养学生描述研究过程和研究结果的能力。8-10 单元为最后一个模块——“呈现报告”，这三个单元从学术论文的文体风格出发培养学生恰当地使用学术语言撰写学术论文的能力。从“探究问题”到“提出问题”“解决问题”和“呈现报告”，四个模块贯穿起来，构建起本教程的整体框架。在循序渐进的学术技能推进下，十个不同学科内容的单元呈现出独到的整体性和系统性。这一构架本身既体现了“研究是一个过程”的理念，又展示出“项目研究教学法”的优势。

本教程希望培养学生学习的自主性、积极性和团队合作精神，建议教师在使用本教程的过程中结合“微课”“翻转课堂”“项目研究教学法”等多种教学模式，鼓励学生多思考，多交流，多合作。

本教程由中国科学院大学教材出版中心资助，提供学习和教学资源，为英语教学提供支持。

鉴于编者水平有限，教程中的不足之处在所难免，敬请广大读者批评指正。

编者

2018 年 11 月

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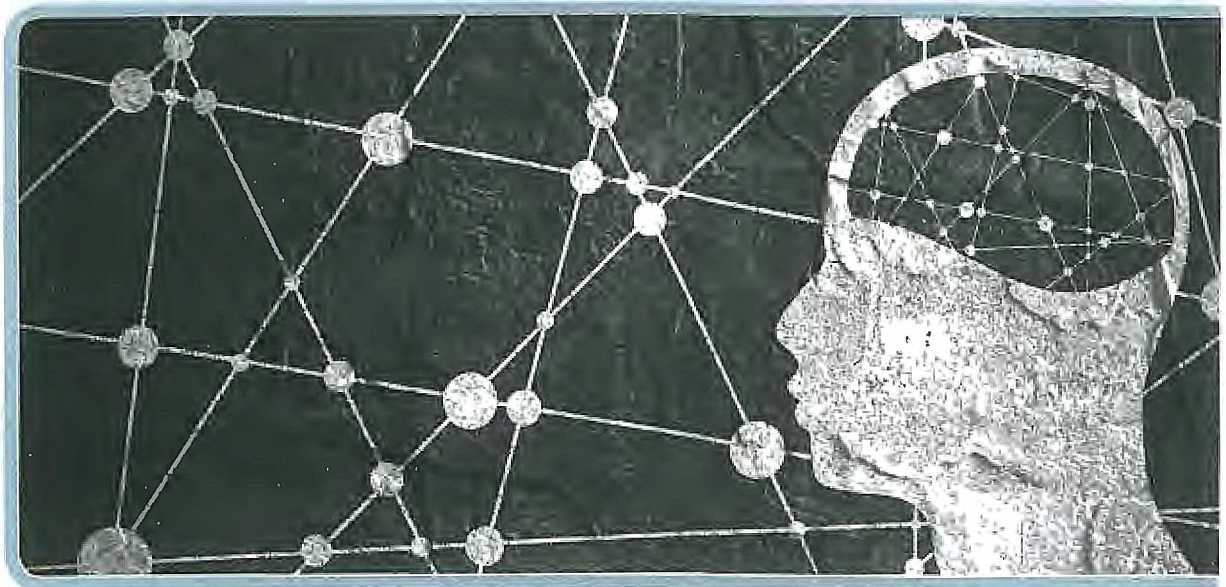
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Unit 1

Psychology

I doubt, therefore I think; I think, therefore I am.

— René Descartes



1 Academic Reading

Text A: Descartes Was Wrong: “A Person Is a Person Through Other Persons”

Academic Reading Skills: Predicting Theme and Identifying Patterns and Structures

2 Academic Writing

Text B: The Impacts and Opportunities of Face Recognition Variability Outside the Lab

Academic Writing Skills: Paraphrasing

1

Academic Reading

Lead-in Questions

1. What is your opinion of the quotation “I think, therefore I am”?
2. How do other people affect your self-perception?

Text A

Descartes Was Wrong: “A Person Is a Person Through Other Persons”

Abeba Birhane

1. According to Ubuntu¹ philosophy, which has its origins in ancient Africa, a newborn baby is not a person. People are born without “ena”, or selfhood, and instead must acquire it through interactions and experiences over time. So the “self”/“other” distinction that’s axiomatic in Western philosophy is much blurrier in Ubuntu thought. As the Kenyan-born philosopher John Mbiti put it in *African Religions and Philosophy* (1975): “I am because we are, and since we are, therefore I am.”
2. We know from everyday experience that a person is partly forged in the crucible of community. Relationships inform self-understanding. Who I am depends on many “others”: my family, my friends, my culture, my colleagues. The self I take grocery shopping, say, differs in her actions and behaviours from the self that talks to my PhD supervisor. Even my most private and personal reflections are entangled with the perspectives and voices of different people, be it those who agree with me, those who criticise, or those who praise me.
3. Yet the notion of a fluctuating and ambiguous self can be disconcerting. We can chalk up this discomfort, in large part, to René Descartes². The 17th-century French

1 Ubuntu: 乌班图，起源于非洲南部的一种伦理概念。

2 René Descartes: 勒内·笛卡尔，法国哲学家、数学家和科学家。

philosopher believed that a human being was essentially self-contained and self-sufficient; an inherently rational, mind-bound subject, who ought to encounter the world outside her head with skepticism. While Descartes didn't single-handedly create the modern mind, he went a long way towards defining its contours.

- 4 Descartes had set himself a very particular puzzle to solve. He wanted to find a stable point of view from which to look on the world without relying on God-decreed wisdoms; a place from which he could discern the permanent structures beneath the changeable phenomena of nature. But Descartes believed that there was a trade-off between certainty and a kind of social, worldly richness. The only thing you can be certain of is your own *cogito*³ – the fact that you are thinking. Other people and other things are inherently fickle and erratic. So they must have nothing to do with the basic constitution of the knowing self, which is a necessarily detached, coherent and contemplative whole.
- 5 Few respected philosophers and psychologists would be identified as strict Cartesian dualists⁴, in the sense of believing that mind and matter are completely separate. But the Cartesian *cogito* is still everywhere you look. The experimental design of memory testing, for example, tends to proceed from the assumption that it's possible to draw a sharp distinction between the self and the world. If memory simply lives inside the skull, then it's perfectly acceptable to remove a person from her everyday environment and relationships, and to test her recall using flashcards or screens in the artificial confines of a lab. A person is considered a standalone entity, irrespective of her surroundings, inscribed in the brain as a series of cognitive processes. Memory must be simply something you *have*, not something you *do* within a certain context.
- 6 Social psychology purports to examine the relationship between cognition and society. But even then, the investigation often presumes that a collective of Cartesian subjects are the real focus of the enquiry, not selves that co-evolve with others over time. In the 1960s, the American psychologists John Darley and Bibb Latané became interested in the murder of Kitty Genovese, a young white woman who had been stabbed and assaulted on her way home one night in New York. Multiple people had witnessed the crime but none stepped in to prevent it. Darley and Latané designed a series of experiments in which they simulated a crisis, such as an epileptic⁵ fit, or

3 *cogito*: 笛卡尔的哲学思想 “我思故我在”

4 dualists: 二元论者

5 epileptic: 癫痫的

smoke billowing in from the next room, to observe what people did. They were the first to identify the so-called “bystander effect”, in which people seem to respond more slowly to someone in distress if others are around.

- 7 Darley and Latané suggested that this might come from a “diffusion of responsibility”, in which the obligation to react is diluted across a bigger group of people. But as the American psychologist Frances Cherry argued in *The Stubborn Particulars of Social Psychology: Essays on the Research Process* (1995), this numerical approach wipes away vital contextual information that might help to understand people’s real motives. Genovese’s murder had to be seen against a backdrop in which violence against women was not taken seriously, Cherry said, and in which people were reluctant to step into what might have been a domestic dispute. Moreover, the murder of a poor black woman would have attracted far less subsequent media interest. But Darley and Latané’s focus makes structural factors much harder to see.
- 8 Is there a way of reconciling these two accounts of the self – the relational, world-embracing version, and the autonomous, inward one? The 20th-century Russian philosopher Mikhail Bakhtin believed that the answer lay in dialogue. We need others in order to evaluate our own existence and construct a coherent self-image. Think of that luminous moment when a poet captures something you’d felt but had never articulated; or when you’d struggled to summarise your thoughts, but they crystallised in conversation with a friend. Bakhtin believed that it was only through an encounter with another person that you could come to appreciate your own unique perspective and see yourself as a whole entity. By “looking through the screen of the other’s soul”, he wrote, “I vivify my exterior.” Selfhood and knowledge are evolving and dynamic; the self is never finished – it is an open book.
- 9 So reality is not simply *out there*, waiting to be uncovered. “Truth is not born nor is it to be found inside the head of an individual person, it is born *between people* collectively searching for truth, in the process of their dialogic interaction,” Bakhtin wrote in *Problems of Dostoevsky’s Poetics* (1929). Nothing simply is itself, outside the matrix of relationships in which it appears. Instead, *being* is an act or event that must happen in the space between the self and the world.
- 10 Accepting that others are vital to our self-perception is a corrective to the limitations of the Cartesian view. Consider two different models of child psychology. Jean Piaget’s theory of cognitive development conceives of individual growth in a Cartesian fashion, as the reorganisation of mental processes. The

developing child is depicted as a lone learner – an inventive scientist, struggling independently to make sense of the world. By contrast, “dialogical” theories, brought to life in experiments such as Lisa Freund’s “doll house study” from 1990, emphasise interactions between the child and the adult who can provide “scaffolding” for how she understands the world.

- 11 A grimmer example might be solitary confinement in prisons. The punishment was originally designed to encourage introspection: to turn the prisoner’s thoughts inward, to prompt her to reflect on her crimes, and to eventually help her return to society as a morally cleansed citizen. A perfect policy for the reform of Cartesian individuals. But, in fact, studies of such prisoners suggest that their sense of self dissolves if they are punished this way for long enough. Prisoners tend to suffer profound physical and psychological difficulties, such as confusion, anxiety, insomnia, feelings of inadequacy, and a distorted sense of time. Deprived of contact and interaction – the external perspective needed to consummate and sustain a coherent self-image – a person risks disappearing into non-existence.
- 12 The emerging fields of embodied and enactive cognition⁶ have started to take dialogic models of the self more seriously. But for the most part, scientific psychology is only too willing to adopt individualistic Cartesian assumptions that cut away the webbing that ties the self to others. There is a Zulu⁷ phrase, “*Umntu ngumuntu ngabantu*”, which means “A person is a person through other persons”. This is a richer and better account, I think, than “I think, therefore I am”.

New Words and Expressions

axiomatic /ˌæksɪəˈmætɪk/ *adj.* something that is
axiomatic does not need to be proved because you
can easily see that it is true 不需证明的, 不言自
明的

blurry /ˈblɜːri/ *adj.* indistinct or hazy in outline 模
糊的

forge /fɔːdʒ/ *vt.* to develop something new,
especially a strong relationship with other people,
groups, or countries 形成, 缔造

crucible /ˈkruːsɪbəl/ *n.* a container in which
substances are heated to very high temperatures 坩
埚, 熔炉

6 enactive cognition (or enactivism): 生成认知

7 Zulu: 南非祖鲁族人 (的)

6 Unit 1

inform /ɪn'fɔ:m/ *vt.* to influence someone's attitude or opinion 影响 (某人的态度或观点)

entangle /ɪn'tæŋɡəl/ *vt.* to involve someone in an argument, a relationship, or a situation that is difficult to escape from 使卷入, 使陷入

disconcerting /dɪskən'sɜ:tɪŋ/ *adj.* making you feel slightly confused, embarrassed, or worried 令人困惑的, 令人尴尬的, 令人不安的

contour /'kɒntʊə/ *n.* the shape of the outer edges of something such as an area of land or someone's body 轮廓; 外形

discern /dɪ'sɜ:n/ *vt.* to notice or understand something by thinking about it carefully (仔细想过之后) 觉察出, 弄清楚, 辨明

fickle /'fɪkəl/ *adj.* someone who is fickle is always changing their mind about people or things that they like, so that you cannot depend on them 三心二意的, 靠不住的, 变化无常的
something such as weather that is fickle often changes suddenly (天气等) 变幻莫测的

erratic /ɪ'ræɪtɪk/ *adj.* something that is erratic does not follow any pattern or plan but happens in a way that is not regular 不规则的, 不确定的, 不稳定的

contemplative /kən'templətɪv, 'kɒntəmplətɪv/ *adj.* spending a lot of time thinking seriously and quietly 沉思的, 冥想的, 默想的

confines /'kɒnfamz/ *n.* limits or borders 范围, 界限

inscribe /ɪn'skraɪb/ *vt.* to carefully cut, print, or write words on something, especially on the surface of a stone or coin (尤指在石头或硬币表面) 雕刻, 印制, 题写

purport /pɜ:'pɔ:t/ *vt.* to claim to be or do something, even if this is not true 声称

bystander /'baɪstændə/ *n.* someone who watches what is happening without taking part 旁观者, 局外人, 看热闹的人

dilute /daɪ'lʊt/ *vt.* to make a quality, belief etc. weaker or less effective 降低(质量); 削弱(信念)

reconcile /'rekənsaɪl/ *vt.* if you reconcile two ideas, situations, or facts, you find a way in which they can both be true or acceptable 使和谐一致; 调和; 协调

luminous /'lu:mɪnəs/ *adj.* shining in the dark 发光的; 夜明的

crystallise /'krɪstəlaɪz/ *v.* if an idea, plan etc. crystallises or is crystallised, it becomes very clear in your mind (使想法、计划等) 变得明朗而具体

vivify /'vɪvəfaɪ/ *vt.* to give new life or energy to something 使生动, 使活跃

solitary /'sɒlətəri/ *adj.* doing something without anyone else with you 单独的, 无伴的

introspection /ɪn'trə'spekʃən/ *n.* the process of thinking deeply about your own thoughts, feelings, or behaviour 内省, 反省

insomnia /ɪn'sɒmniə/ *n.* if you suffer from insomnia, you are not able to sleep 失眠

consummate /'kɒnsəmeɪt/ *vt.* to make something complete, especially an agreement 实现, 完成

embody /ɪm'bɒdi/ *vt.* to be a very good example of an idea or quality 代表, 体现 (思想或品质)

Building Your Vocabulary

Task 1

Complete the following sentences using the expressions in the box below. Change the form where necessary.

solitary	scaffold	inscribe	contour
purport	wipe away	axiomatic	contextual

1. Nowadays, there are hundreds of insurance policies that _____ to provide cyber insurance coverage.
2. Even modest price competition can easily _____ a third or more of operating income if grocers use lower prices to win customers.
3. Oracle Mobile Cloud aims to provide a more _____ user experience, which enables chatbots to switch seamlessly between unstructured conversation and personalised, structured data exchange.
4. In a beautiful mid-19th century painting by Rosa Bonheur, it's possible to see asymmetrical "mountains" in the _____ of her sheep's hindquarters.
5. He looked pleased: a(n) _____ man, by the look of him, one who cherished his privacy.
6. Each step you take provides _____ which will enable you to build the essay a little further.
7. It is _____ that all the sites that are infected should be treated if there is to be any hope of cure.
8. The memoirs of these survivors are rarely _____ in the chroniclers' sentimental journeys.

Understanding the Text

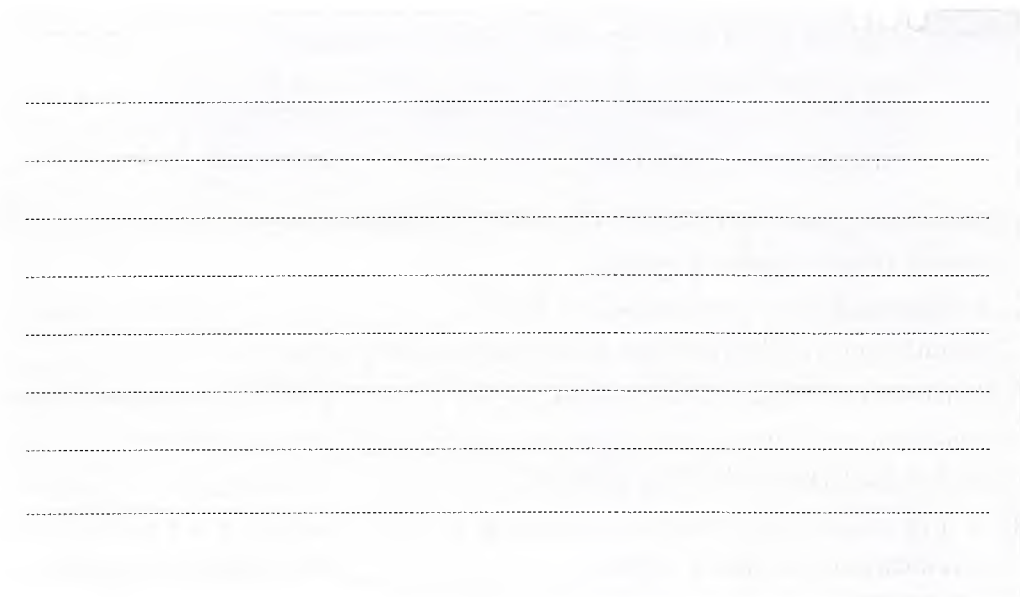
Task 2

Choose the statement that best summarises the theme of Text A.

- A The "self"/ "other" distinction has a long-term influence in academia.
- B Self-perception develops through impacts from others.
- C Descartes was wrong, due to the "self"/ "other" distinction on selfhood.

Task 3

Read Paragraphs 3–4 and analyse the pattern the writer has used to explain Descartes' opinion on selfhood.

A large rectangular area with horizontal dashed lines for writing. The lines are evenly spaced and extend across the width of the area. The background is a light, textured grey.

Academic Reading Skills

Predicting Theme and Identifying Patterns and Structures

A piece of writing for general purposes is composed of a title (in some cases, a title and subtitles), an introduction, a body and a conclusion, and each of these should echo the theme. For research that involves reading a huge number of academic papers, researchers should optimise their reading strategy, thereby saving time and increasing efficiency. To this end, a quick scan of the paper's title (and the subtitles), the abstract, the introduction (particularly the final part of the introduction) and the conclusion, while skipping the body, would present a general idea of the paper. And to identify the theme or the main ideas of a paper sometimes would even shed light upon important details, such as research methodology, major problem(s) encountered and the conclusion reached. By this time, researchers should decide whether the paper benefits their research and whether it is necessary to go on reading the rest or not.

Authors choose from a variety of patterns and structures to organise the information for readers.

Chronological sequence

Authors can proceed with their writing in the order of time. Look for time markers, such as “in 1989” and “last winter”, at the start of a sentence while perusing the text. Longer time markers, usually in the form of adverbial clauses, tend to be placed toward the end to avoid a top-heavy sentence structure. For instance, “The Chinese economy has soared since the introduction of the epoch-making policy of reform and opening-up in 1978”. With those time markers, readers can understand the chronological order of the article better. For example:

Research spanning many decades has shown nutrients in the gastrointestinal tract can shape animals' flavour preferences. One of the earliest findings of this effect dates back to the 1960s, when Garvin Holman of the University of Washington reported hungry rats preferred consuming a liquid paired with food injected into the stomach rather than a solution coupled with a gastric infusion of water. More recently Ivan de Araujo, a neuroscientist at the Icahn School of Medicine at Mount Sinai, and his colleagues have shown calories can trump palatability: Their work has demonstrated mice prefer consuming bitter solutions paired with a sugar infusion injected in the gut rather than

a calorie-free sweet solution.

Cause & Effect

This is fairly self-explanatory. Do beware of the tendency of the English language that places effect before cause, which is traceable to Aristotelian deductive thought. This may explain why many English writers will state the conclusion, the topic sentence, or the effect first, and then provide the evidence, the rationale, the cause and other subordinate information. For example:

One of the reasons that games like Fortnite are so addicting is that they play on the human emotional system, comprised of a long-standing set of psychological adaptations that has features going back millions of generations and across thousands of kinds of species of animals. Darwin himself was the first person to really make a strong case for the evolved function and motivational nature of the human emotional system.

Comparison & Contrast

Comparisons and contrasts are extensively employed in writing. Due to the fact that the English language is a hypotactic language, meaning that logical connections between sentences are usually exposed via transitional devices, such as “in addition”, “therefore”, “in contrast”, etc., comparisons and contrasts are usually easy to identify through commonly used transitional expressions. For example:

Encoding refers to the temporary registration of sensations and thoughts into short-term memory, a kind of “buffer” or RAM that can hold information up to 30 seconds. For any event we experience, including this one, we’re not taking in every detail. From moment to moment, what our brain encodes is a function of what we’re paying attention to, and what has emotional significance to us. Those details are called central details. In contrast, what we’re not paying attention to, or has little or no significance to our brain at the time, are called peripheral details. Those are encoded poorly or not at all.

Classification

The writer, in order to facilitate reader understanding or arrive at a conclusion, usually needs to define which category a piece of random information belongs to, and whether it is quantitative or qualitative, prescriptive or descriptive, instructive or argumentative, etc. Pieces of information of the same nature or purpose should be combined into the same category. For example:

Abundant evidence links personality with emotion via coping. Alternatively, personality can be viewed as an emergent property of responses to the experience of emotion. Dispositions to control, approach, escape, and avoid one's emotional experience underlie diverse traits, including positive and negative urgency, trait emotional approach and avoidance, alexithymia, and emotional expressiveness.

Problem & Solution

Critical essays sometimes will identify the problems of a theory and propose solutions to these problems. In other words, problems are almost always identified and explained before their potential solutions are proposed. Keep this cognitive sequence in mind when browsing through a text for this relationship. For example:

A new study published in Computers in Human Behaviour shows how attitudes about communicating online – such as whether a person believes texting will lead to miscommunication, is easier than face-to-face communication or is an important part of maintaining relationships – can explain “text intensity”, or how dependent that person is on text messaging. According to Dr. Andrew Ledbetter, there are five attitudes people can hold about online communication. People vary on how much they hold each of these five attitudes. Those variations can come from the way a person grew up communicating with their family or that person's competence or skill at communicating with others.

Readers experience a variety of text structures. They select specific comprehension strategies that fit a particular text based on knowledge of how the information is organised. Readers can anticipate what information will be revealed in a selection when they understand text structure. Understanding the pattern of a text helps readers organise ideas for synthesising and summarising. Here are some guiding questions to understand the pattern or structure of a text.

- ✓ Skim the article for titles, subtitles, headings, and key words. After scanning the text, how do you think the author organises the information?
- ✓ Which framework does this author use to organise the information? Chronological? Cause & Effect? Problem & Solution? Comparison & Contrast?
- ✓ Does the author use a combination of different patterns?
- ✓ How does the author organise the text to be “reader-friendly”?
- ✓ Which text features help you collect information from the article?

Task 1 Predict the theme of Text A.

By scanning the title and the introduction of Text A, it can be inferred that the theme of the passage is that _____.

- A the Cartesian notion of “I think, therefore I am” is perfectly justified
- B neither the Cartesian nor the Ubuntu philosophy is reasonable
- C the Cartesian philosophy makes more sense than the Ubuntu one given that the former finds its roots in the superior Western culture
- D the Ubuntu philosophy differs from the Cartesian one in that the former doesn’t consider selfhood as inborn

Task 2 Match the following paragraphs with their roles in Text A.

Paragraph	Role in Text A
Para. 1: _____	A The omnipresence of the Cartesian view in the actual world
Para. 2: _____	B One explanation of a social psychology experiment
Para. 3: _____	C Lead-in to the theme
Para. 4: _____	D Brief introduction of the Cartesian view
Para. 5: _____	E Examples stressing the benefits of harmonising the two views
Para. 6: _____	F An alternative explanation to the social psychology experiment
Para. 7: _____	G Reiteration of the author’s preference towards one of the two views
Para. 8: _____	H An intuitive, common-sense-based understanding of the Ubuntu view
Para. 9: _____	I Reconciliation of the two views
Para. 10: _____	J Application of the reconciliatory approach to understanding reality and truth
Para. 11: _____	K Examples showing the negative consequences of leaving the differences of the two views unattended to
Para. 12: _____	L Uneasiness on the understanding of selfhood is attributable to the Cartesian view

Task 3

Read the following passage and discuss the patterns or structures the author has used to organise the information.

Black Lies

Deception is so common that not only humans, but even animals engage in it. For instance, while apes often simply take food from weaker counterparts, they have also been shown to employ deception. When they can steal food by reaching through opaque instead of see-through tunnels, they often reach for the opaque tunnels so that their competitors cannot detect their actions. These cases of deception are exploitative, as the deceiving apes strategically mislead their counterparts for personal benefits (e.g. tasty food). But what determines whether or not people (and apes) engage in such black lies?

An obvious factor that influences whether people deceive is whether they think that they will get caught. For example, low chances of being detected increase deception of taxpayers. Such behaviour is rational, as being detected reduces the gain one can expect. Imagine the used car dealer who considers lying about a car's history of accidents to charge a higher price for the car (i.e. the gain). If the lie is easily detected (e.g. if the car has bumps and scratches), the likelihood to sell the car decreases. Consequently, the salesman will be honest.

A common assumption is that, rationally, deceiving for higher, rather than lower gains is more beneficial. Surprisingly however, this is not what psychological research finds. People seem to cheat equally often when both high and low gains are at play. Research suggests that cheating does not only depend on materialistic gains, but also on psychological costs that deception inflicts on the deceiver. On the one hand, deceiving for larger gains is more attractive than deceiving for smaller gains. On the other hand, deceiving for larger gains carries larger psychological costs. Psychological costs – the internal discomfort that people experience when doing something against their beliefs or values – depend on the magnitude of a lie. In one experiment, participants were paid according to the outcome they secretly rolled with a die. When asked about their outcome, they were more likely to commit “smaller” deceptions (i.e. reporting 5 instead of 4) than “bigger” ones (reporting 6 instead of 1). Thus, it seems that the psychological cost of telling a lie increases with the magnitude of the lie.

The psychological costs of a lie are closely linked to what one thinks about oneself. In general, people want to think that they are honest. Telling big lies and deceiving others is incompatible with this image. Telling somewhat smaller lies that are “almost true” is easier to reconcile with a positive image of oneself. Because generating a plausible justification for one's lie (e.g. “I almost rolled a 6 with my die”) is often a crucial part

of deception, limiting people's ability to come up with explanations for their lying increases honesty. Furthermore, measures that highlight that one wants to be a good person increase subsequent honesty. For example, signing on top of a self-report form (e.g. tax returns) increases the attention to the moral self. Consequently, people cheat less when completing the form. In other words, the human desire to view oneself as a moral person can be utilised to deter deception.

Taken together, people try to exploit others with black lies. Whether people engage in black lies depends on whether something can be gained through the deception, whether they will get caught, and whether psychological costs occur.

Yet, there is another important factor: the relationship with the deceived. Research shows that cheating socially distant others is more acceptable. However, people more frequently deceive close others. A possible explanation is that there is often more to gain from deceiving close others, and more to lose from revealing unpleasant truths. However, we suggest that this is not the only reason. Deception is not only driven by exploitative motives but can also result from affiliative motives. Therefore, we next discuss cases of deception that result from the motivation to forge a positive relationship with the deceived or to please the deceived.

Task 4

Read the following passage and choose the best answer to each exercise from the choices given below.

Chimpanzees Play the Ultimatum Game

Darby Proctor et al

Humans often make decisions that seem irrational from an economic perspective. For instance, they may engage in behaviour that actually decreases their absolute wealth. One explanation for these decisions is that humans are not only concerned with their own rewards but also the rewards of others. Human reactions to reward distributions have been extensively studied by means of experimental economics tasks, in particular the ultimatum game⁸ and the dictator game⁹. In the ultimatum game (UG), one individual (the proposer) is asked to split a quantity of money with another individual

8 ultimatum game: 最后通牒博弈, 一种由两名参与者进行的非零和博弈。在这种博弈中, 一名提议者向另一名响应者提出一种分配资源的方案, 如果响应者同意这一方案, 则按照这种方案进行资源分配; 如果不同意, 则两人都什么也得不到。

9 dictator game: 独裁者博弈, 对最后通牒博弈进行修改, 取消响应者对提议者所提要求的否决权, 分配者就可以被叫作“独裁者”。

(the respondent). If the respondent accepts the offer, both players are rewarded using the proposed split. If the respondent rejects the offer, then neither player is rewarded. The dictator game (DG) is a variant of the UG in which the respondent has no chance to reject the offer and thus all of the proposer's offers are "accepted".

Proposers in both the UG and DG generally go against their own short-term interests in offering the partner more than the minimum possible amount of money. In UGs, people from Western cultures typically offer around 50% of the available money, even in anonymous one-shot games that lack any future interaction. In DGs, people still offer more of the money than a purely self-interested model would suggest, but offers are lower than in UGs. The reasons why humans typically offer more than self-interested models would predict are twofold. First, humans may be concerned with the welfare of others and thus behave more generously out of an altruistic motivation. Second, they may anticipate refusals of inequitable reward distributions during UGs and make larger offers to ensure that they are accepted, thus serving their own self-interest. Whereas either of these reasons is sufficient to drive human behaviour in these tasks, they may also work in concert.

However, cultural norms of fairness vary across study populations. For example, the Lamelara of Indonesia¹⁰ typically offer more than a fair share (mean 58%), presumably because they are culturally dependent on large-scale cooperation (to hunt whales) and thus have mechanisms in place to share surplus resources. In contrast, the Hadza of Tanzania¹¹, who are hunter-gatherers that share food with group members because of cultural expectations and the fear of ostracism¹², make the lowest offers of any study population, and these offers are often rejected. This likely occurs because of the specific experimental setting of the UG, which may reduce the fear of being ostracised, allowing the Hadza participants to follow their self-interest. In all cases, a given culture's degree of cooperation, sharing, and punishment influences offers in economic games. What remains unclear is how other primates, including one of our closest living relatives, the chimpanzee (*Pan troglodytes*¹³), respond to these types of situations. Studying other primates may shed light on the evolutionary basis for the human tendency towards "fair" distributions.

10 the Lamelara of Indonesia: 印度尼西亚的拉买拉拉捕鲸人

11 the Hadza of Tanzania: 坦桑尼亚的哈德萨狩猎猎人

12 ostracism: 排挤, 孤立。陶片放逐法 (Ostracism) 是古希腊雅典等城邦实施的一项政治制度。雅典公民可以在陶片上写上那些不受欢迎以及极具社会威望、广受欢迎、最可能成为僭主的人的名字, 并通过投票表决将企图威胁雅典民主制度的政治人物予以政治放逐。

13 *Pan troglodytes*: 黑猩猩的学名

16 Unit 1

1. A scan of the title and the above introduction, taken from a psychological paper, indicates that the theme of the entire passage is most probably about _____.
 - A the economic incentives when humans make decisions
 - B the balancing of short-term and long-term interests by proposers in Ultimatum and Dictator Games
 - C the probe into the evolution of the human sense of fairness via the study of chimpanzees and other primates
 - D the debate on the inherent selfishness of the human species

2. How does the underlined sentence structure-wisely relate to the preceding paragraph?
 - A It introduces individual differences to generalisations.
 - B It leads to conclusions from supporting evidence.
 - C It presents a contradiction to a research finding.
 - D It serves as an illustration of a consensus.

3. Which one of the following hypothetical scenarios best matches the writing pattern of the second and the third paragraphs?
 - A Despite local activists' ardent and successful campaign urging the authority to loosen restrictions on Internet censorship, the overall political landscape of this particular region has not exactly displayed a marked improvement on a decade ago.
 - B The crime rates in all parts of the Town of Wisteria are below the national average. Such reassuring generality notwithstanding, the crime rate in the downtown area is relatively high due to a lack of nightly patrol, whereas the suburbia is almost immune to illegalities.
 - C John, a heavy smoker, has received an unexpectedly high overall score in his annual physical check-up. In particular, his lungs are rated positively by all the tests; so are his dental hygiene. This surprise may be attributable to his regular and conscientiously-executed physical exercise and frequent visits to the dentists.
 - D ABC Middle School is acknowledged as a highly diverse institution. However, circumstances vary between the junior and senior middle school divisions. The former is known for the students' highly diverse talents – artistic, academic, or otherwise; whereas the latter, probably faced with exam pressures, is far from diverse by any measure.

Academic Writing

The Impacts and Opportunities of Face Recognition Variability Outside the Lab

Brad Duchaine

- 1 Individual differences in psychological characteristics are taken into account in many situations in daily life. Lifeguards need to be strong swimmers, pilots must have good vision and courtrooms treat eyewitness testimony cautiously when witnesses were tired or under the influence of drugs. Similarly, individual differences in face recognition are relevant in a number of occupations and situations.
- 2 Police work is one occupation that obviously benefits from excellent face recognition ability. In the last few years, London's Scotland Yard¹⁴ has assembled a group of 200 police officers who have especially strong face recognition and used them to great effect. For example, after the London riots of 2011, the police had thousands of hours of low-quality video from security cameras, and most officers were able to recognise very few of the suspects and face recognition software also fared poorly. In contrast, one police constable was able to identify 180 suspects in the videos because of his excellent face recognition ability. Scotland Yard also engages a group of super-recognisers to view live footage from security cameras during large festivals and gatherings like the Notting Hill Carnival in order to identify known offenders.
- 3 Inspired by the success in London, the police department in St. Petersburg, Florida contacted me in order to test the face recognition abilities of their 750 employees. Laura Germine at Harvard Medical School and I are developing a method that we will use to test all the officers and staff, and the department will then consider face recognition ability when assigning people to tasks. People with extraordinary face recognition may be used in a manner similar to the super-recognisers in London,

14 Scotland Yard: 苏格兰场（英国首都伦敦警务处总部及其代称）

whereas officers with especially poor face recognition can focus on tasks that do not demand strong face recognition. Face recognition ability may also become a factor that departments consider when hiring new officers.

- 4 Face recognition abilities are also highly relevant to security personnel who inspect identification photos. Face matching to IDs is difficult even for professionals with normal face recognition, with one recent study showing passport control officers accept fraudulent photos 14 percent of the time. This study also found that, despite years of experience inspecting passport photos, passport officers are no better than student participants at matching faces.
- 5 Each time I pass through airport security or customs, I wonder about the face recognition abilities of the person checking my ID. Is their face recognition good enough that they would be likely to notice a fraudulent photo that resembles the ID holder? Given the large number of Transportation Security Administration officers, it seems a near certainty that a significant number of them spend hours each day checking IDs even though their face recognition is so poor they should be assigned to other jobs.
- 6 Thousands of studies have investigated the accuracy of and influences on person recognition by eyewitnesses. It is well known that accuracy in eyewitness identification is poor, and retrospective analyses show that eyewitness identification errors are an important source of miscarriages of justice. However, the role that individual differences in face recognition ability play has only recently been examined. Not surprisingly, people who score better on face recognition tests are also better at eyewitness identification. Given the important role that eyewitness identification often plays in prosecutions, assessments of the face recognition ability of eyewitness should be carried out to allow police, judges and juries to better gauge the credibility of an eyewitness's identification of a suspect.
- 7 The military could also benefit from consideration of face recognition ability. Several prosopagnosics¹⁵ have told me that their time in the military was extremely challenging because uniforms and short hair deprived them of cues they normally rely on. If prosopagnosics were identified when they enlist, accommodations could be made for them. In addition, although this review has focused on face identification, variability also exists for other aspects of face processing that are

15 prosopagnosics: 面孔失认症患者，面孔失认症是一种神经心理性疾病，表现为对熟悉面孔的识别能力降低或丧失。

important for military personnel such as emotion recognition and other social judgements. Soldiers often need to engage with local populations in face-to-face interactions, often in the absence of a common language, and in such situations properly reading the face is critical.

- 8 In summary, psychological research has revealed that face recognition ability varies greatly. Face recognition ability impacts a number of occupations and consequential situations, and consideration of these differences holds promise as a means to help organisations dedicated to law enforcement and security function more efficiently.

Academic Writing Skills

Paraphrasing

There are three standard ways to incorporate academic sources into your paper: quoting, paraphrasing, and summarising. When you read through academic papers or articles, paraphrasing or summarising effectively in your notes for your current or future research can not only save your time in collecting different ideas but also help you avoid plagiarism in using them properly in your own article.

Paraphrasing refers to rewriting a given sentence or an excerpt using your own words. When you need to use other people's ideas or thoughts, you can paraphrase them to integrate them into your own writing as well as to give the original writer proper credit. In addition to using different words, you can use different grammar (e.g. different parts of speech). A proper paraphrase will demand the change of structure, particularly for longer statement(s). Otherwise, it might be considered as plagiarism of structure. Usually, the change of structure will be first considered before you use different words or different grammar.

The main purpose of paraphrasing has to do with being able to use someone else's ideas properly while you incorporate them into your own texts. Of course, it is required that any writer acknowledges the original source by using the proper citation format.

Read the following example to analyse the differences between appropriate and inappropriate paraphrase.

Original sentence: Human reactions to reward division are often studied by means of the ultimatum game, in which both partners need to agree on a distribution for both to receive rewards.

Inappropriate paraphrase: People's responses to the division of rewards are usually researched through the ultimatum game, and in such a game both players should reach consensus on a distribution for both to be rewarded. (This paraphrase only changes a few details of wording, for example, converting "human" to "people", or "studied" to "researched", while keeping the grammatical structure largely intact. Such technique is far from sufficient to write up a proper paraphrase.)

Appropriate paraphrase: In an ultimatum game, if both players are to be rewarded, consensus is required of both of them on how the distribution should be conducted. It is through this game that research attempts to understand how human beings respond to reward distribution.

Paraphrasing consists of two processes mainly: One is close reading, and the other rewriting. Specifically, the following steps can help you paraphrase a text effectively.

- Step 1 Read the text you want to paraphrase until you fully understand its meaning, the relationship between different information, and the author's purpose or intention.
- Step 2 Reorganise the information in a new or different order.
- Step 3 Rewrite its points in your own words and expressions.
- Step 4 Check your paraphrasing against the original.
- Step 5 Include a citation.

Task 1 Paraphrase the following sentences from Text B.

1. People with extraordinary face recognition may be used in a manner similar to the super-recognisers in London, whereas officers with especially poor face recognition can focus on tasks that do not demand strong face recognition. Face recognition ability may also become a factor that departments consider when hiring new officers. (Para. 3)

Your paraphrase: _____

2. This study also found that, despite years of experience inspecting passport photos, passport officers are no better than student participants at matching faces. (Para. 4)

Your paraphrase: _____

3. Soldiers often need to engage with local populations in face-to-face interactions, often in the absence of a common language, and in such situations properly reading the face is critical. (Para. 7)

Your paraphrase: _____

Task 2

Identify the better one of the two paraphrases of the underlined text, and justify your choice.

Original text:

However, cultural norms of fairness vary across study populations. For example, the Lamelara of Indonesia typically offer more than a fair share (mean 58%), presumably because they are culturally dependent on large-scale cooperation (to hunt whales) and thus have mechanisms in place to share surplus resources. In contrast, the Hadza of Tanzania, who are hunter-gatherers that share food with group members because of cultural expectations and the fear of ostracism, make the lowest offers of any study population, and these offers are often rejected. This likely occurs because of the specific experimental setting of the UG, which may reduce the fear of being ostracised, allowing the Hadza participants to follow their self-interest.

Paraphrase 1:

The Lamelara of Indonesia have put in place arrangements for distributing excess resources as their culture dictates cooperation among a wide community of people for activities like whale-hunting. As a result, offers made by the Lamelara tend to exceed 50%, at an average of 58%. This is one example showing how ideas of fairness differ from culture to culture.

Paraphrase 2:

Nevertheless, the idea of fairness differs from study population to study population. For instance, the Lamelara of Indonesia usually give more than half of the amount (average: 58%), possibly owing to the fact that they rely on massive cooperation to hunt whales. As a result, they have measures in place to distribute surplus resources.

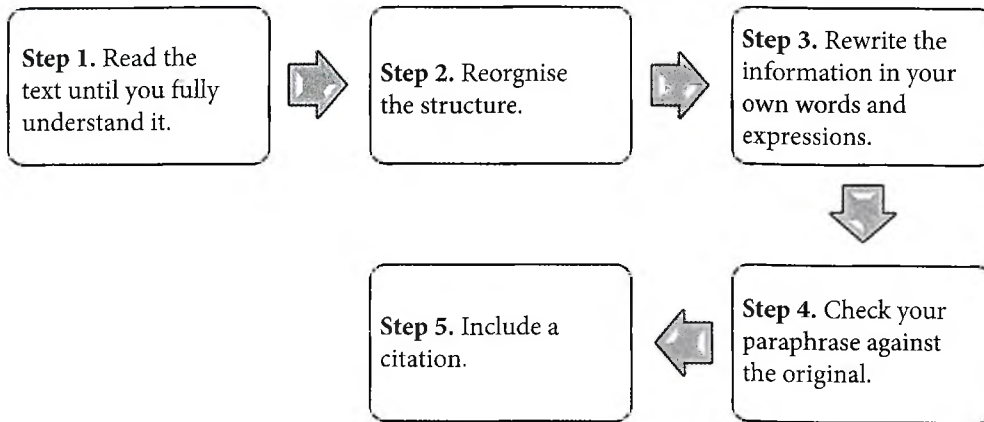
Explain your reason(s) for choosing Paraphrase 1 or Paraphrase 2:

Task 3 Follow the suggested steps to paraphrase the following text.

Original text:

Intelligence quotient (IQ) is generally touted as a gift predicting exceptional outcomes in many domains including educational attainment and income level. However, there are conflicting studies in the literature which point to an association between gifted IQ, particularly high verbal ability, and various mental and immunological outcomes. (Karpinski et al, 2018)

Writing steps:



Your paraphrase: _____

Task 4 Paraphrase the following paragraphs in your own words.

Paragraph 1:

Visioning is important especially in the difficult times. There are times in every relationship when it is so easy to get swamped by the negative emotions of disappointment, frustration, anger, and fear. The complaints filling our mind can even cause an atmosphere of distress, which can even prompt us to consider leaving the relationship. Once we envision, we see possibilities and opportunities, we put an action plan together, we start working, and then our positivity can rise.

Your paraphrase: _____

Paragraph 2:

Reading a road map upside-down and generating synonyms for the word “brilliant” are two very different skills. But each is a measurable indicator of general intelligence, a construct that includes problem-solving abilities, spatial manipulation and language acquisition. Scientists generally agree that intelligence can be captured by psychometric tests. But the study of intelligence is dogged by questions of just how much IQ contributes to an individual’s success and well-being, how genes and environment interact to generate smarts and why the average IQ score rose throughout the world during the twentieth century.

Your paraphrase: _____

Paragraph 3:

Humans often make decisions that seem irrational from an economic perspective. For instance, they may engage in behaviour that actually decreases their absolute wealth. One explanation for these decisions is that humans are not only concerned with their own rewards but also the rewards of others. Human reactions to reward distributions have been extensively studied by means of experimental economics tasks, in particular the ultimatum game and the dictator game.

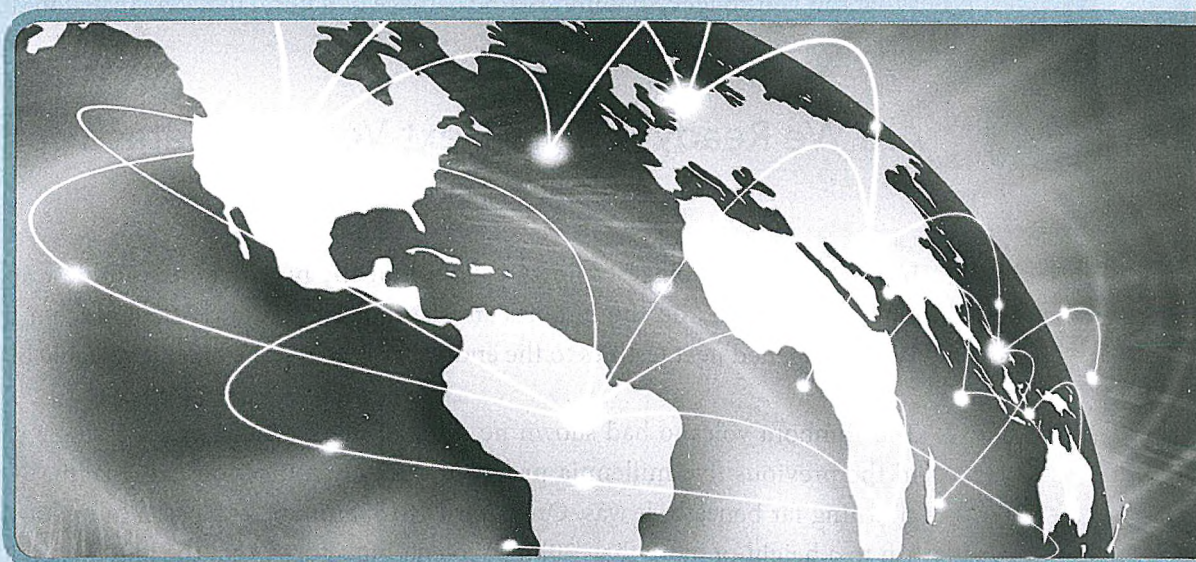
Your paraphrase: _____

Unit 2

Geoscience

Our imagination is struck only by what is great; but the lover of natural philosophy should reflect equally on little things.

— Alexander von Humboldt



1 Academic Reading

Text A: Are We Ready for the Next Volcanic Catastrophe?

Academic Reading Skills: Making Inferences

2 Academic Writing

Text B: Hot Mantle Rising

Academic Writing Skills: Summarising

1

Academic Reading

Lead-in Questions

1. Can you name some major natural disasters in history?
2. What can you suggest to minimise the damage of natural disasters (e.g. flooding)?

Text A

Are We Ready for the Next Volcanic Catastrophe?

Bill McGuire

- 1 In April 1815, the biggest known eruption of the historical period blew apart the Tambora volcano¹, on the Indonesian island of Sumbawa, 12,000km from the UK. What happened next testifies to the enormous reach of the biggest volcanic blasts.
- 2 The Tambora volcano had shown no signs of life for 1,000 years; a single eruption in the previous five millennia provided the only indication that magma² was still churning far beneath. It was. On 5 April 1815, a titanic explosion hurled a cloud of ash to a height of more than 30km. Violent, but short-lived, the blast lasted just two hours, after which the volcano returned to a state of brooding menace. According to the lieutenant governor, Thomas Stamford (later Sir Stamford) Bingley Raffles, to whom volcanologists are indebted for his accounts of the eruption, the detonation was so loud that it was mistaken across Java for cannon fire, causing consternation among the British troops, which had ousted the Dutch and French forces just a few years earlier.
- 3 But the blast was small beer in comparison with what followed. After five days of relative calm, the climactic phase of the eruption began with a colossal explosion that

1 Tambora volcano: 印度尼西亚的坦博拉火山。1815 年该火山爆发, 为近代记载中最严重的火山爆发。

2 magma: 岩浆

launched a towering column of ash to the edge of space. For four or five days, utter blackness reigned across the island as the hurricane blasts of hot ash and scalding gas – known as pyroclastic³ flows – scoured the flanks of the volcano of everything and everyone, and drifts of ash metres thick entombed what few signs of life remained. When the explosions ceased and the darkness finally lifted, the view revealed was a vision of Tolkien's Mordor⁴; a grey landscape within which nothing lived or moved. The top 500m of the volcano was gone, blasted into smithereens, and replaced by a 6km-wide maw from which steam spiralled skywards. Communities on the flanks of the volcano had vanished, along with the lives of around 12,000 men, women and children. These, perhaps, were the lucky ones, as a further 60,000 survivors of the eruption succumbed slowly and agonisingly to famine or disease.

- 4 But the consequences were not confined to this Indonesian backwater. The explosion was heard 2,600km away in Sumatra, while giant rafts of floating pumice⁵ – some kilometres in length – clogged shipping routes for years. The 50 cubic kilometres or so of ash ejected over the course of the eruption returned to earth in the following days and weeks, leaving a thick covering as far away as Borneo, 500km to the north. In addition to the ash, an estimated 200 million tonnes of microscopic sulphur⁶ particles pumped into the stratosphere⁷, spread outwards from Sumbawa to form a giant aerosol⁸ veil that enclosed the planet and acted as a block to incoming sunlight.
- 5 The consequences for the developed societies of the northern hemisphere were dire. A dry, sulphurous fog draped itself across the landscape of eastern North America, causing temperatures to plunge and bringing unprecedented summer cold. In New York State, snow fell in June, while the bitter cold and killing frosts wiped out crops and halved the length of the growing season across much of the region. On the other side of the Atlantic, Europe saw summer temperatures down by 2°C compared to the average for the decade; the unseasonal cold accompanied by incessant rains and – into the following winter – by unusually powerful storms. Analysis of climate records reveals that 1816, the so-called “year without a summer”, was the second coldest in the northern hemisphere of the past six centuries.

3 pyroclastic: 火山碎屑的

4 Tolkien's Mordor: 英国作家托尔金《指环王》三部曲中的“魔都”

5 pumice: 浮岩, 浮石 (一种火山岩)。

6 sulphur: 硫

7 stratosphere: [气] 平流层

8 aerosol: [化] 气溶胶; 喷雾剂; 悬浮微粒。

- 6 The alleged cultural implications of this “volcano weather” for Europe are somewhat whimsical. The brilliant, gas-charged sunsets have been declared by some to have provided the inspiration for some of J. M. W. Turner’s⁹ more flamboyant skies. In a similar vein, the damp and gloom of the 1816 summer has been charged with setting the scene for both Lord Byron’s¹⁰ grim vision *Darkness*, and Mary Shelley’s¹¹ gothic novel *Frankenstein*. For the less well-to-do of Europe, however, the Tambora eruption brought nothing less than hunger, disease and death. Widespread harvest failure resulted in the most serious famine for more than a hundred years, doubling the price of grain and spawning bread riots and widespread civil unrest. Such was the degree of breakdown of food supply that economic historian John Post has called the episode “the last, great subsistence crisis in the Western world”. Malnourished and weakened, the starving succumbed rapidly to disease, with typhus¹² in particular rife. Many tens of thousands are thought to have died across the continent, including more than 40,000 in Ireland alone.
- 7 How would we fare if faced with a Tambora-sized eruption today? Is it even something we could feasibly prepare for? Received wisdom has it that globalisation would make it easier to cope. Should the European harvest fail, so the thinking goes, we can always buy our food from elsewhere. The very interconnectedness of world markets may, however, make things worse – the collapse of food production across Europe, parts of North America and perhaps elsewhere, could result in global shortages which in turn would drive a dramatic rise in the cost of food commodities. At the same time, the intense worldwide competition for food supplies, scarce as a consequence of the harvest failures, could drastically reduce the range of products available in the UK, interfere with supply and distribution, and bring about a collapse of the supermarkets’ ultra-sensitive, time-critical, stock-control systems, leaving their shelves increasingly depleted. While the less well-off could be priced out of purchasing even staple foodstuffs, panic buying by those who can afford it could quickly empty the stores.
- 8 On top of this, harvest disruption in response to volcano weather might extend far beyond Europe, and might – in ensuing decades – be exacerbated by the consequence of

9 J. M. W. (Joseph Mallord William) Turner: 约瑟夫·马洛德·威廉·透纳, 英国画家, 西方艺术史上最为杰出的风景画家之一。

10 Byron: 拜伦, 英国 19 世纪初期浪漫主义诗人。

11 Mary Shelley: 玛丽·雪莱, 英国小说家, 被誉为“科幻小说之母”。

12 typhus: 斑疹伤寒

rampant climate change. In spite of our modern farming methods and distribution systems, the ramifications could be far more severe than we expect. It is also worth considering that while the Tambora blast was approximately 1,000 times bigger than the 2010 Icelandic eruption, it was a minor hiccup in comparison with the greatest volcanic explosions of history. The Toba eruption¹³ that excavated the world's largest volcanic crater in Sumatra, around 74,000 years ago, for example, injected hundreds of times more sulphur gases into the stratosphere than Tambora. The severe "volcanic winter" that followed probably lasted for several years and saw a third or more of the earth covered with snow and ice and the wholesale dieback of vegetation.

- 9 So, if a Tambora-scale scenario would be bad news, far worse could be lying in wait. While we can't stop the next Tambora, nor handle its potential impacts on the climate and the harvest, we can ensure that contingency plans are in place to keep everyone adequately fed until the sulphur veil dissipates and temperatures return to normal. In the UK at present, contingency food supplies probably amount to little more than a few weeks' worth. Some serious policy changes are needed if a future volcanic blast is not to bring about another subsistence crisis.
- 10 Hazardous a guess about when and where the next Tambora will explode is far from an exact science. Eruptions on such a scale seem to happen, on average, a few times every millennium and one estimate holds that there is a 1 in 10 chance of a comparable event in the next 50 years. The earth does not, however, operate to a timetable so such an eruption is equally likely to occur in any single year. There is even a chance that climate change may have a hand to play. Looking back at previous episodes of dramatic climate warming provides us with plenty of robust evidence for a vigorous volcanic response, most notably as our world heated up rapidly at the end of the last Ice Age. The reaction is most pronounced at ice-covered volcanoes, where melting reduces the weight acting on the volcanoes beneath, facilitating eruptions and even promoting the production of more magma. Coastal volcanoes may also be brought to eruption as the increased load of water, due to climbing sea levels, bends the crust¹⁴ around the margins of the oceans, squeezing magma upwards like toothpaste out of a tube.
- 11 No volcano erupts without warning signs, caused by rising magma triggering

13 The Toba eruption: 多巴火山爆发。多巴火山位于印尼苏门答腊岛北部，曾于约 74,000 年前有过一次超级爆发。

14 crust: (地球的) 地壳

earthquake swarms and inflating the ground surface. The problem is that out of our world's 1,300 or more active and potentially active volcanoes, we monitor only a few hundred. The Tambora eruption reinforces the unofficial volcanological axiom: The longer the wait, the bigger the bang. That rule of thumb is borne out by the fact that fully half of the biggest eruptions since 1800 originated at volcanoes that had previously been dormant throughout history. What we should be keeping a special watch on then, in order to prepare ourselves for the next arrival of Vulcan's¹⁵ shock troops, are those seemingly innocuous volcanoes that have kept their heads down for centuries or even millennia. While there are too many candidates to keep a serious eye on, the numbers can be narrowed down by focusing on those that have been recently "restless"; perhaps best regarded as the volcanologists' term for "bubbling under". Beyond that, though, it's anyone's guess.

New Words and Expressions

eruption /ɪ'rʌpʃən/ *n.* an occasion when a volcano suddenly throws out burning rocks, smoke, etc. 喷发, 爆发

testify /'testɪfaɪ/ *v.* to show clearly that something is the case 证明, 证实

churn /tʃɜ:n/ *v.* if water, mud etc. churns, or if something churns it, it moves about violently (水、烂泥等) 剧烈翻腾

hurl /hɜ:l/ *vt.* to throw something violently and with a lot of force 猛投, 用力掷

brooding /'bru:dn/ *adj.* mysterious and threatening 神秘莫测的; 阴森逼人的

indebted /m'detɪd/ *adj.* very grateful to someone for the help they have given you (对某人十分) 感激的

detonation /,detə'neɪʃən/ *n.* an explosion; the action of making something explode 爆炸, 爆发

consternation /,kɒnstə'neɪʃən/ *n.* a feeling of worry, shock, or fear 惊愕; 惊惶失措

oust /aust/ *vt.* to force someone out of a position of power, especially so that you can take their place (尤指为取而代之) 强迫 (某人) 放弃职权, 把……撵走

climactic /klaɪ'mæktɪk/ *adj.* forming a very exciting or important part of an event or story, especially near the end of it 高潮的, 形成高潮的

colossal /kə'lɒsəl/ *adj.* used to emphasise that something is extremely large 巨大的, 庞大的

scour /skaʊə/ *vt.* to clean something by rubbing its surface hard with rough material 冲刷, 冲洗

smithereens /,smɪðə'ri:nz/ *n. pl.* very small pieces 碎片

spiral /'spaɪərəl/ *vi.* to move in a continuous curve that gets nearer to or further from its central point as it goes round 螺旋式上升 [下降]

15 Vulcan: 伏尔甘, 罗马神话中的火与工匠之神。

confined /kən'faɪnd/ *adj.* existing in or affecting only a particular place or group (指空间)有限的, 受限制的

clog /klɒg/ *vt.* to block something or become blocked 阻塞, 塞住

enclose /ɪn'kləʊz/ *vt.* to surround something, especially with a fence or wall, in order to make it separate (尤指用篱笆或围墙)将……围起来

plunge /plʌndʒ/ *vi.* if a price, rate etc. plunges, it suddenly decreases by a large amount (价格、比率等)暴跌, 骤降

unprecedented /ʌn'presɪdɪntɪd/ *adj.* never having happened before, or never having happened so much 空前的, 前所未有的

incessant /ɪn'sesənt/ *adj.* continuing without stopping 持续不断的, 没完没了的

implication /ɪmplɪ'keɪʃən/ *n.* a suggestion that is not made directly but that people are expected to understand or accept 含意, 暗示

whimsical /'wɪmzɪkəl/ *adj.* unusual or strange and often amusing 古怪的; 异想天开的

flamboyant /flæm'bɔɪənt/ *adj.* brightly coloured and easily noticed 色彩艳丽的

episode /'epɪsəʊd/ *n.* an event or a short period of time during which something happens 一段经历; 一段时期

feasible /'fi:zəbəl/ *adj.* a plan, idea, or method that is feasible is possible and is likely to work (计划、想法或方法)可行的, 可实行的, 行得通的

intense /ɪn'tens/ *adj.* having a very strong effect or felt very strongly 剧烈的, 强烈的

staple /'sterpəl/ *adj.* forming the greatest or most important part of something 主要的; 最重要的

exacerbate /ɪg'zæsəbeɪt/ *vt.* to make a bad situation worse 使恶化; 使加重

rampant /'ræmpənt/ *adj.* if something bad, such as crime or disease, is rampant, there is a lot of it and it is very difficult to control (犯罪、疾病等)猖獗的, 肆虐的, 失控的

ramification /ræmɪfɪ'keɪʃən/ *n.* an additional result of something you do, which may not have been clear when you first decided to do it 衍生后果, 派生影响

excavate /'ekskeɪveɪt/ *vt.* to find by digging in the ground 发掘, 挖出

scenario /sə'ni:əriəʊ/ *n.* a situation that could possibly happen 可能发生的事, 可能出现的情况

dissipate /'dɪsɪpeɪt/ *v.* to gradually become less or weaker before disappearing completely, or to make something do this (使某事物)消散, 消失

hazard /'hæzəd/ *vt.* to say something that is only a suggestion or guess and that might not be correct 冒昧提出; 大胆猜测

trigger /'trɪgə/ *vt.* to make something happen very quickly, especially a series of events 引发, 激发

dormant /'dɔ:mənt/ *adj.* not active or not growing at the present time but able to be active later 蛰伏的, 休眠的

innocuous /ɪ'nɒkjuəs/ *adj.* not offensive, dangerous, or harmful 无冒犯之意的; 不危险的; 无害的

Building Your Vocabulary

Task 1

Match the words in Column A with their appropriate explanations in Column B.

Column A	Column B
1 _____ testify	A connotation, meaning
2 _____ confine	B happening immediately after other events
3 _____ scenario	C demonstrate, prove
4 _____ feasible	D huge, enormous
5 _____ facilitate	E block, set a limit to
6 _____ reinforce	F incident, experience
7 _____ colossal	G make it stronger or more intense
8 _____ episode	H make it easier or more likely to happen
9 _____ ensuing	I a postulated sequence of possible events
10 _____ implication	J capable of being done

Understanding the Text

Task 2

Choose the statement that best summarises the theme of Text A.

- A Tambora volcano is the most destructive eruption known in history.
- B Dormant volcanoes like Tambora may bring lethal damage once they are active.
- C The Tambora volcano eruption provides a lesson we need to learn.

Task 3

Paraphrase the following sentences.

- Analysis of climate records reveals that 1816, the so-called “year without a summer”, was the second coldest in the northern hemisphere of the past six centuries. (Para. 5)

2. For the less well-to-do of Europe, however, the Tambora eruption brought nothing less than hunger, disease and death. (Para. 6)

3. No volcano erupts without warning signs, caused by rising magma triggering earthquake swarms and inflating the ground surface. (Para. 11)

4. That rule of thumb is borne out by the fact that fully half of the biggest eruptions since 1800 originated at volcanoes that had previously been dormant throughout history. (Para. 11)

Academic Reading Skills

Making Inferences

Making inferences is a language learning strategy. Learners can use available information to guess the meaning or usage of unknown or unfamiliar information, to predict outcomes, and to compensate for missing information in a language task. Context at both the sentence and discourse levels can be utilised by learners to make inferences. Several types of inferences are illustrated below.

Type 1: Gap-filling inferences

Readers can make this type of inferences by referring to their life experience and prior knowledge. For example, when reading the first three paragraphs of Text A, by referring to our prior knowledge, we can infer that the consequences of the eruption of the Tambora Volcano were disastrous.

Type 2: Text-connecting inferences

Readers make inferences by analysing the relationship between sentences. By referring to existing information, readers can understand the implicit cause-effect relationship in a sentence. For instance:

By the 1960s, Holmes' idea began to gain more credibility as scientists increased their understanding of the ocean floor via mapping, discovered its mid-ocean ridges and learned more about its age. (Based on this part, we can infer that scientists' understanding of the ocean floor made Arthur Holmes' viewpoint accepted.)

Type 3: Coherence inferences

To fully understand a sentence, readers need to infer what pronouns in the sentence refer to. For instance:

Communities on the flanks of the volcano had vanished, along with the lives of around 12,000 men, women and children. These, perhaps, were the lucky ones, as a further 60,000 survivors of the eruption succumbed slowly and agonisingly to famine or disease. (Based on the above sentences, we can infer that "these" refers to "around 12,000 men, women and children", rather than "60,000 survivors of the eruption".)

Type 4: On-line inferences

On-line inferences are drawn by finding out the statements that are supported by details or examples. For instance:

According to the lieutenant governor, Thomas Stamford (later Sir Stamford) Bingley Raffles, to whom volcanologists are indebted for his accounts of the eruption, the detonation was so loud that it was mistaken across Java for cannon fire, causing consternation among the British troops, which had ousted the Dutch and French forces just a few years earlier. (We can infer that the author includes this piece of information to show the 1815 eruption at Tambora was intimidating.)

Task 1

Choose the best answer to the following questions based on the information in Text A.

1. What can be inferred from the effects of the Tambora volcano eruption?
 - A The effects were far-reaching and long-term.
 - B The effects were soon under control by scientists' efforts.
 - C The effects were confined to Indonesian backwater.
 - D Developed countries were more influenced by the eruption.

2. What can be inferred from Paragraph 7?
 - A With advanced technologies, we can cope with a Tambora-sized eruption.
 - B Globalisation can prepare us well for disastrous volcano eruptions.
 - C Wealthy people get more resources than poor people in disastrous situations.
 - D The harvest failures could reduce the products available in the UK.

3. What can be inferred from Paragraph 10?
 - A Scientists can predict when and where the next Tambora will explode.
 - B Records of climate warming may help scientists predict next eruption.
 - C Coastal volcanoes are more likely to erupt than other types of volcanoes.
 - D Volcano eruptions may promote the production of more magma.

4. What can be inferred from Paragraph 11?

- A Inactive volcanoes may bring catastrophic consequences once they erupt.
- B Since 1800, more and more inactive volcanoes have shown warning signs of eruptions.
- C Scientists should focus on those volcanoes that have been recently restless.
- D The number of active volcanoes outweighs that of inactive volcanoes.

Task 2 Choose the best answer based on the following reading materials.

Reading material 1:

About 450,000 years ago Britain was connected to France by a long, rocky, chalk ridge, approximately 32km long, behind which was a great lake, likely dotted with icebergs, with ice stretching across what is now the North Sea. "It would have been a dramatic landscape," said Sanjeev Gupta, professor of earth science at Imperial College London and co-author of the research. The English Channel itself would have been dry, except for small rivers, while the surrounding land would have been forbidding. "It would have been cold, grey, rocky, with very, very sparse vegetation, like Svalbard or Siberia," he said. But, whether as a result of melting of the ice sheet or some other reason, it seems this dam-like ridge began to overflow.

What does "*It*" in Line 3 of this reading material refer to?

- A Britain.
- B The ridge.
- C The English Channel.
- D The North Sea.

Reading material 2:

... sea levels are indeed rising at faster rates each year. "The rate of sea-level rise is increasing, and that increase is basically what we expected," says Steven Nerem, a remote-sensing expert at the University of Colorado Boulder who is leading the reanalysis. He presented the as-yet-unpublished analysis on 13 July in New York City at a conference sponsored by the World Climate Research Programme and the International Oceanographic Commission, among others. Nerem's team calculated that the rate of sea-level rise increased from around 1.8 millimetres per year in 1993 to roughly 3.9 millimetres per year today as a result of global warming. In addition to the satellite calibration error, his analysis also takes into account other factors that

have influenced sea-level rise in the last several decades, such as the eruption of Mount Pinatubo in the Philippines in 1991 and the recent El Niño weather pattern.

What can be inferred from the above reading material?

- A Sea levels are rising at faster rates annually.
- B Steven Nerem is specialised in remote-sensing.
- C Factors influencing sea-level rise are complicated.
- D The study finding is widely supported by many scholars.

Reading material 3:

Researchers created a computer programme called SEAGLAS that combined several climate simulations to forecast US climate until 2100, assuming greenhouse gas emissions keep ramping up. Then, using data from previous studies on how temperature and rainfall affect several economic factors – including crop yields, crime rates and energy expenditures – SEAGLAS predicted how the economy of each of the 3,143 counties in the United States would fare. By the end of the century, some counties may see their gross domestic product decline by more than 20 percent, while others may actually experience more than a 10 percent increase in GDP. This could make for the biggest transfer of wealth in US history, says study co-author Solomon Hsiang, an economist at the University of California, Berkeley.

What can be inferred from the above reading material?

- A Greenhouse gas emissions will hit their highest emission levels in 2100.
- B Climatic information may help predict the economy of each county in the US.
- C Some counties will suffer the gross domestic product decline by the end of the century.
- D Few greenhouse gas emissions may increase the gross domestic product of the United States.

2

Academic Writing

Text B

Hot Mantle Rising

Oliver Shorttle

- 1 The earth's history has been punctuated by vast magmatic episodes. These events are preserved in the geological record as large igneous provinces – areas of the earth's surface flooded by millions of cubic kilometres of lava that erupted in short periods of time. Thought to be triggered by upwelling plumes¹⁶ of hot mantle¹⁷, the size and frequency of these volcanic episodes may have been greater in the past, fuelled by the hotter mantle of the earth's Archaean eon¹⁸. During this time, a distinctive type of igneous rock – komatiite¹⁹ – formed from magmas with high eruption temperatures that cooled to grow long needle-like olivine crystals. The rarity of komatiite eruptions more recently in the earth's history is taken as evidence of the mantle's slow cooling. However, komatiitic lavas that formed 89 million years ago from the volcanic outpourings of the nascent Galápagos plume have been found in the Tortugal suite of the Caribbean large igneous province. Writing in *Nature Geoscience*, Trela et al demonstrate that these lavas formed from anomalously hot mantle with a temperature similar to that which produced the ancient Archaean komatiites, challenging our view of the earth's thermal structure and history.
- 2 Our present picture of the earth's thermal structure is anchored by observations from different depths within the planet. The chemistry and temperature of primitive erupted lavas can be used to reconstruct the temperature of the upper mantle, whereas seismic observations of temperature-dependent mineral phase changes can be used to infer temperatures at the base of the upper mantle. By taking into account the influence of pressure on temperature, these estimates can then be extrapolated to

16 plumes: 地幔柱，一股上升的圆柱状固态物质的热塑性流体。

17 mantle: 地幔，地质学专业术语，是指地壳下面、地核上面的地球中间层。

18 Archaean eon: 太古时代，该时代距离我们久远，是地质发展史中最古老的时期。

19 komatiite: 科马提岩，一种在高温下形成的喷出岩。

infer temperatures at greater depths, down to the core-mantle boundary. Applying this normalisation to the temperature estimates made at various mantle depths, and accounting for phase changes, leads to a coherent picture in which ambient mantle potential temperature is approximately 1,300°C. Present-day variations about this typical upper-mantle temperature are of the order 200°C.

- 3 Trela and colleagues use geochemical measurements of primitive lavas erupted above the early Galápagos plume – and now preserved in Costa Rica – to reconstruct mantle temperatures. They find that the lavas formed from a mantle source that was 200°C warmer than the hottest regions of the present-day mantle and 400°C above its ambient temperature. It is in the context of these present-day mantle temperatures that the observations from the Tortugal suite are remarkable. These lavas formed just 89 million years ago and are extremely young with respect to the major period of komatiite eruptions, 2.5 billion years ago during the Archaean. The existence of such hot eruptions implies that mantle domains with Archaean-like temperatures are still formed or preserved in the deep earth. Such long-term preservation of hot mantle domains resonates with a recent idea that lower mantle temperatures have long been close to the steady state, in contrast to the model of progressive mantle cooling that was previously assumed.
- 4 Trela et al characterise hot komatiitic lavas that erupted relatively recently, in geological terms, above the early Galápagos plume. The existence of such hot and relatively young lavas implies that regions of the earth's mantle with extreme temperatures are still present, and that this heat can be preserved in ascending mantle plumes and drive melting and volcanism at the present day.

Academic Writing Skills

Summarising

As academic readers, you need to learn how to summarise appropriately, as it helps you take notes when you process audio and written information. To avoid plagiarism, your own words should be used to write summaries. Before learning how to write qualified summaries, you need to figure out the differences between summarising and paraphrasing.

In a summary, it is necessary to include the gist or main idea of a text but exclude details and examples. As summarising aims to condense a text to its most important ideas, a summary should be shorter than the original text, about 1/3 of the original text in terms of length. Paraphrasing means using your own words to match the source text in meaning. Specifically, you need to change the words or phrases of a text but maintain the meaning of the original text. Paraphrasing should be about the same length as the original text.

Skills for writing a summary

1. Read the original text carefully and
2. Take notes on the following:
 - A The source of the original text (author's first/last name, title, date of publication, volume number, place of publication, publisher, URL, etc.)
 - B The main idea of the original text (use your own words)
 - C Important supporting points (arrange the points in a logical manner)
3. The first sentence of the summary should include the in-text citation of the source as well as the main idea of the original passage. There are different requirements for summaries at different length. For a summary containing just one paragraph, use a separate sentence to discuss each supporting point. Give 1–2 explanations for each supporting point, and the explanations should be concise and concluded from the original text. For a multi-paragraph summary, each paragraph should discuss a supporting point. Give each supporting point, as the topic sentence, at the beginning of each paragraph.
4. To avoid plagiarism, use your own words when summarising.
5. Transitional words and expressions should be used to show how your summary is organised.
6. The sources of the original texts should be acknowledged in your summaries.

Language conventions for writing a summary

Your own language should be used to summarise other researchers' ideas. To prevent plagiarism, you must cite the sources of the ideas you use properly.

Reporting verbs, such as "argue", "state", and "claim", will be written in the simple present tense. Some examples are listed in the following box:

1. In this article, _____ (author's name) (year) argues that _____ (main idea). For example:
In his article, Davis (2017) describes how Britain separated from the European mainland.
2. _____ (author's name) (year) argues that _____.
(main idea). For example:
Ravilious (2017) argues that dams may produce negative consequences.
3. According to _____ (author's name) (year), _____
(main idea). For example:
According to Briney (2017), there are three driving forces for the movement of tectonic plates.
4. (main idea) _____ (author's name, year). For example:
Hot and relatively young lavas suggest that regions of the earth's mantle with extreme temperatures still exist (Shorttle, 2017).

Task 1

Identify the sentence option which best expresses the essential information in each of the following sentences excerpted from Text B.

1. The chemistry and temperature of primitive erupted lavas can be used to reconstruct the temperature of the upper mantle, whereas seismic observations of temperature-dependent mineral phase changes can be used to infer temperatures at the base of the upper mantle. (Para. 2)
 - A The temperatures of the upper mantle and its base can be obtained through different methods.
 - B The chemistry and temperature of primitive lavas can be used to infer temperatures at the base of the upper mantle.
 - C Seismic observations of temperature-dependent mineral phase changes can be used to infer the temperature of the upper mantle.

2. Writing in *Nature Geoscience*, Trela et al demonstrate that these lavas formed from anomalously hot mantle with a temperature similar to that which produced the ancient Archaean komatiites, challenging our view of the earth's thermal structure and history. (Para. 1)
 - A Our view of the earth's thermal structure and history is challenged by the fact that komatiitic lavas and the ancient Archaean komatiites were formed by hot mantle with similar temperatures.
 - B According to our knowledge of the earth's history and structure, the temperature of komatiitic lavas is similar to that of the ancient Archaean komatiites.
 - C Trela et al challenged our view of the earth's thermal structure and history by demonstrating that komatiitic lavas and the ancient Archaean formed simultaneously.

3. Thought to be triggered by upwelling plumes of hot mantle, the size and frequency of these volcanic episodes may have been greater in the past, fuelled by the hotter mantle of the earth's Archaean eon. (Para. 1)
 - A In the past, the hotter mantle of the earth's Archaean eon may have made the size of volcanic episodes greater; now, the upwelling plumes of hot mantle are making it greater.
 - B The hotter mantle of the earth's Archaean eon and the upwelling plumes of hot mantle make the size and frequency of the volcanic episodes greater than in the past.
 - C In the past, the size and frequency of the volcanic episodes, which were supposedly a result of upwelling plumes of hot mantle, were possibly greater.

Task 2

Complete the following summarising statements by citing researchers properly.

1. Links between subduction and Yellowstone volcanism are also the gist of a third recent study _____ (Quan Zhou et al).

2. _____ (MacDonald et al), an even greater proportion (37%) of the Indo-Gangetic Basin groundwater system suffers from high levels of arsenic.

3. _____ (Ying Zhou) for depths above 700km imply that mantle

upwelling and downwelling indeed coexist.

4. Indeed, _____ (Nelson and Grand) recently report having traced the Yellowstone plume from the core-mantle boundary upwards.

Task 3 Use one sentence only to summarise each of the following paragraphs.

1. Scientists have uncovered the largest volcanic region on earth – two kilometres below the surface of the vast ice sheet that covers west Antarctica. The project, by Edinburgh University researchers, has revealed almost 100 volcanoes – with the highest as tall as the Eiger, which stands at almost 4,000 metres in Switzerland. Geologists say this huge region is likely to dwarf that of east Africa's volcanic ridge, currently rated the densest concentration of volcanoes in the world. (Robin McKie, 2017)

Your summary:

2. The monster El Niño weather pattern of 2014–2016 caused tropical forests to burn up 3 billion tonnes of carbon, according to a new analysis. That is equivalent to nearly 20% of the emissions produced during the same period by burning fossil fuels and making cement. Measurements taken by NASA's Orbiting Carbon Observatory-2 (OCO-2) satellite, which measures the level of carbon dioxide in the atmosphere, suggest that El Niño boosted emissions in three ways. A combination of high temperatures and drought increased the number and severity of wildfires in southeast Asia, while drought stunted plant growth in the Amazon rainforest, reducing the amount of carbon it absorbed. And in Africa, a combination of warming temperatures and near-normal rainfall increased the rate at which forests exhaled CO₂. The overall jump in emissions from tropical forests was roughly three times the annual average carbon output from deforestation and land-use change globally between 2006 and 2015. (Gabriel Popkin, 2017)

Your summary:

Read the following material and summarise the three primary driving forces for the movement of the earth's tectonic plates within 100 words.

Task 4

Scientists today have a better understanding of the make-up of the tectonic plates, the driving forces of their movement, and the ways in which they interact with one another. A tectonic plate itself is defined as a rigid segment of the earth's lithosphere²⁰ that moves separately from those surrounding it.

There are three main driving forces for the movement of the earth's tectonic plates. They are mantle convection, gravity, and the earth's rotation. Mantle convection is the most widely studied method of tectonic plate movement and it is very similar to the theory developed by Holmes in 1929. There are large convection currents of molten material in the earth's upper mantle. As these currents transmit energy to the earth's asthenosphere²¹ (the fluid portion of the earth's lower mantle below the lithosphere) new lithospheric material is pushed up toward the earth's crust.

Evidence of this is shown at mid-ocean ridges where younger land is pushed up through the ridge, causing the older land to move out and away from the ridge, thus moving the tectonic plates.

Gravity is a secondary driving force for the movement of the earth's tectonic plates. At mid-ocean ridges the elevation is higher than the surrounding ocean floor. As the convection currents within the earth cause new lithospheric material to rise and spread away from the ridge, gravity causes the older material to sink toward the ocean floor and aid in the movement of the plates. The earth's rotation is the final mechanism for the movement of the earth's plates but it is minor in comparison to mantle convection and gravity.

20 lithosphere: 岩石圈

21 asthenosphere: 软流圈

As the earth's tectonic plates move they interact in a number of different ways and they form different types of plate boundaries. Divergent boundaries are where the plates move away from each other and new crust is created. Mid-ocean ridges are an example of divergent boundaries. Convergent boundaries are where the plates collide with one another causing the subduction of one plate beneath the other. Transform boundaries are the final type of plate boundaries and at these locations no new crust is created and none is destroyed. Instead the plates slide horizontally past one another. No matter the type of boundary though, the movement of the earth's tectonic plates is essential in the formation of the various landscape features we see across the globe today.

(Amanda Briney, 2018)

Your summary:

Unit 3

Physics

To raise new questions, new possibilities, to regard old problems from a new angle, requires creative imagination and marks real advance in science.

— Albert Einstein



1 Academic Reading

Text A: Theorists, Experimentalists and the Bias in Popular Physics

Academic Reading Skills: Determining Differences Between Facts and Opinions

2 Academic Writing

Text B: Observation of Gravitational Waves from a Binary Black Hole Merger

Academic Writing Skills: Reporting and Synthesising

1

Academic Reading

Lead-in Questions

1. Physics, more than any other discipline, depends on two categories of scholars: the theorist and the experimentalist. Do you know any eminent theoretical physicists or experimental physicists?
2. What are the differences that distinguish a theorist from an experimentalist?

Text A

Theorists, Experimentalists and the Bias in Popular Physics

Ashutosh Jogalekar

- 1 Most people with more than a passing interest in physics will tell you who came up with the idea of quarks – Murray Gell-Mann¹.
- 2 Now gather around the same crowd which knows about Gell-Mann and ask them who Henry Kendall, Jerome Friedman and Richard Taylor² are. It's very likely that you will draw mostly blank stares.
- 3 Yet “coming up with the idea” was as far as Gell-Mann went in 1964 when he and George Zweig³ independently developed the concept. Without the 1968 experiments of Kendall, Friedman and Taylor at the Stanford Linear Accelerator Center (SLAC), quarks would have remained a mere theory, a will-o-wisp whose existence was

1 Murray Gell-Mann: 默里·盖尔曼，美国物理学家，提出了夸克模型，并因此获得了1969年诺贝尔物理学奖。

2 Henry Kendall, Jerome Friedman and Richard Taylor: 1969年诺贝尔物理学奖获得者——肯德尔、弗里德曼和泰勒三位物理学家。

3 George Zweig: 乔治·茨威格，美国物理学家及神经生物学家，与默里·盖尔曼分别提出夸克模型。

confidently postulated but never proven.

- 4 Similar themes proliferate throughout the popular view of physics. Everyone knows Paul Dirac⁴ who conjectured the existence of the positron⁵, but how many know Carl Anderson⁶ and his collaborator Seth Neddermeyer⁷ who actually found it? People similarly know about Wolfgang Pauli⁸ and Enrico Fermi⁹ stating the requirement for a ghostly particle called the neutrino¹⁰ in the 30s, but ask popular science enthusiasts if they are aware of the dogged pursuit of the neutrino by Raymond Davis¹¹ for over 30 years and you will likely see knitted brows. Finally, even today, a schoolchild would likely know Einstein's prediction of the bending of starlight by the gravitational field of a star, but Arthur Eddington's¹² verification of this fact would be little known.
- 5 I started mulling over this vivid gap between the public's appreciation of theorists vs experimentalists on reading a post by physics professor Chad Orzel who, taking a cue from my post about famous American physicists, makes the cogent point that while American theorists lagged behind their European counterparts until the post-war years, they were almost equal to the Europeans even in the 1920s. His point is that we often tend to overemphasise the role of theory over experiment.
- 6 Now there's no doubt that physicists themselves would be the first ones to recognise the value of experimentalists; for instance Anderson, Davis and the Kendall-Friedman-Taylor trio were all recognised by Nobel Prizes. But their recognition in the public mind ranges from vague to non-existent. This gap in perception is especially startling given the singular importance of experiment in physics and all of science, a central paradigm that has been the centerpiece of the scientific method since Galileo (apocryphally) dropped iron balls from the leaning tower of Pisa.

4 Paul Dirac: 保罗·狄喇克, 英国理论物理学家, 量子力学的奠基者之一。

5 positron: 正电子

6 Carl Anderson: 卡尔·安德森, 美国物理学家, 因发现正电子获 1936 年诺贝尔物理学奖。

7 Seth Neddermeyer: 塞思·尼特迈耶尔, 美国物理学家, μ 介子的发现者之一。

8 Wolfgang Pauli: 沃尔夫冈·泡利, 美籍奥地利裔物理学家, 因泡利不相容原理获 1945 年诺贝尔物理学奖。

9 Enrico Fermi: 恩里科·费米, 美籍意大利裔物理学家, 在现代物理理论和实验物理学方面均有重大贡献。

10 neutrino: 中微子

11 Raymond Davis: 雷蒙德·戴维斯, 美国物理学家, 因在探测宇宙中微子领域做出的贡献获 2002 年诺贝尔物理学奖。

12 Arthur Eddington: 阿瑟·埃丁顿, 英国天文学家、物理学家、数学家。

Richard Feynman¹³ paid a sparkling tribute to the supremacy of experiment when he said:

“In general we look for a new law by the following process. First we guess it. Then we compute the consequences of the guess to see what would be implied if this law that we guessed is right. Then we compare the result of the computation to nature, with experiment or experience, compare it directly with observation, to see if it works. If it disagrees with experiment it is wrong. In that simple statement is the key to science. It does not make any difference how beautiful your guess is. It does not make any difference how smart you are, who made the guess, or what his name is – if it disagrees with experiment it is wrong.”

- 7 An even more pointed and roaring tribute to experiment came from the utterly self-assured king of experimental physics, Ernest Rutherford¹⁴. His opinion of theoreticians was that “they play games with their symbols, but we turn out the real facts of Nature”. And he is said to have admonished the capable students working under his tutelage – nine of whom won Nobel Prizes – to not “let me catch anyone talking about the Universe”.
- 8 Rutherford was the ultimate experimentalist and Feynman was the ultimate theorist, but Feynman was well aware of how a well-conceived experiment is really the only thing that can make or break a theory. Ironically, the public devaluing of experimentalists applies to Feynman’s own work. The theory of quantum electrodynamics which he developed is perhaps the most accurate theory in physics. As one example, it can calculate the magnetic moment of the electron correctly to an unprecedented 15 decimal places. But we would never have known this if it weren’t for the experimentalists who devised increasingly ingenious experiments to measure the parameter. Yet everyone has heard of Feynman, but who has heard of Lamb, Kusch or Foley?
- 9 It seems to me that there are at least two important reasons why the public, in spite of tacitly appreciating the all-important role of experiment in physics, fails to give experimentalists their due. First is the sheer success of theoretical physics in uncovering the deepest mysteries of the universe through armchair speculation. Nobody can fail to gasp in awe at an Einstein or Bohr¹⁵ who, working with a few

13 Richard Feynman: 理查德·费曼, 美籍犹太裔物理学家, 1965 年诺贝尔物理学奖得主。

14 Ernest Rutherford: 欧内斯特·卢瑟福, 英国物理学家, 原子核物理学之父, 被公认为继法拉第之后最伟大的实验物理学家。

15 Bohr: 玻尔, 丹麦物理学家, 哥本哈根学派的创始人, 1922 年获诺贝尔物理学奖。

facts and pencil and paper, divine grand operating principles for the cosmos in short order.

- 10 Compared to their efforts based on pure thought, the corresponding efforts of experimentalists who get down on their knees, liberally coat their hands with grease and spend most of their time soldering electronic circuits and fashioning precision machine parts on a lathe sounds humdrum and boring. Yet this mundane work is an essential step toward the grand finale of hard factual discovery. Even the rare combination of theorist and experimentalist appreciates this; for instance, in spite of his pioneering contributions to theory, Fermi always said that his first love was experiment and he could often be found performing the most mundane of tasks.
- 11 To be fair though, it's hard not to admire theorists when many experimentalists, as ingenious as their contraptions are, "simply" validate things which the theorists have already said. Anderson might have discovered the positron, but Dirac invented it first. Eddington might have observed deflected starlight, but Einstein simply plucked it out of thin air based on what seemed like magical speculation.
- 12 Firstly however, it's very important to realise that all the awe for Einstein which we rightly feel comes only after the fact, after a thousand increasingly demanding tests of general relativity have established the veracity of the theory beyond any doubt. As Feynman said, no matter how pretty the theory looks and no matter how brilliant its creator sounds, it is no more than a hypothesis until it's verified. Einstein unverified would have been a mystic. Fortunately the public seems to have gradually woken up to the straitjacket that ugly, grease-and-solder experiment imposes on elegant theory. This is most apparent in the decline of popular versions of string theory; after a period of breathless ascendancy by its proponents, the public seems to increasingly realise the gaping chasm between theory and experiment which the string theoretical framework constantly displays. String theory in fact is the perfect test of the ability of an informed public to distinguish between fact and speculation, and so far the signs seem promising.
- 13 Secondly, there are also outstanding example of discoveries made by experimenters which really had no theoretical precedent. That is what makes Rutherford and Faraday¹⁶ the two greatest experimental physicists in history. Rutherford discovered the atomic nucleus in 1908, but it took 30 years for physicists to develop a concrete

16 Faraday: 法拉第, 英国物理学家、化学家, 由于电磁学方面的伟大贡献, 被称为“电学之父”和“交流电之父”。

theory of the nucleus. Similarly Faraday discovered the seamless relationship between electricity and magnetism – one of the very few examples of unification by experiment – but it took until after his death for Maxwell¹⁷ to come up with his pioneering theory of electromagnetism. Experimentalists often follow in the steps of theorists, but the instances in which they lead the way are as full of creativity and achievement as the work of an Einstein, Bohr or Feynman. And even when they follow, they are the ones who bridge the gap between idea and hard fact.

- 14 The other big reason why for the public seems to downplay the key role of experiments is the bias in physics popularisation toward theory. And here at least part of the blame must be laid at the feet of experimentalists themselves. For instance, if we ponder over who the leading physics popularisers in the last 20 years are, the names that come to our minds include Brian Greene¹⁸, Lisa Randall¹⁹, Leonard Susskind²⁰, Brian Cox²¹ and Sean Carroll²². Almost no experimenter makes the list; Anil Ananthaswamy²³ is one of those rare writers who has shined a light on the heroic efforts of experimenters in validating cutting-edge theories. In a previous post I mentioned how the public has been fed increasingly exotic and speculative physics fare that tends to influence their opinion about what they consider are the most important fields in physics. Cosmology and quantum theory rank high on their list, condensed matter physics and biophysics rank low. But condensed matter theory still ranks higher than condensed matter experiment. Observational cosmology still takes a backseat to speculations about the Big Bang. This has to change.
- 15 If we want to improve the public visibility of experimentalists and place experimentalists in their rightful place in the pantheon of popular physics, the main initiative would have to come from experimentalists themselves. There is no doubt that experimental physics has seen some amazing advances in the last two decades, so there's certainly no dearth of stories to tell. For instance, just last year²⁴ the Nobel Prize in Physics

17 Maxwell: 麦克斯韦, 英国物理学家、数学家, 经典电动力学创始人。

18 Brian Greene: 布赖恩·格林, 美国物理学家, 弦理论领域的科学家。

19 Lisa Randall: 莉萨·兰德尔, 美国理论物理学家。

20 Leonard Susskind: 李奥纳特·苏士侃, 美国理论物理学家, 弦理论的创始人之一。

21 Brian Cox: 布赖恩·考克斯, 英国粒子物理学家。

22 Sean Carroll: 肖恩·卡罗尔, 美国理论物理学家。

23 Anil Ananthaswamy: 阿尼尔·阿南塔斯瓦米, 《物理学的边缘》(*The Edge of Physics*) 一书的作者。

24 本文写于 2013 年, 因此这里指的是 2012 年。

went to Serge Haroche²⁵ and David Wineland²⁶ who have achieved amazing feats in trapping ions and atoms and verifying some of the most bizarre predictions of quantum mechanics. Yet where are the books which elaborate on these successes? Three years ago²⁷ the physics Nobel again went to experimenters who used the simplest and most ingenious methods to create graphene²⁸. Still there are no vividly written books about these experiments. There are plenty of other motifs, from the observation of supernovae²⁹ and x-ray astronomy to the manipulation of single DNA molecules using lasers, which can be productively captured in popular physics books. In addition, the manipulation of these tools to plumb the depths of nature's secrets is every bit as exciting to its practitioners as calculating the curvature of space-time is to its own. It's up to those who deftly wield this machinery to convey their personal pleasure of finding things out to the public.

- 16 Experiment is the ultimate arbiter of science and it's a pity that the current popular physics literature does not reflect this all-important fact. Experimenters and their journalist friends need to now pick up the baton and run with it. They need to communicate to the public why ion traps³⁰ are as engrossing as Lie groups³¹, why even the most elegant mathematical edifice can crumble in the face of confounding experimental evidence, why, in Rutherford's words, "the theorists play games with their symbols while they are the ones who turn out the real facts of Nature".

New Words and Expressions

postulate /'pɒstjʊleɪt/ *vt.* to suggest that something might have happened or be true 提出（理论等），假定，假设

proliferate /prə'lɪfəreɪt/ *vi.* if something

proliferates, it increases quickly and spreads to many different places 激增，扩散

conjecture /kən'dʒektʃə/ *vt.* to form an idea or opinion without having much information to base

25 Serge Haroche: 塞尔日·阿罗什，法国物理学家，2012年因为研究能够量度和操控个体量子系统的突破性实验方法，与美国物理学家大卫·维因兰德共同获诺贝尔物理学奖。

26 David Wineland: 大卫·维因兰德，美国物理学家，2012年与法国物理学家阿罗什共同获诺贝尔物理学奖。

27 本文写于2013年，因此这里指的是2010年。

28 graphene: 石墨烯

29 supernovae: [天] 超新星

30 ion traps: 离子阱

31 Lie groups: 李群，一种代数结构。

it on 推测, 猜想

verification /ˌverɪfɪˈkeɪʃən/ *n.* the act or state of being verified 证明; 证实

mull /mʌl/ *vt.* to think about a problem, plan etc. for a long time before making a decision 认真琢磨, 反复思考 (问题、计划等)

cogent /kəʊdʒənt/ *adj.* if a statement is cogent, it seems reasonable and correct 令人信服的, 有说服力的

paradigm /ˈpærədaɪm/ *n.* a very clear or typical example of something 示例, 范例, 样式

tribute /ˈtrɪbjʊt/ *n.* something that you say, do, or give in order to express your respect or admiration for someone (向某人表示敬慕的) 致敬 (行为)

admonish /ədˈmɒnɪʃ/ *vt.* to tell someone severely that they have done something wrong (严正) 批评, 责备, 警告, 告诫

tutelage /ˈtju:tʃlɪdʒ/ *n.* when you are taught or looked after by someone 受指导 (期); 受监护 (期)

ingenious /mˈdʒɪniəs/ *adj.* an ingenious plan, idea, or object works well and is the result of clever thinking and new ideas (计划、主意或物体) 精巧的, 巧妙的

tacitly /ˈtæsɪtli/ *adv.* in a way that can be understood without being openly expressed 心照不宣地, 默认地

divine /dɪˈvaɪn/ *vt.* to discover or guess something 发现; 猜出

humdrum /ˈhʌmdrʌm/ *adj.* boring and ordinary, and having no variety or interest 单调的, 刻板的, 乏味的

mundane /mʌnˈdeɪn/ *adj.* ordinary and not interesting or exciting 平凡的, 平淡的

contraption /kənˈtræpʃən/ *n.* a piece of equipment or machinery that looks funny, strange, and unlikely to work well 奇怪的机械装置, 怪模怪样的玩意儿

validate /ˈvælɪdeɪt/ *vt.* to prove that something is true or correct, or to make a document or agreement officially and legally acceptable 证实; 使生效; 使合法化

pluck /plʌk/ *vt.* to pull something quickly in order to remove it 揪, 扯, 拉

veracity /vəˈræsɪti/ *n.* the fact of being true or correct 真实 (性), 正确 (性)

ascendancy /əˈsendənsi/ *n.* a position of power, influence, or control 优势; 支配 (地位)

proponent /prəˈpəʊnənt/ *n.* someone who supports something or persuades people to do something 支持者; 拥护者; 鼓吹者

pantheon /ˈpænθiən/ *n.* a religious building that is built in honour of all gods 万神殿, 万神庙

dearth /dɜːθ/ *n.* a situation in which there are very few of something that people want or need 缺乏

verify /ˈverɪfaɪ/ *vt.* to discover whether something is correct or true 核对, 核实

curvature /ˈkɜːvətʃə/ *n.* the state of being curved, or the degree to which something is curved 弯曲 (的形态); 曲率, 曲度

baton /ˈbætn, -tɒn/ *n.* a short light stick that is passed from one person to another during a race 接力棒

edifice /ˈedɪfɪs/ *n.* a building, especially a large one (尤指宏伟的) 建筑

confounding /kənˈfaʊndɪŋ/ *adj.* surprising or confusing (令人) 困惑的, 惊惶的

Building Your Vocabulary

Task 1

Complete the following sentences using the words in the box below. Change the form where necessary.

A contraption B ingenious C tacit D conjecture E mundane
 F humdrum G veracity H admonish I delicacy J supremacy
 K practitioner L tribute M proponent N curvature O postulate
 P verify Q proliferate R manipulation S confound T ultimate

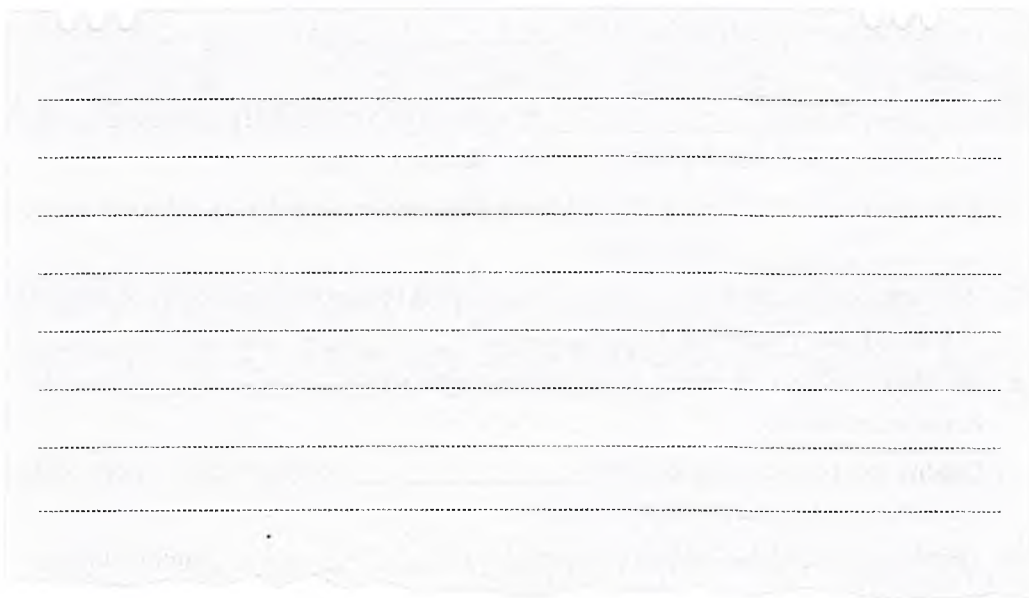
1. Nor can I _____ how far I strayed north or south from my course.
2. The renowned theoretical physicist has for years been a(n) _____ of real-life, NASA-led interstellar travel.
3. My children's teeth never had to have all this _____ on them.
4. I would address you frankly and _____ you to go no more into such places.
5. Even people who _____ a creative God usually acknowledge that his existence shifts the big question rather than resolving it.
6. This video of one of his last, and most impressive, stunts was released as a(n) _____ to his memory.
7. We came up with a(n) _____ plan that would light a fire in the belly of the digital revolution.
8. He often receives inquiries from sellers eager to _____ that their items are authentic.
9. Quality can be found in the most _____ works of man – even within the rusting gears of a motorcycle engine.
10. There were three main groups competing for _____ among them.
11. Despite the increase in recognition of autism spectrum disorders in Western countries, the real cause of the disease continues to _____ and confuse scientists.
12. Accordingly, it tends to face less criticism on the basis of _____ and more on its moral implications.
13. But to recognise that the soul of a man is unknowable is the _____ achievement of wisdom.
14. I anticipate no difficulty, though it requires some thought in _____.
15. Do you seriously think of becoming a(n) _____ of medicine?

Understanding the Text

Task 2 Choose the statement that best summarises the theme of Text A.

- A Theorists have obviously contributed much more to the development of modern physics.
- B The significance of experimentalists deserves the attention of both scientists and the public.
- C It is unnecessary to distinguish the discrepancy between theorists and experimentalists, since the nature of their studies varies.

Read Paragraphs 9–14 and write a summary of this section by using writing skills on summarising you've learned in Unit 2. Your summary should be no more than 100 words.

A rectangular piece of white paper with horizontal dashed lines for writing. The paper has a slightly textured, torn-edge appearance at the top and bottom. There are 10 dashed lines in total, providing space for a summary.

Academic Reading Skills

Determining Differences Between Facts and Opinions

To determine the validity of an argument, we need to examine the evidence (facts) or conclusions (opinions) presented before we decide to what extent we are ready to accept the author's arguments, opinions, or conclusions. How to examine facts or opinions and how to tell the difference between them are the crucial components of critical reading.

A fact can be proven by observation or experiment, while an opinion is a statement of personal feeling, value, belief, interpretation, or judgement.

Recognising different facts and opinions is an important skill in academic reading. In research articles, writers often show their own point of view on an issue when they synthesise different findings and ideas, when they comment on the data from their own studies, or when they explain their own research results and findings. Usually, in order to express their positions or opinions, writers may use certain words or expressions, including view markers (such as “to assume”, “to believe”, “to suggest”, “analysis”, and “judgement”), contrast markers (such as “however” and “yet”), qualifying markers (such as “always”, “should”, and “likely” to show certainty or probability), and assessment markers (such as “mistake”, “profound”, and “extremely” to show their evaluation).

Task 1

Read the following excerpts from Text A and identify each sentence as a fact (F), an opinion (O), or both a fact and an opinion (F/O).

1. Now there's no doubt that physicists themselves would be the first ones to recognise the value of experimentalists; for instance Anderson, Davis and the Kendall-Friedman-Taylor trio were all recognised by Nobel Prizes. (Para. 6)
2. This gap in perception is especially startling given the singular importance of experiment in physics and all of science, a central paradigm that has been the centerpiece of the scientific method since Galileo (apocryphally) dropped iron balls from the leaning tower of Pisa. (Para. 6)

3. Ironically, the public devaluing of experimentalists applies to Feynman's own work. (Para. 8)
4. The theory of quantum electrodynamics which he developed is perhaps the most accurate theory in physics. (Para. 8)
5. Yet this mundane work is an essential step toward the grand finale of hard factual discovery. (Para. 10)
6. Even the rare combination of theorist and experimentalist appreciates this; for instance, in spite of his pioneering contributions to theory, Fermi always said that his first love was experiment and he could often be found performing the most mundane of tasks. (Para. 10)
7. For instance, just last year the Nobel Prize in Physics went to Serge Haroche and David Wineland who have achieved amazing feats in trapping ions and atoms and verifying some of the most bizarre predictions of quantum mechanics. (Para. 15)
8. Still there are no vividly written books about these experiments. (Para. 15)

Read the following introduction section of a research article on
Task 2 nuclear power. Discuss the tone of the underlined sentences with the words and phrases in bold.

(1) Few other products of human ingenuity generate **stronger** demand for global governance than civilian nuclear technology. It is therefore reassuring to see that the present nuclear safety regime is relatively closely regulated. (2) An **elaborate** set of institutions, regulations, and practices aim at safeguarding millions of tons of radioactive material and a vast number of nuclear facilities. Historically, after major nuclear disasters, safety improvements were always made concerning both reactor design and safety governance.

The Chernobyl accident, for instance, led to many of the nuclear safety regulations and practices currently in place. The Fukushima Daiichi accident gave rise to a number of preventive measures, in Japan and beyond, to improve the response to nuclear disasters and to further reinforce nuclear safety in nuclear reactors.

While the evolution of nuclear safety mechanisms occurred incrementally (in response to accidents) it was always shaped by concerns of national sovereignty. (3) Even though governments realise the urgency of strengthening international measures, for nuclear safety, they tend to rely predominantly on voluntary engagement.

(4) In this paper, we argue that the governance of nuclear safety should be turned into a more robust regime, including elements of international supervision, monitoring and verification. In other words, instead of working on a by-accident approach (i.e. in response to accidents), we need to improve nuclear safety by design.

Read the following passage and analyse how the writer expresses facts and opinions. Underline the expressions for facts and highlight the expressions for opinions.

Task 3

Two Cultures of Physics: Theorists vs Experimentalists

Mary Miller

I had a fascinating breakfast meeting with Rocky Kolb while he was in Berkeley recently giving a talk. Rocky formerly led the particle astrophysics group at Fermi lab and is now chair of astronomy and astrophysics at the University of Chicago. He's also a great public speaker and a bit of a celebrity: Rocky was Dr. December in the Stud Muffins of Science Calendar, circa 1996. One of the things we talked about over eggs and pancakes was the dual personality of physics. More than any other scientific discipline, physics depends on two varieties of scientists: the theorist and the experimentalist. Rocky is a theorist, but probably the most famous theoretical physicist was Einstein. His relativity theories, written early in the last century, kept the experimentalists who build observatories and particle accelerators busy for decades trying to confirm his theoretical predictions. But experimentalists like nothing better than coming up with observations about the universe, matter, or the inner workings of particles that catch theorists with their mathematical pants down. A sign about the doorway of condensed matter physicist Sid Nagel reads: Here is where theories come to die. Rocky's take: "We're smarter, better looking and generally taller." But the shorter ones do occasionally exact their revenge. It happened in 1999 when Saul Perlmutter from the Lawrence Berkeley Lab announced that, billions of years after the Big Bang, the universe was actually accelerating rather than slowing down and collapsing under the weight of gravity. No theorist had predicted this finding and collectively they've been scratching their heads ever since trying to

explain it. “This is the first time in 3,000 years of cosmology that theorists are playing catch-up to the experimentalists. In the past there were more theories than observations could confirm or refute, but now there are more observations that the theories cannot explain,” Rocky told me. It’s certainly an interesting time to be a cosmologist and a sideline observer.

Task 4

Read the following excerpt from a research paper on physics education. Distinguish the writer’s points of view from those of other scholars.

Many may think that it is a joke when one says that the most substantive problem about which students frequently complain is that they fail to understand physics. Indeed they do because it does not make sense to them at all. A physicist, Redish states that they (the instructors) are often surprised by how their students seem to know so little mathematics despite successful performance in mathematics classes. The reason is that the symbols used in physics, unlike mathematics, are not arbitrarily chosen and thereby represent certain physical quantities and are loaded by certain physical meanings. It follows that physics students fail to attach the physical meanings to the symbols of equations and formulae. In other words, the problems of physics are not like those of mathematics and one cannot solve a physics problem like solving a purely mathematical one. Thus, a difficulty met in physics education is that the students are not capable of interpreting the symbols in equations.

In another paper, Smigiel and Sonntag state similar problems in physics education in France. According to them, a majority of teachers just concentrate on mathematical calculations rather than on actual scientific concepts and hence students cannot comprehend the meaning behind the formulae.

In the aforementioned paper Redish describes a model for the use of mathematics in sciences: First of all a scientist discerns a physical system to be described. And then the first step comes, i.e. he maps the physical structure into a mathematical model. Secondly, in the process step, to transform the initial description he is involved in some mathematical manipulations. In the third, he interprets his results in terms of physical terms again and finally evaluates whether the results fit to the physical system chosen at the beginning.

Although Redish takes this description to be a description of the use of mathematics

in science, it can be seen as a description of the methodology of science. Each of these steps is controversial and still being discussed in the philosophy of science today. For the present purpose, however, I leave the question whether this description represents science at all. At least it is sufficient to state that similar descriptions can be found in many textbooks.

2

Academic Writing

Text B

Observation of Gravitational Waves from a Binary Black Hole Merger

B. P. Abbott et al

(LIGO Scientific Collaboration and Virgo Collaboration)

- 1 In 1916, the year after the final formulation of the field equations of general relativity, Albert Einstein predicted the existence of gravitational waves. He found that the linearised weak-field equations had wave solutions: transverse waves of spatial strain that travel at the speed of light, generated by time variations of the mass quadrupole moment of the source. Einstein understood that gravitational-wave amplitudes would be remarkably small; moreover, until the Chapel Hill conference in 1957 there was significant debate about the physical reality of gravitational waves.
- 2 Also in 1916, Schwarzschild published a solution for the field equations that was later understood to describe a black hole, and in 1963 Kerr generalised the solution to rotating black holes. Starting in the 1970s theoretical work led to the understanding of black hole quasinormal modes, and in the 1990s higher-order post-Newtonian calculations preceded extensive analytical studies of relativistic two-body dynamics. These advances, together with numerical relativity breakthroughs in the past decade, have enabled modeling of binary black hole mergers and accurate predictions of their gravitational waveforms. While numerous black hole candidates have now been identified through electromagnetic observations, black hole mergers have not previously been observed.
- 3 The discovery of the binary pulsar system PSR B1913+16 by Hulse and Taylor and subsequent observations of its energy loss by Taylor and Weisberg demonstrated the existence of gravitational waves. This discovery, along with emerging astrophysical

understanding, led to the recognition that direct observations of the amplitude and phase of gravitational waves would enable studies of additional relativistic systems and provide new tests of general relativity, especially in the dynamic strong-field regime.

- 4 Experiments to detect gravitational waves began with Weber and his resonant mass detectors in the 1960s, followed by an international network of cryogenic resonant detectors. Interferometric detectors were first suggested in the early 1960s and the 1970s. A study of the noise and performance of such detectors, and further concepts to improve them, led to proposals for long-baseline broadband laser interferometers with the potential for significantly increased sensitivity. By the early 2000s, a set of initial detectors was completed, including TAMA 300 in Japan, GEO 600 in Germany, the Laser Interferometer Gravitational-Wave Observatory (LIGO) in the United States, and Virgo in Italy. Combinations of these detectors made joint observations from 2002 through 2011, setting upper limits on a variety of gravitational-wave sources while evolving into a global network. In 2015, Advanced LIGO became the first of a significantly more sensitive network of advanced detectors to begin observations.
- 5 A century after the fundamental predictions of Einstein and Schwarzschild, we report the first direct detection of gravitational waves and the first direct observation of a binary black hole system merging to form a single black hole. Our observations provide unique access to the properties of space-time in the strong-field, high-velocity regime and confirm predictions of general relativity for the nonlinear dynamics of highly disturbed black holes.

Academic Writing Skills

Reporting and Synthesising

When you read through many research papers, you try to understand what previous studies have done in the field you're interested in. As you read them closely and take notes either by paraphrasing or summarising, you will find similar or different information mentioned in those sources. After a careful analysis of the sources, you expect to combine the varied information in your own research paper, not only to demonstrate your understanding of prior research, but to show the proper and acceptable way to incorporate the sources into your writing without committing plagiarism. This process of combining the ideas of two or more sources into your research paper (in the Introduction section or the Literature Review section, in particular) is called synthesising.

Here is a three-step process on how to synthesise effectively.

Step 1 Analysing

A systematic preliminary analysis of the sources will help. You will find related sources and read through them critically to locate the relevant parts shared by those sources. You will develop your own perspectives and interpretations of the sources through critical reading.

Step 2 Summarising

After analysing and critical reading, you will summarise the relevant parts in each source briefly and concisely. And your summary can help you to compare similarities and/or contrast differences of the related sources.

Step 3 Integrating

In order to combine the summaries of the sources, you will explore different ways to organise the information, which will demonstrate your own voice or your own perspective. Strategies to organise the information in the Introduction section may include:

- A climactic order (from those most distantly related to your study to those closely related);
- B chronological order (according to the time when the research was conducted, or the time when the paper was published);
- C comparison and contrast (to show similarities or differences of the sources);
- D classification (according to different approaches, different solutions,

different theoretical premises or frameworks, or different variables or factors).

Use of cohesive devices is another helpful strategy to consider in order to combine the sources together in a logical way. With different ways of organisation, you may use different transitional links to indicate the logical relationships (comparison or contrast, for example) among the combined information.

Task 1

Discuss the organisation strategy of synthesising in the following excerpts from Text B.

Excerpt 1:

Also in 1916, Schwarzschild published a solution for the field equations that was later understood to describe a black hole, and in 1963 Kerr generalised the solution to rotating black holes. Starting in the 1970s theoretical work led to the understanding of black hole quasinormal modes, and in the 1990s higher-order post-Newtonian calculations preceded extensive analytical studies of relativistic two-body dynamics. (Para. 2)

Excerpt 2:

Experiments to detect gravitational waves began with Weber and his resonant mass detectors in the 1960s, followed by an international network of cryogenic resonant detectors. Interferometric detectors were first suggested in the early 1960s and the 1970s. A study of the noise and performance of such detectors, and further concepts to improve them, led to proposals for long-baseline broadband laser interferometers with the potential for significantly increased sensitivity. (Para. 4)

Task 2

Discuss the pattern of information organisation used in the following samples of syntheses from the Introduction section of research papers.

Sample 1:

To date, only a few studies have been related to the auralisation of railway noise. In the sound quality of traction noise of starting vehicles was assessed using synthesised sounds. In train pass-bys have been auralised based on a combination of filtered and resynthesised binaural recordings. Within the SILENCE project, a software called VAMPPASS was developed which features audio synthesis capabilities for railway vehicle pass-bys.

Sample 2:

The degree of auditory distraction and the associated decline in cognitive performance within an open-plan office sound environment has been shown to be affected by the degree of uncontrolled (or, to-be-ignored) audition of irrelevant sounds: the so-called irrelevant sound effect (ISE). In this regard, what makes an irrelevant sound stream distracting, consistently over time, has been encapsulated within the so-called changing-state hypothesis. Within the changing-state hypothesis, the sound stream can be thought in terms of segments, which vary in their acoustic-perceptual properties over time (speech being a prime example). For such segments, the extent or the degree of distraction increases with the extent to which the segments change state within the sound stream.

Sample 3:

At Hanford, 54 million gallons of waste were stored in 177 underground tanks. The oldest, single-shell tanks were built between 1943 and 1964, with designed service lives of 10 to 20 years. Out of these 177 tanks, 67 have or are suspected to have leaked up to 1 million gallons into the environment, with first leaks confirmed in 1959. Double-shell carbon-steel tanks were built starting in 1968 to provide better confinement. Waste was then pumped from single-shell to double-shell tanks, yet 2.8 million gallons were still stored in single-shell tanks in 2012. Moreover, leaks have also been discovered between shells of double-shell tanks. Construction of a facility to immobilise the high level waste using similar approaches to those used at Savannah River began in 2002. However, due to various unresolved technical problems and work stoppages, the estimated cost to construct this treatment and immobilisation facility has tripled from 4.3 billion to 13.4 billion dollars, and its scheduled completion date slipped by nearly a decade to 2019. Completion of clean-up activities is not expected before 2050. Clean-up efforts for all the waste sites are projected to cost more than 280 billion dollars.

Sample 4:

The adaptive optics (AO) technology is a powerful tool to overcome the effect of the random optical disturbance, and has been applied in many fields. Up to now, it has been shown that the AO technology can largely improve the performance of the atmospheric non-coherent optical communications. Some researchers also recommend the AO technology to improve the atmospheric coherent optical communications. Belmonte and Kahn have analysed theoretically the performances of the free-space coherent optical links when the AO technology is adopted.

Sample 5:

Small rooms represent enclosed acoustic spaces that have interior volumes in the range from a few cubic metres to a few hundred cubic metres. In the acoustical sense, one

has to deal with rooms having sizes of the same order of magnitude with the wave length at the frequencies in question or rooms in which the early reflections by walls, ceilings, and room objects arrive within milliseconds of the direct sound. Small sizes cause that a room response is dominated by modal behaviour; thus, acoustics of small rooms are often characterised by irregular sound distributions at low frequencies. A shape of a small room is rectangular usually; therefore, attempts have been made to classify room's low frequency sound distribution with regard to its aspect ratio. Common metrics have relied on the homogeneous distribution and from these optimal aspect ratios have been found. However, to avoid a flutter echo and other unwanted artifacts typical for rectangular rooms, enclosures with nonrectangular shapes are often recommended. Another acoustic issues in small rooms stem from some amplification or attenuation of sound at certain frequencies because it causes the so-called boomy sound and unwanted sound colouration. These effects and improper reverberation properties may prevent the correct perception of sound in rooms where speech, music, listening or recording is part of normal use. Therefore, the acoustic behaviour of small spaces has been extensively studied when they were used as performance studios, studio control rooms, listening rooms, audio program assessment rooms or small conference and lecture rooms.

Task 3

Synthesise the following two groups of information from different sources in physics.

Group 1

From M. Fernandez, 2017:

The artificial neural networks are able to predict the behaviour of the nuclear power system with good accuracy.

From J. Lee, 2017:

Advanced nuclear energy systems will utilise remotely-handled facilities, and the issue of safeguardability should be addressed.

Your synthesis of the information from the two sources above:

Group 2

From P. Wang et al, 2015:

Photonic crystals (PCs) are periodic nanostructures designed to control the motion of photons in a similar way that periodic semiconductor crystals affect the motion of electrons.

From Q. Zhang & G. Cao, 2011:

Nanotechnology opens a door to tailoring materials and creating various nanostructures for use in dye-sensitised solar cells.

Your synthesis of the information from the two sources above:

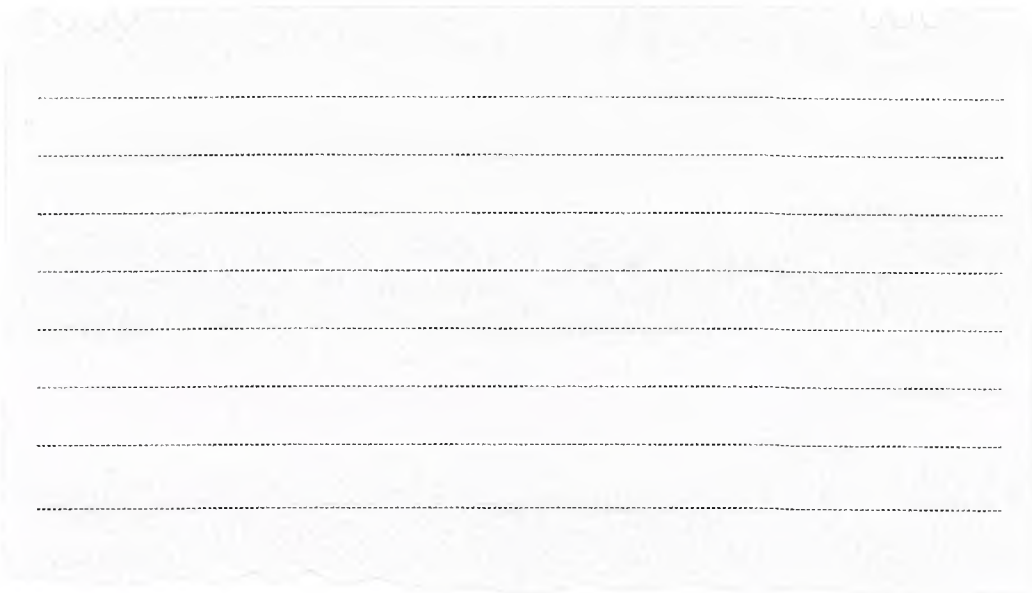
Write a synthesis to review previous research on photonics applied
Task 4 in image recognition for quality control. Writing requirements
and a writing checklist are provided as follows.

Writing requirements:

1. Your synthesis should review at least four research papers or other sources.
2. You will write at least 100 words.
3. Your synthesis should contain a reference list.

Writing checklist:

1. Search for relevant literature on this topic.
2. Read through the literature to find out similarities or differences among those previous studies.
3. Synthesise the previous research in your own words (at least 100 words).
4. Organise your review with a proper pattern or structure, using cohesive devices to indicate the organisation.
5. Follow a specific format to include citations in your statement.

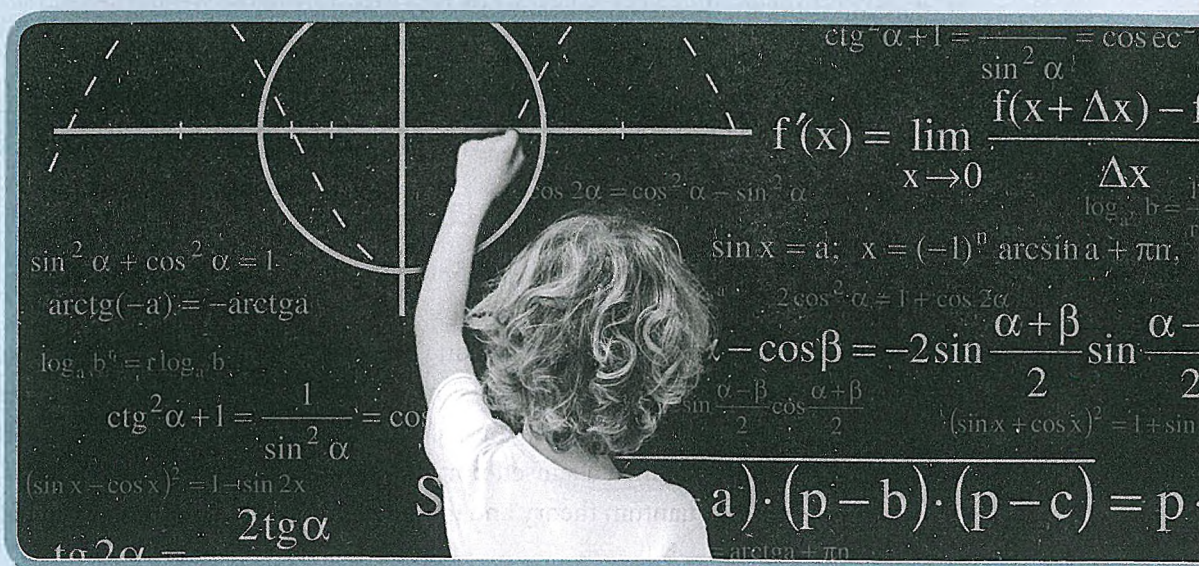
A large rectangular area with horizontal dashed lines, intended for writing the literature review. The lines are evenly spaced and cover the entire width of the box.

Unit 4

Mathematics

Beauty is the first test; there is no permanent place in the world for ugly mathematics.

— G. H. Hardy



1 Academic Reading

Text A: Magic Numbers: Can Maths Equations Be Beautiful?

Academic Reading Skills: Evaluating Facts or Evidence

2 Academic Writing

Text B: Seeing as Understanding: The Importance of Visual Mathematics for Our Brain and Learning

Academic Writing Skills: Choosing Effective Evidence to Support a Thesis

1

Academic Reading

Lead-in Questions

1. How do you think mathematicians are contributing to human progress?
2. Do you think mathematics is beautiful? Why or why not?

Text A

Magic Numbers: Can Maths Equations Be Beautiful?

Ian Sample

- 1 Paul Dirac had an eye for beauty. In one essay, from May 1963, the British Nobel laureate referred to beauty nine times. It makes four appearances in four consecutive sentences. In the article, he painted a picture of how physicists saw nature. But the word *beauty* never defined a sunset, nor a flower, or nature in any traditional sense. Dirac was talking quantum theory and gravity. The beauty lay in the mathematics.
- 2 What does it mean for maths to be beautiful? It is not about the appearance of the symbols on the page. That, at best, is secondary. Maths becomes beautiful through the power and elegance of its arguments and formulae; through the bridges it builds between previously unconnected worlds. When it surprises. For those who learn the language, maths has the same capacity for beauty as art, music, a full blanket of stars on the darkest night.
- 3 “The slow movement of the Mozart clarinet concerto¹ is a really beautiful piece of music, but I don’t print off a page of the score and put that on my wall. It’s not about that. It’s about the music and the ideas and the emotional response,” says Vicky Neale, a mathematician at Oxford University. “It’s the same with a piece of

¹ Mozart clarinet concerto: 莫扎特单簧管协奏曲

mathematics. It's not how it looks; it's about the underlying thought processes.”

- 4 Brain scans of mathematicians show that gazing at formulae considered beautiful by the beholder elicits activity in the same emotional region as great art and music. The more beautiful the formula, the greater the activity in the medial orbitofrontal cortex². “So far as the brain is concerned, maths has beauty just like art. There is common neurophysiological ground,” says Sir Michael Atiyah, an honorary professor of mathematics at Edinburgh University.
- 5 Ask mathematicians about the most beautiful equation and one crops up time and again. Written in the 18th century by the Swiss mathematician, Leonhard Euler³, the relation is short and simple: $e^{i\pi} + 1 = 0$. It is neat and compact even to the naive eye. But the beauty comes from a deeper understanding: Here the five most important mathematical constants are brought together. Euler's formula marries the world of circles, imaginary numbers and exponentials.
- 6 The beauty of other formulae may be more obvious. With $E=mc^2$, Albert Einstein built a bridge between energy and mass, two concepts that had previously seemed worlds apart. Maggie Aderin-Pocock, the space scientist, declared it the most beautiful equation and she is in good company. “Why is it so beautiful? Because it comes to life. Now energy will have mass and mass can be put into energy. These four symbols capture a complete world. It's difficult to imagine a shorter formula with more power,” says Robbert Dijkgraaf, director of the Institute for Advanced Study in Princeton, where Einstein was one of the first faculty members.
- 7 “One of the reasons there's almost an objective beauty in mathematics is that we use the word *beautiful* also to indicate the raw power in an idea. The equations or results in mathematics that are seen to be beautiful are almost like poems. The power per variable is something that is part of the experience. Just seeing a huge part of mathematics or nature being described with just a few symbols gives a great sense of elegance or beauty,” Dijkgraaf adds. “A second element is you feel its beauty is reflecting reality. It's reflecting a sense of order that's out there as part of the laws of nature.”

2 orbitofrontal cortex: 眶额皮层，位于大脑额叶前下方的前额皮层，是人类情绪产生的主要神经机制。

3 Leonhard Euler: 莱昂哈德·欧拉，瑞士数学家、自然科学家。

- 8 The power of an equation to connect what seem like completely unrelated realms of mathematics comes up often. Marcus du Sautoy, a maths professor at Oxford, has more than a soft spot for Riemann's formula. Published by Bernhard Riemann⁴ in 1859 (the same year Charles Darwin stunned the world with *On the Origin of Species*), the formula reveals how many primes⁵ exist below a given number, where primes are whole numbers divisible only by themselves and 1, such as 2, 3, 5, 7 and 11. While one side of the equation describes the primes, the other is controlled by zeros.
- 9 "This formula turns these indivisible prime numbers, into something completely different," says du Sautoy. "On the one side, you've got these indivisible prime numbers and then Riemann takes you on this journey to somewhere completely unexpected, to these things which we now call the Riemann zeros. Each of these zeros gives rise to a note – and it's the combination of these notes together which tell[s] us how the primes on the other side are distributed across all numbers."
- 10 More than 2,000 years ago, the ancient Greek mathematician, Euclid⁶, solved a numerical puzzle so beautifully that it still makes Neale smile every time it comes to mind. "When I think about beauty in mathematics, my first thoughts are not about equations. For me it's much more about an argument, a line of thinking, or a particular proof," she says.
- 11 Euclid proved there are infinitely many prime numbers. How did he do it? He began by imagining a universe where the number of primes was not infinite. Given a big enough blackboard, one could chalk them all up.
- 12 He then asked what happened if all these primes were multiplied together: $2 \times 3 \times 5$ and so on, all the way to the end of the list, and the result added to the number 1. This huge new number provides the answer. Either it is a prime number itself, and so the original list was incomplete, or it is divisible by a smaller prime. But divide Euclid's number by any prime on the list and always there is a 1 left over. The number is not divisible by any prime on the list. "It turns out you reach an absurdity, a contradiction," says Neale. The original assumption that the number of primes is finite must be wrong.

4 Bernhard Riemann: 伯恩哈德·黎曼，德国数学家，在数学分析和微分几何方面做出了重要贡献，开创了黎曼几何，并为后来爱因斯坦的广义相对论提供了数学基础。

5 primes: 质数，又称素数，指大于 1 且不能被 1 和它自身之外的其他自然数整除的自然数。

6 Euclid: 欧几里得，古希腊数学家，著有《几何原本》一书。

- 13 “The proof for me is really beautiful. It takes some thinking to get your head around it, but it doesn’t involve learning lots of difficult concepts. It’s surprising that you can prove something so difficult in such an elegant way,” Neale adds.
- 14 Behind beautiful processes lie beautiful mathematics. Well, some of the time. Hannah Fry, a lecturer in the mathematics of cities at UCL spent years staring at the Navier-Stokes equations. “They’re a single mathematical sentence that is capable of describing the miraculously beautiful and diverse behaviour of almost all of the earth’s fluids,” she says. With a grasp of the formulae, we can understand blood flow in the body, make boats glide through the water, and build awesome chocolate enrobers⁷.
- 15 In his 1963 essay, Dirac elevated beauty from an aesthetic response to something far more profound: a route to the truth. “It is more important to have beauty in one’s equations than to have them fit experiment,” he wrote, continuing: “It seems that if one is working from the point of view of getting beauty in one’s equations, and if one has really a sound insight, one is on a sure line of progress.” Shocking at first pass, Dirac captured what is now a common sentiment: When a beautiful equation seems at odds with nature, the fault may lie not with the maths, but in applying it to the wrong aspect of nature.
- 16 “Truth and beauty are closely related but not the same,” says Atiyah. “You are never sure that you have the truth. All you are doing is striving towards better and better truths and the light that guides you is beauty. Beauty is the torch you hold up and follow in the belief that it will lead you to truth in the end.”
- 17 Something approaching faith in mathematical beauty has led physicists to draw up two of the most compelling descriptions of reality: supersymmetry and string theory⁸. In a supersymmetric universe, every known type of particle has a heavier, invisible twin. In string theory, reality has 10 dimensions, but six are curled up so tight they are hidden from us. The mathematics behind both theories are often described as beautiful, but it is not at all clear if either is true.
- 18 There is a danger here for mathematicians. Beauty is a fallible guide. “You can

7 enrobers: 浸挂糖衣机

8 supersymmetry and string theory: 超对称是费米子 (fermions) 和玻色子 (bosons) 之间的一种对称性, 该对称性至今尚未在自然界中观测到。弦理论认为所有的亚原子粒子都并非小点, 而是类似于橡皮筋的弦。

literally be seduced by something that is not correct. This is a risk,” says Dijkgraaf, whose institute motto, “Truth and Beauty”, features one naked and one dressed woman. “Sometimes I feel that physicists, like Odysseus, must tie themselves to the mast of the ship so they are not seduced by the Sirens of mathematics.”

- 19 It may be that mathematicians and scientists are the only groups that still use the word “beautiful” without hesitation. It is rarely employed by critics of literature, art or music, who perhaps fear it sounds superficial or kitschy.
- 20 “I’m very proud that in mathematics and science the concept of beauty is still there. I think it’s an incredibly important concept in our lives,” says Dijkgraaf. “The sense of beauty we experience in maths and science is a multidimensional sense of beauty. We don’t feel it’s in any conflict with being deep, or interesting, or powerful, or meaningful. For the mathematician, it’s all captured by that one word.”

New Words and Expressions

laureate /'lɔːriət/ *n.* someone who has been given an important prize or honour, especially the Nobel Prize 得奖人, 获奖者

consecutive /kən'sekjʊtɪv/ *adj.* consecutive numbers or periods of time follow one after the other without any interruptions 相继的, 连续的

underlying /ˌʌndə'laɪ-ɪŋ/ *adj.* the cause, idea etc. that is the most important, although it is not easily noticed 基本的, 根本的

elicit /ɪ'lisɪt/ *vt.* to succeed in getting information or a reaction from someone, especially when this is difficult 探得, 引出

faculty /'fækəlti/ *n.* all the teachers in a university 全体教职员

stun /stʌn/ *vt.* to surprise or upset someone so much that they do not react immediately 使目瞪口呆

口呆, 使大吃一惊

glide /glɑɪd/ *vi.* to move smoothly and quietly, as if without effort 悄然滑行

aesthetic /i:s'θetɪk, es-/ *adj.* connected with beauty and the study of beauty 美的, 美学的

compelling /kəm'pelɪŋ/ *adj.* very interesting or exciting, so that you have to pay attention 引人注目的

fallible /'fælɪbəl/ *adj.* able to make mistakes or be wrong 难免有错误的, 会犯错的

seduce /sɪ'dju:s/ *vt.* to make someone want to do something by making it seem very attractive or interesting to them 迷惑, 使入迷

kitschy /'kɪtʃi/ *adj.* something that appeals to popular or lowbrow taste and is often of poor quality (艺术作品) 浅薄鄙俗的

Building Your Vocabulary

Task 1

Complete the following sentences using the expressions in the box below. Change the form where necessary.

underlying	compelling	diversity	fallible
multidimensional	indivisibility	elicit	give rise to

1. The book engagingly relates mathematics to _____ real-life problems in biology and contemporary society and shows how mathematical tools can be used to gain insight into these modern, common problems to provide effective, real solutions.
2. By analysing the most elegant and intricate pattern known to mathematics, Lisi has uncovered a(n) _____ relationship among all the universe's particles and forces, including gravity.
3. In all demonstrative sciences the rules are certain and infallible; but when we apply them, our _____ and uncertain faculties are very apt to depart from them, and fall into error.
4. As part of a study into the mathematical understanding of engineering students, a questionnaire has been developed which seeks to _____ from students their concept images attached to key mathematical concepts.
5. In mathematical system theory, a(n) _____ system or m-D system is a system in which not only one dependent variable exists (like time), but there are several independent variables.
6. The movie *Hidden Figures* tells the story of Katherine Johnson, Dorothy Vaughan and Mary Jackson. These black women represented dozens, if not hundreds, of black female "computers" whose largely unrecognised intellectual contributions after World War II _____ the US space programme.
7. Business leaders are understanding the importance of a cohesive, collaborative and creative working environment and luckily, are acknowledging that this can be done most effectively with a(n) _____ workforce.
8. The concept of the interdependence and _____ of human rights, as originally conceived, refutes any suggestion of a hierarchy of rights.

Understanding the Text

Task 2 Choose the statement that best describes the attitude of the author.

- A The beauty of mathematics lies in the elegant appearance of the symbols.
- B There is always beautiful mathematics behind beautiful processes.
- C The beauty we perceive in mathematics is not at odds with its power and meaning.

Task 3 Identify supporting details for the following questions.

1. Why do we feel an almost objective beauty in mathematics?

2. According to the author, what is the relationship between truth and beauty?

3. Why does the author say “beauty is a fallible guide”?

Academic Reading Skills

Evaluating Facts or Evidence

Evidence is the concrete facts used to support a claim. Ideally, evidence is something everyone agrees on, or something that anyone could, with sufficient training and equipment, verify for himself/herself. Evaluating evidence is an essential skill for you to read critically in a scholarly environment.

Before you evaluate the evidence, you should:

- ✓ identify the point the author is trying to prove (the claim),
- ✓ identify the specific facts the author gives to support the claim (the evidence), and
- ✓ explain how the evidence is supposed to relate to the claim (the relationship).

Once you have identified the claim, the evidence, and their relationship, you are on much stronger grounds for evaluating the evidence.

When evaluating evidence, you go beyond simply describing what it is and how it relates to the claim. You also say whether it is good or bad. Obviously, this is an essential step in evaluating the overall quality of an argument. If the evidence fails for any reason, the argument fails and the claim is not proven. To determine whether the evidence is proper or not, you should examine whether it is:

- ✓ relevant: having a definite relationship to the claim the author is trying to prove
- ✓ sufficient: enough to make the argument convincing
- ✓ reliable: from credible and authoritative sources
- ✓ current: the latest and most updated information

Task 1 Answer the following questions based on Text A.

1. What types of evidence does the author use to prove that mathematical equations can be beautiful?

2. How do you evaluate the neuroscience of mathematical beauty as a piece of evidence in this article?

3. How does the supersymmetry and string theory (Para. 17) fit into the author's overall idea of mathematical beauty?

Task 2

Read the following excerpts on the importance of mathematics and discuss the type(s) of evidence used.

Excerpt 1:

"The language of Nature is mathematics." How much or little we comprehend maths is exactly how much or little we comprehend Nature. Comprehension is probably a function of brain wiring; some are better at it than others. Progress in commerce, science, and technology all depends upon maths comprehension; we may have had to learn calculus to do physics problems, but at the first, Newton's study of physics led him (and Leibniz) to develop calculus! George Boole in the 1800s developed a branch of algebra for logic manipulations; every computer in existence relies on them now.

Excerpt 2:

I totally agree with those who talk about the everyday practical applications of maths. However, those are man-made necessities. I think the most fascinating context in which we see maths is Nature. Consider the Fibonacci numbers. Tree branching, leaf placement, artichokes, pineapples and pine cones, the spiral of a conch shell – all reflect Fibonacci numbers. When we look to atomic structure, DNA, etc., there are more mathematical echoes. Maths is part of the universe.

Excerpt 3:

A recent Gallup poll concluded that Americans consistently rate mathematics the most valuable subject they took in school, ahead of English, science, and history. Specifically, 34% of those polled in both 2002 and 2013 rated mathematics the most important subject. English, meaning English, reading, and literature, came in second, with 21% in 2013 rating English the most important. Between 2002 and 2013, incidentally, science jumped from 4% to 12%.

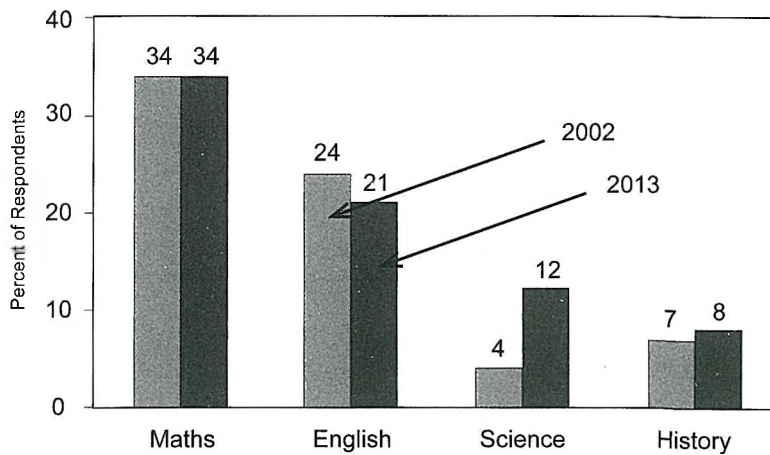


Figure 1 Percentage of responses to Gallup polls taken in 2002 and 2013

Task 3

Read the following report on the neuroscience of mathematical beauty and answer the questions.

The latest neuroscience of aesthetics suggests that the experience of visual, musical, and moral beauty all recruit[s] the same part of the “emotional brain”: field A1 of the medial orbitofrontal cortex (mOFC).

But what about mathematics? Plato believed that mathematical beauty was the highest form of beauty since it is derived from the intellect alone and is concerned with universal truths. Similarly, the art critic Clive Bell noted:

“Art transports us from the world of man’s activity to a world of aesthetic exaltation. For a moment, we are shut off from human interests; our anticipations and memories are arrested; we are lifted above the stream of life. The pure mathematician rapt in his studies knows a state of mind which I take to be similar, if not identical. He feels an emotion for his speculations which arises from no perceived relation between them and the lives of men, but springs, inhuman or super-human, from the heart of an abstract science. I wonder, sometimes, whether the appreciators of art and of mathematical solutions are not even more closely allied.”

A new study suggests that Bell might be right. Semir Zeki and [his] colleagues recruited 16 mathematicians at the postgraduate or postdoctoral level as well as 12 non-mathematicians. All participants viewed a series of mathematical equations in the fMRI scanner and were asked to rate the beauty of the equations as well as their understanding of each equation. After they were out of the scanner, they filled out a questionnaire in which they reported their level of understanding of each equation as well as their emotional experience viewing the equations.

This equation was most consistently rated as beautiful (Leonhard Euler’s identity):

$$e^{i\pi} + 1 = 0$$

Other winners of the equation beauty contest included the Pythagorean identity, the identity between exponential and trigonometric functions derivable from Euler’s formula for complex analysis, and the Cauchy-Riemann equations. In contrast, this equation was most consistently rated as ugly (Srinivasa Ramanujan’s infinite series for $1/\pi$):

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{k=0}^{\infty} \frac{(4k)!(1103 + 26390k)}{(k!)^4 396^{4k}}$$

Other low-rated equations included Riemann’s functional equation, the smallest number expressible as the sum of two cubes in two different ways, and an example of an exact sequence where the image of one morphism equals the kernel of the next.

Looking at the brain scans, the researchers found that the experience of mathematical beauty was related to the same part of the brain that has been found in prior studies to be associated with the experience of visual, musical, and moral beauty: field A1 of the medial orbitofrontal cortex (mOFC). What's more, the stronger the reported intensity of the experience, the stronger the brain activation in this area.

Interestingly, understanding and beauty were far from perfectly correlated. Even the non-mathematical participants found some equations more beautiful than others, even though they didn't understand the equations. The activation of field A1 of the mOFC was particularly related to ratings of beauty, not understanding.

1. What argument is the author trying to prove?

2. What evidence does the author use to prove the argument?

3. What findings does the brain scan study produce?

Task 4

Read the above text again and evaluate whether we can draw the following conclusions from the findings in the text. Why or why not?

1. The mOFC is solely responsible for processing mathematical beauty.

2. The mOFC is dedicated to processing beauty.

3. The cluster of neurons in field A1 of the mOFC caused the experience of mathematical beauty.

Academic Writing

2

Seeing as Understanding: The Importance of Visual Mathematics for Our Brain and Learning

Text B

Jo Boaler et al

- 1 Good mathematics teachers typically use visuals, manipulative and motion to enhance students' understanding of mathematical concepts, and the US national organizations for mathematics, such as the National Council for the Teaching of Mathematics (NCTM) and the Mathematical Association of America (MAA) have long advocated for the use of multiple representations in students' learning of mathematics. But for millions of students in US mathematics classes, mathematics is presented as an almost entirely numeric and symbolic subject, with a multitude of missed opportunities to develop visual understanding. Students who display a preference for visual thinking are often labeled as having special educational needs in schools, and many young children hide their counting on fingers, as they have been led to believe that finger counting is babyish or just wrong. This short paper, collaboration between a neuroscientist and mathematics educators, shares stunning new evidence from the science of the brain, showing the necessity and importance of visual thinking – and, interestingly, finger representations – to all levels of mathematics.
- 2 In recent years, scientists have developed a more nuanced understanding of the ways our brains work when we study and learn mathematics. Our brains are made up of “distributed networks”, and when we handle knowledge, different areas of the brain light up and communicate with each other. When we work on mathematics, in particular, brain activity is spread out across widely-distributed networks, which include two visual pathways: the ventral and dorsal visual pathways. Neuroimaging has shown that even when people work on a number calculation, such as 12×25 ,

with symbolic digits (12 and 25), our mathematical thinking is grounded in visual processing (Figure 2).

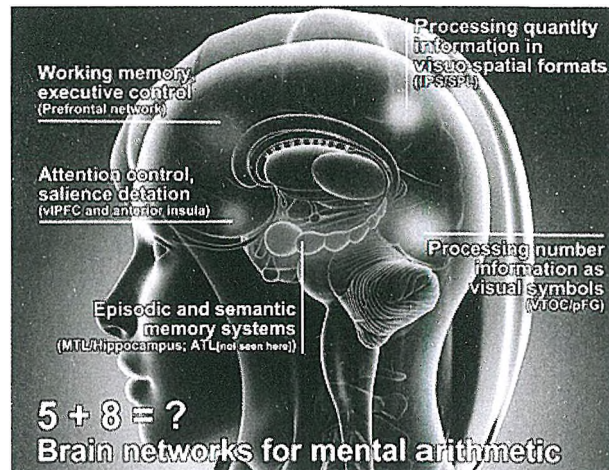


Figure 2 Distributed networks

- 3 A widely distributed brain network underpins the mental processing of mathematics knowledge. The area of the brain shown in green [IPS/SPL], which is part of the dorsal visual pathway, has reliably been shown to be involved when both children and adults work on mathematics tasks. This area of the brain particularly comes into play when students consider visual or spatial representations of quantity, such as a number line. A number line representation of number quantity has been shown in cognitive studies to be particularly important for the development of numerical knowledge and a precursor of children's academic success.
- 4 Researchers even found that after four 15-minute sessions of playing a game with a number line, differences in knowledge between students from low-income backgrounds and those from middle-income backgrounds were eliminated.
- 5 The researchers in the study highlighted the importance of students learning numerical knowledge through linear representations and visuals. This is just one of many studies that show that visual mathematics problems help students and raise achievement. The brain research sheds light on this, as it is showing that the dorsal visual pathway is the core brain region for representing the knowledge of quantity.
- 6 One yet-to-be published study from our colleagues at Stanford, with children between the ages of 8 and 14, showed that as children get older they develop part of the ventral visual pathway, shown in orange [VTOC/pFG] in Figure 2, and the brain

becomes more sensitive and specialized in representing visual number forms. The study also showed an important and increased interaction between the two visual pathways. This indicates that as children learn and develop, the brain becomes more interactive, connecting the visual processing of symbolic number forms, such as the number 10, with visuo-spatial knowledge of quantity, such as an array of dots or another visual representation. Different areas of the brain are involved when we think mathematically, including the frontal networks shown in red and purple [those on the left of Figure 2], the medial temporal lobe and, importantly, the hippocampus – the horseshoe shaped area in red [bottom left]. The important point that we want to stress in this paper is that the neurobiological basis of mathematics cognition involves complicated and dynamic communication between the brain systems for memory, control and detection and the visual processing regions of the brain.

- 7 A compelling and rather surprising example of the visual nature of mathematical activity in the brain comes from a new study on the ways that the brain uses representations of fingers, well beyond the time and age that people use their fingers to count. The different studies on the brain's use of finger representations give fascinating insights into human learning and clear implications for mathematics classrooms.
- 8 Berteletti and Booth studied one specific region of our brain that is dedicated to the perception and representation of fingers, known as the somatosensory finger area of the brain. Remarkably brain researchers know that we “see” a representation of our fingers in our brains, even when we do not use fingers in a calculation. Berteletti and Booth found that when 8–13 year olds were given complex subtraction problems, the somatosensory finger area lit up, even though the students did not use their fingers.
- 9 We “see” a representation of fingers in our brains when we calculate.
- 10 The researchers also found that this finger representation area was involved to a greater extent with more complex problems that involved higher numbers and more manipulation. Penner-Wilger found that even university students' somatosensory knowledge of fingers predicts their calculation scores. She also found that finger perceptions in Grade 1 predict performance on number comparison and estimation in Grade 2. Researchers assess whether children have a good awareness of their fingers by touching the finger of a student – without the student seeing which

finger is touched – and asking them which finger is being touched to perceive and represent their own fingers, they develop better representations of their fingers, which leads to higher mathematics achievement. Researchers found that when 6-year-olds improved the quality of their finger representation they improved in arithmetic knowledge, particularly subitizing, counting and number ordering. Remarkably the 6-year-old's finger representation was a better predictor of future mathematics success than their scores on tests of cognitive processing. Subitizing is the process of estimating small quantities such as 1, 2 or 3 without counting.

Academic Writing Skills

Choosing Effective Evidence to Support a Thesis

One of the most important skills in academic writing is to use effective evidence to prove or disprove a claim. Evidence is the facts, examples, or sources used to support a claim. In the sciences, this might be data retrieved from an experiment or a scientific journal article. In the humanities, it may be a quotation from a text, published information from academic critics, or a theory that supports your claims. Evidence can be separated into two categories – primary and secondary sources.

What are primary sources?

Primary sources are first-hand experiences, accounts, observations, reports, or narratives. Primary sources could include diaries, letters, contemporary newspapers, or eyewitness accounts of events. Official documents (e.g. the Canadian Charter of Rights and Freedoms), data collected from surveys, and lab results are also primary sources. In the humanities, the text you are writing about is also considered your primary text. So, for example, if you are writing a paper on Macbeth, then the play is your primary source. In the sciences, primary sources are also the results of an experiment that have been peer-reviewed and published in an academic journal.

What are secondary sources?

Secondary sources are critiques written by academics and scholars. These sources are considered secondary because they examine primary sources to present an argument or support a point of view; as such, they may be selective with their evidence or insert themselves in a debate occurring among a number of scholars. In the sciences, reviews, which are surveys of articles that demonstrate an understanding of a field, are considered secondary. It is a good idea to be aware of the bias in secondary sources when employing them as evidence.

Among the forms of evidence you might draw from are:

- ✓ graphs, charts, tables, or figures
- ✓ statistics
- ✓ experiments or studies done by peer-reviewed sources
- ✓ surveys conducted by reputable sources

- ✓ interviews
- ✓ quotes or paraphrases from primary sources
- ✓ quotes or paraphrases from secondary sources

To be considered effective, your supporting evidence should have a definite relationship to the claim you want to prove (*relevant*), be enough to make the argument convincing (*sufficient*), originate from credible and authoritative sources (*reliable*), and cite the latest and most updated information (*current*).

Task 1 Answer the following questions based on Text B.

1. What is the authors' main claim in Text B?

2. What forms of evidence do the authors use to prove their claim?

3. How do you evaluate the authors' evidence in terms of effectiveness?

Task 2

Identify the evidence the author used to support his argument in the following passage.

Why Math Works

Mario Livio

Mathematics is unreasonably effective in two distinct ways, one I think of as active and the other as passive. Sometimes scientists create methods specifically for quantifying real-world phenomena. For example, Isaac Newton formulated calculus for the purpose of capturing motion and change, breaking them up into infinitesimally small frame-by-frame sequences. Of course, such active inventions are effective; the tools are, after all, made to order. What is surprising, however, is their stupendous accuracy in some cases. Take, for instance, quantum electrodynamics, the mathematical theory developed to describe how light and matter interact. When scientists use it to calculate the magnetic moment of the electron, the theoretical value agrees with the most recent experimental value – measured at 1.00115965218073 in the appropriate units in 2008 – to within a few parts per trillion!

Even more astonishing, perhaps, mathematicians sometimes develop entire fields of study with no application in mind, and yet decades, even centuries, later physicists discover that these very branches make sense of their observations. Examples of this kind of passive effectiveness abound. French mathematician Évariste Galois, for example, developed group theory in the early 1800s for the sole purpose of determining the solvability of polynomial equations. Very broadly, groups are algebraic structures made up of sets of objects (say, the integers) united under some operation (for instance, addition) that obey specific rules (among them the existence of an identity element such as 0, which, when added to any integer, gives back that same integer). In 20th-century physics, this rather abstract field turned out to be the most fruitful way of categorizing elementary particles – the building blocks of matter. In the 1960s physicists Murray Gell-Mann and Yuval Ne'eman independently showed that a specific group, referred to as $SU(3)$, mirrored a behavior of subatomic particles called hadrons – a connection that ultimately laid the foundations for the modern theory of how atomic nuclei are held together.

The study of knots offers another beautiful example of passive effectiveness. Mathematical knots are similar to everyday knots, except that they have no loose ends. In the 1860s Lord Kelvin hoped to describe atoms as knotted tubes of ether. That misguided model failed to connect with reality, but mathematicians continued to analyze knots for many decades merely as an esoteric arm of pure mathematics.

Amazingly, knot theory now provides important insights into string theory and loop quantum gravity – our current best attempts at articulating a theory of space-time that reconciles quantum mechanics with general relativity. Similarly, English mathematician Hardy’s discoveries in number theory advanced the field of cryptography, despite Hardy’s earlier proclamation that “no one has yet discovered any warlike purpose to be served by the theory of numbers.” And in 1854 Bernhard Riemann described non-Euclidean geometries – curious spaces in which parallel lines converge or diverge. More than half a century later Einstein invoked those geometries to build his general theory of relativity.

A pattern emerges: Humans invent mathematical concepts by way of abstracting elements from the world around them – shapes, lines, sets, groups, and so forth – either for some specific purpose or simply for fun. They then go on to discover the connections among those concepts. Because this process of inventing and discovering is man-made – unlike the kind of discovery to which the Platonists subscribe – our mathematics is ultimately based on our perceptions and the mental pictures we can conjure. For instance, we possess an innate talent, called subitizing, for instantly recognizing quantity, which undoubtedly led to the concept of number. We are very good at perceiving the edges of individual objects and at distinguishing between straight and curved lines and between different shapes, such as circles and ellipses – abilities that probably led to the development of arithmetic and geometry. So, too, the repeated human experience of cause and effect at least partially contributed to the creation of logic and, with it, the notion that certain statements imply the validity of others.

Read the following reasons why we should study mathematics in college or university. Provide evidence to support each claim.

Task 3

Make sure your evidence is relevant, sufficient, reliable and current.

Why Study Mathematics

Reason 1: Studying mathematics sharpens your analytical skills.

Evidence: _____

Reason 2: Academic research in other subject areas relies on quantitative and statistical reasoning.

Evidence: _____

Reason 3: Many career paths require mathematical skills.

Evidence: _____

Reason 4: Mathematics is worth studying simply for its elegance.

Evidence: _____

Write a proposal to your university leaders to make your case for the importance of offering statistics courses to all undergraduate students irrespective of their majors. Your writing should be no more than 300 words.

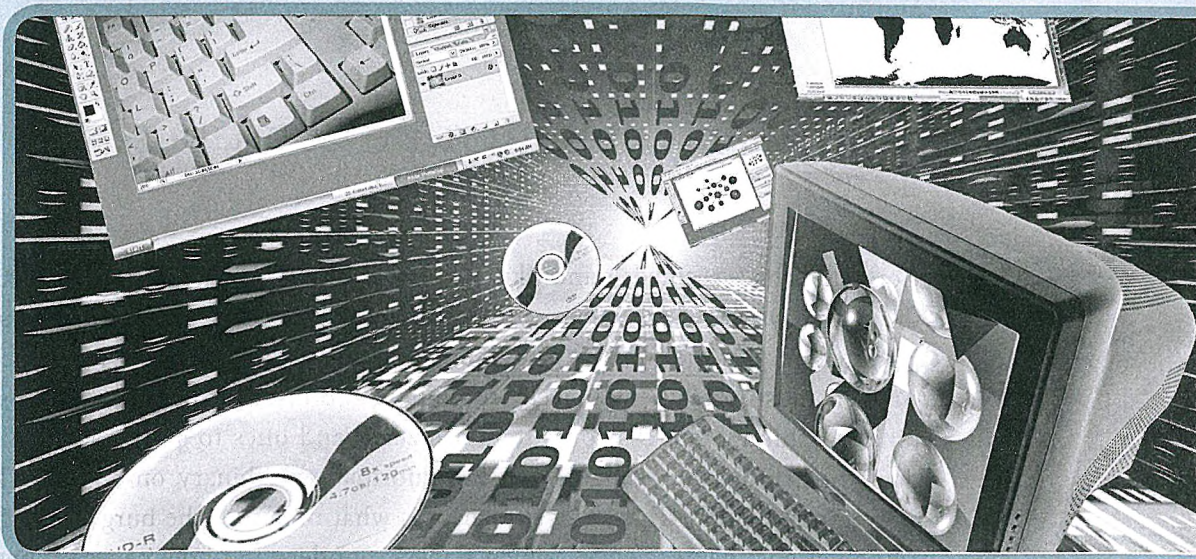
Task 4

Unit 5

Computer Science

I am in the camp that is concerned about super intelligence. First the machines will do a lot of jobs for us and not be super intelligent. That should be positive if we manage it well. A few decades after that, though, the intelligence is strong enough to be a concern.

— Bill Gates



1 Academic Reading

Text A: Can Computers Be Conscious?

Academic Reading Skills: Evaluating Arguments

2 Academic Writing

Text B: ROBUST Path Strategy Evaluator

Academic Writing Skills: Identifying Limitations and Indicating Research Gaps

1

Academic Reading

Lead-in Questions

1. Do you think that AI will someday beat human beings and threaten humanity's existence? Why or why not?
2. Do you think AI is able to experience emotions and have consciousness the way humans do?

Text A

Can Computers Be Conscious?

Max Miller

- 1 Computers have seemed “mind-like” to people since they were invented in 1950s. In the early days they were widely called “electronic brains” for their ability to process information. But the similarity between computers and brains isn't just superficial: At their most fundamental levels, computers and brains process data in a similar binary fashion. Whereas computers use zeros and ones to store and manipulate data, the neurons in our brains transmit information in binary, on/off spikes known as action potentials. This basic similarity is what underlies the burgeoning field of computational neuroscience¹, which hopes to understand how neuronal networks give rise to processes like memory and facial recognition so that they might be replicated in intelligent machines.
- 2 But artificial intelligence has progressed slower than many had initially hoped. Yes, AI may have solved the game of checkers, but this is a far cry from being able to simulate consciousness. The central problem remains: We have no real understanding of how the brain gives rise to the mind, of how neurons and action potentials create consciousness.

1 computational neuroscience: 计算神经科学，又称为神经信息学，是一门了解大脑如何运作的科学。

- 3 Instead of trying to build thinking machines from the ground up, several major projects have recently turned to a new approach: replicating virtual brains through reverse-engineering². By studying the neural networks in the brain, scientists have constructed computer-based models that mirror the brain's complex biological networks. In turn, they can then run experiments on these brain-like computers in order to learn about how the brain thinks.
- 4 Henry Markram is the South African neuroscientist who heads the Blue Brain Project at the Ecole Polytechnique Federale de Lausanne in Switzerland. For 15 years, Markram and his team collected data from the neocortexes³ of rats' brains with the hopes of integrating it into a 3D model. If they could accurately recreate the behaviours and structures of a biological brain, their computer simulation should shed light on both normal cognition and disorders like depression and schizophrenia⁴. In its trial stages the project successfully recreated a single neocortical column of a two-week-old rat, which contains about 10,000 neurons. Of course, this sample is infinitesimally small compared to the 100 billion neurons in a human brain. But this project is all a matter of scaling. "Technologically, in terms of computers and techniques to acquire data, it will be possible to build a model of the human brain within 10 years," Markram told *Discover* magazine last year.
- 5 But will this full-scale model teach us how to re-create consciousness, or perhaps even become conscious itself? "It's really difficult to say how much detail is needed for consciousness to emerge," said Markram. "I do believe that consciousness is an emergent phenomenon. It's like a shift from a liquid to a gas ... It's like a machine that has to run fast enough and suddenly it's flying." In other words, they can't know for sure until the model is finished.
- 6 Even if the model can learn and reason, that doesn't guarantee that it will be a truly intelligent being. Many people studying AI have equated problem-solving with thinking, but thinking is different from reasoning, says Yale computer scientist David Gelernter. To demonstrate this, he points to daydreaming and free association. "Free association is a kind of thinking also. My mind doesn't shut off, but I'm certainly not solving problems; I'm wandering around."

2 reverse-engineering: 逆向工程（又称逆向技术），是一种产品设计技术再现过程，即对一目标产品进行逆向分析及研究，从而演绎并得出该产品的处理流程、组织结构、功能特性及技术规格等设计要素，以制作出功能相近但又不完全一样的产品。

3 neocortexes: 新（大脑）皮层

4 schizophrenia: [医]精神分裂症

- 7 “The field of Artificial Intelligence had studied only the very top end of the spectrum and still tends to study only the very top end,” says Gelernter. “It tends to say, what is thinking? It’s this highly focused, wide awake, alert, problem-solving state of mind. But not only is that not the whole story, but the problem – the biggest unsolved problem that has tended to haunt philosophy of mind, cognitive psychology, and AI – is creativity.”
- 8 The general consensus is that creativity is the ability to invent new analogies, to connect two things that are not obviously related. And this invention of analogy relies not on analytic problem-solving thought but on letting your mind drift from one thought to another in a sort of free-associative state, says Gelernter. “Creativity doesn’t operate when your focus is high,” Gelernter writes in an essay for *Edge*. “Only when your thoughts have started to drift is creativity possible. We find creative solutions to a problem when it lingers at the back of our minds, not when it monopolises attention by standing at the front.”
- 9 So how can computers create new analogies? The answer probably has something to do with emotion, says Gelernter. “Emotion is what allows us to take two thoughts or ideas that seem very different and connect them together, because emotion is a tremendously subtle kind of code or tag that can be attached to a very complicated scene.” We tend to think of emotions in discrete terms, like *happy*, *sad*, and *angry*, but they’re really much more subtle than that. “If I say, ‘What is your emotion on the first really warm day in April or March when you go out and you don’t need a coat and you can smell the flowers blooming and there may be remnants of snow but you know it’s not going to snow anymore and there’s a certain springiness in the air, what do you feel?’” Gelernter asks. “It’s not that you feel happy exactly. There are a million kinds of happiness. It’s a particular shade of emotion.” Though there may not be an exact word to describe this nuanced emotion, the mind can recognise it and can connect two very different scenes that may have inspired the same emotion.
- 10 The other difficulty with emotion – and the reason why computers won’t ever be able to experience emotions the way humans do – is that they are produced by an interaction between the brain and the body working together. “When you feel happy, your body feels a certain way, your mind notices, and the resonance between body and mind produces an emotion,” Gelernter explains. Until computers can simulate this experience, they will never be truly intelligent.

New Words and Expressions

binary /'bʌməri/ *adj.* the binary system: a system of counting, used in computers, in which only the numbers 0 and 1 are used (计算机运算系统) 二进制

manipulate /mə'nɪpjʊleɪt/ *vt.* to work skilfully with information, systems, etc. to achieve the result that you want (熟练地) 操作, 使用 (信息、系统等)

burgeoning /'bɜ:dʒənɪŋ/ *adj.* growing, increasing, or developing very quickly 迅速增长 (或发展) 的

simulate /'sɪmjʊleɪt/ *vt.* to make or produce something that is not real but has the appearance or feeling of being real 模拟, 模仿

infinitesimal /ˌɪnfɪnɪ'tesɪməl/ *adj.* extremely small 无限小的; 极微小的

emergent /ɪ'mɜ:dʒənt/ *adj.* (only before noun) in an early state of development 新兴的, 发展初期的

analogy /ə'nælədʒi/ *n.* something that seems similar in two situations, processes, etc. 类似处, 相似处

haunt /hɔ:nt/ *vt.* to cause problems for someone over a long period of time 长期困扰

consensus /kən'sensəs/ *n.* an opinion that everyone in a group agrees with or accepts 共同意见, 一致看法, 共识

monopolise /mə'nɒpəlaɪz/ *vt.* to use a lot of someone's time or attention 占用 (某人大量时间或注意力)

subtle /'sʌtl/ *adj.* not easy to notice or understand unless you pay careful attention 难以察觉 (理解) 的; 微妙的, 细微的

discrete /dɪ'skri:t/ *adj.* clearly separate 分开的, 分离的, 离散的

remnant /'remnənt/ *n.* a small part of something that remains after the rest of it has been used, destroyed, or eaten 残余部分, 剩余部分

nuanced /'nju:ənst/ *adj.* slightly different 有细微差别的

resonance /'rezənəns/ *n.* the special meaning or importance that something has for you because it relates to your own experiences (因与亲身经历有关而产生的) 共鸣

Building Your Vocabulary

Task 1 Complete the following sentences using the words in the box below. Change the form where necessary.

emergent	monopolise	simulate	burgeoning
infinitesimally	underlie	haunt	

1. Although in both emotions sympathetic symptoms are present, different autonomic-somatic patterns _____ aggression and anxiety, respectively.
2. In addition, the results will closely _____ the process that occurs during real trading, in which traders frequently re-optimize their systems to bring them up-to-date with fundamental or technical changes in the trading market.
3. Unresolved issues often come back later to _____ us. Many of these challenges can be the result of early childhood experiences.
4. It's very unlikely that the _____ field of nano-biotechnology will be reserved solely for medical uses.
5. In the absence of existing human "herd" immunity to this virus, only immunisation provides a significant hope of suppressing the long-term impact of this newly _____ virus.
6. Thus, virtually all her time and energy is _____ by the children
7. The action of municipal, county, or state school boards or boards of education is small, _____ small in comparison with the number of districts.

Understanding the Text

Task 2 Choose the statement that best summarises the theme of Text A.

- A Although computers and brains are quite similar, it is still difficult for AI to simulate human consciousness.
- B Thinking is different from reasoning. AI's ability of learning and reasoning doesn't guarantee that it will be a truly intelligent being.
- C Without the help of emotion, it is difficult for AI to invent new analogies and become creative.

Read Paragraphs 6–10 in Text A and write a brief summary for the excerpt. Your summary should be no longer than one third of the length of the original excerpt (around 300 words).

A rectangular piece of white paper with horizontal ruling lines, resembling notebook paper. The paper has a slightly wavy, torn edge at the bottom. It is positioned below the task instructions and is intended for the student to write their summary on.

Academic Reading Skills

Evaluating Arguments

What are arguments?

An argument consists of two parts; one part (premise or evidence) supposedly provides a reason or a proof for thinking that the other part (the claim or the conclusion) is true. The core of critical thinking lies in the ability to identify, construct and evaluate arguments.

Some people may have some misunderstanding of the word “argument”. In the context of critical thinking, it does not refer to quarrels, complaints or any disapproval. Here an argument means the presentation of evidence to support a claim, or:

$$\text{Argument} = \text{Evidence} + \text{Claim}$$

Speaking of arguments, some pairs of terms are always used. Depending on usage and context, “claim” is the synonym of “conclusion”, “proposition”, and “opinion”. And “evidence” is the synonym of “reason”, “premise”, and “proof”. In an academic context, evidence may come from such sources as fieldwork and research.

Notes: Sometimes, the word “argument” will be used to refer only to a premise, as in “That’s a good argument for your conclusion.”

How to support a claim with evidence?

For critical readers, the analysis and evaluation of arguments is as important as it is for writers, who need to advance and support their claims or conclusions with evidence, which is also called the process of reasoning. Therefore, understanding the following three accepted ways to support ideas will help readers to evaluate whether the writer’s process of reasoning is relevant and sufficient.

First, presenting and explaining statistics. Statistics will be used to convey information in a numerical form (referred to as data). Statistics are convincing when they are used in combination with an explanation of why the numbers are significant.

Second, providing appropriate examples. Examples can support the writer’s contention that a general statement is true. Examples can provide specifics and details in support of

a claim, as well as vivid descriptions which will capture and retain reader's attention.

The third accepted way is quoting expert opinions. Expert opinions are usually based on factual evidence to interpret other facts.

These three commonly used ways to support claims can be used separately or in combination.

Determine which of the following passages contain arguments.

Task 1 Remember that an argument occurs when evidence is offered as a reason for believing that a claim is true.

1. Asked whether there will ever be computers as smart as people, the US mathematician and sci-fi author Vernor Vinge replied: "Yes, but only briefly." He meant that once computers get to this level, there's nothing to prevent them getting a lot further very rapidly.
2. One by one, computers take over domains that were previously considered off-limits to anything but human intellect and intuition. We now have machines that have trumped human performance in such domains as chess, trivia games, flying, driving, financial trading, face, speech and handwriting recognition – the list goes on.
3. By default, then, we seem to have no reason to think that intelligent machines would share our values. The good news is that we probably have no reason to think they would be hostile, as such: Hostility, too, is an animal emotion.
4. There are real dangers from AI but they tend to be economic and social in nature. Clever AI will create tremendous wealth for society, but will leave many people without jobs. Unlike the industrial revolution, there may not be jobs for segments of society as machines may be better at every possible job.

Task 2 Read the following excerpts and analyse how the evidence supports the claim(s).

Excerpt 1:

The way technologies are designed can solve or create new problems. For example, by making robots look like humans or cute animals, we may develop emotional affinity toward the machines. This could help promote trust with users – but perhaps also overtrust? Could we become co-dependent and be overattached to robots, causing a problem when they're not around?

Excerpt 2:

The giants of the net are rapidly advancing the art of artificial intelligence, teaching online services to recognise images, understand natural languages, and even carry on conversations – the kinds of artificial intelligence that will empower robots to tackle ever-more complex tasks. Using the AI that Google and Facebook use to identify photos on the net, researchers have already built machines that can teach themselves to screw on a bottle cap. “Today’s technology is different than what we’ve seen in the past,” says Martin Ford, the author of the recent book *Rise of the Robots: Technology and the Threat of a Jobless Future*. “The technology is taking on cognitive tasks. We now have machines and algorithms that can, at least in a limited sense, think.”

Excerpt 3:

In recent years, several object recognition models (including biologically inspired approaches) have shown what appears to be impressively high performance on this test – better than 60% correct, suggesting that these approaches, while still well below human performance, are at least heading in the right direction.

Task 3 Read the following passage and identify the evidence the author has provided in order to support the claim(s).

We should consider AI a threat for two reasons, but there are approaches we can take to minimise that threat. The first problem is that AI is often trained using a combination of logic and heuristics, and reinforcement learning. The logic and heuristics part has reasonably predictable results: We program the rules of the game or problems into the computer, as well as some human-expert guidelines, and then use the computer’s number-crunching power to think further ahead than humans can. This is how the

early chess programs worked. While they played ugly chess, it was sufficient to win.

Reinforcement learning, on the other hand, is more opaque.

We have the computer perform the task – playing Go, for example – repetitively. It tweaks its strategy each time and learns the best moves from the outcomes of its play. In order not to have to play humans exhaustively, this is done by playing the computer against itself. AlphaGo has played millions of games of Go – far more than any human ever has.

The problem is that AI will explore the entire space of possible moves and strategies in a way humans never would, and we have no insight into the methods it will derive from that exploration. In the second game between Lee Sedol and AlphaGo, the AI made a move so surprising – “not a human move” in the words of a commentator – that Lee Sedol had to leave the room for 15 minutes to recover his composure.

This is a characteristic of machine learning. The machine is not constrained by human experience or expectations. Until we see an AI do the utterly unexpected, we don't even realise that we had a limited view of the possibilities. AIs move effortlessly beyond the limits of human imagination.

In real world applications, the scope for AI surprises is much wider. A stock trading AI, for example, will reinvent every single method known to us for maximising return on investment. It will find several that are not yet known to us. Unfortunately, many methods for maximising stock returns – bid support, co-ordinated trading, and so on – are regarded as illegal and unethical price manipulation. How do you prevent an AI from using such methods when you don't actually know what its methods are? Especially when the method it's using, while unethical, may be undiscovered by human traders – literally, unknown to humankind? It's farcical to think that we will be able to predict or manage the worst case behaviour of AIs when we can't actually imagine their probable behaviour.

2

Academic Writing

Text B

ROBUST Path Strategy Evaluator

Angie Shia, Farokh B. Bastani, and I-Ling Yen

- 1 When a swarm of autonomous, distributed robots are deployed for a mission, the robots generally have various tasks to complete. Normally, task completion optimisation can be achieved if the robots are fully functional, the tasks are predetermined, and the robots have *a priori* knowledge of a deterministic and bounded environment. However, in dynamic, hostile environments, the robots may encounter many situations that can prevent them from achieving optimality or completing certain tasks. These situations include various types of damages or death, sensor malfunctions, entering areas that may affect the functionality of various robotic components, etc. Since there is no *a priori* knowledge of the environment, the robots may face situations that are not found in their knowledge base and they will need to dynamically determine appropriate subtasks to cope with such adverse scenarios. For example, if in a search and rescue mission, the robots discover some victims and need to move the injured ones or transport them to another location, they should avoid any areas that may harm the victims, such as extreme weather conditions, thick bushes, chemical spills, radiation, quarantine zones, etc.
- 2 To handle these situations, the robots must have an adaptive software system that can proactively cope with changes. This adaptive system should emulate the intelligence of human reasoning and common sense, and exhibit collective learning behaviours such as those observed in biological creatures such as swarms of bees or ants. The system should also include feedback for producing self-organisation and nonlinear dynamic behaviours with multiple strategies to choose from, depending on the situation at hand. Such a strategy evaluator (SE) should not simply be a long list of hard-coded heuristics rules since trying to encompass all possible situations is in general intractable. Further, in a dynamic environment, it is not even feasible since it cannot be known what to include in the knowledge base. A new method for strategy

selection must be developed so that these rules can be minimised.

- 3 There has been a lot of research on dynamic strategy selection using various techniques. Our work is similar to that of Tan, et al, and Sun, et al, in that we utilise the synergistic effects of neural network[s] and model free reinforcement learning for dynamic, hostile environments. However, there are some differences both in the approach and considerations between our systems and theirs. First, Tan, et al, and Sun, et al, both create a single complex neural network consisting of a large number of nodes. Tan's network ranges anywhere from 30 to 200 nodes to represent various sensors and input data. This can take a long time to converge and the error introduced can be cumbersome to trace. We have created a network of simple networks, where each simple network contains just six nodes. Each simple network learns only one component. The output of a simple network can also be an input to another simple network or can be fed into a consolidating heuristic function. Each learning phase for our modified reinforcement learning algorithm can accept a different, dynamic set of simple networks. This creates a flexible architecture that enables a robot to not only learn quickly, but also determine which component to learn and when to learn. Our architecture enables a robot to dynamically take a component online/offline so as to keep only the critical components online when necessary. Another difference between our approach and [that of] Tan, et al, is that their case study is an extension of the submarine mine avoidance problem addressed by Gordon and Subramanian where the focus is on a single entity learning whereas our study focuses on multi-robot learning, particularly swarm robots. Currently, there is active research on parallel (model free) reinforcement learning for multi-robots. Many of these techniques require tight integration and/or message broadcast through radio communication between robots. In our study, our concern is on swarm learning in dynamic, hostile environments. Autonomous and adaptive robots operating in such environments cannot guarantee sustainment of tight integration and radio communication requirements, such as those described in Wang, et al's or Ground and Kudenko's techniques. As per our previous work, we aim at designing a technique that does not assume that the robots can communicate or be able to maintain even partial communication via radio, vision, stigmergy, or any other communication devices. This technique must also not assume that the robots will be tightly integrated or be at a close range constantly. Furthermore, some studies have shown that parallel reinforcement learning has worse performance than the single robot reinforcement learning in the context of foraging.

- 4 It is through these considerations that we present in this paper a path strategy

evaluator (PSE) that learns an optimal path between two points in any given environment. By optimal path, we consider not just the distance, but also how to minimise damage to each robot (especially its critical functionalities) and also to enhance the likelihood that it will succeed in its mission, with minimal impositions on the functionality of the robots. Our evaluation shows that this PSE is able to learn a dynamic environment and its effect on a robot's critical components and output an optimal path for the robot. The contribution of this paper is a PSE that:

- ✓ learns the functionality of a robot's components under various environmental conditions;
 - ✓ identifies what it considers as “good” regions, namely, those that enable it to minimise damage and achieve its goal;
 - ✓ constructs an optimal path.
- 5 The rest of the paper is organised as follows: Section 2 models the environment, its conditions and the swarm robots and presents a detailed specification of the problem. It then presents our solution to the problem through our adaptive artificial neural network (ANN) and reinforcement learning based technique along with a case study. Section 3 illustrates a platform and implementation dependency free ontology to specify input parameters. Section 4 presents a detailed illustration of our ANN and Q-learning based algorithms. Section 5 describes our implementation and presents our test evaluation results. Section 6 analyses the various issues pertaining to reinforcement learning and swarm systems. It also discusses how our system can be extended for various other scenarios. Finally, Section 7 concludes the paper and outlines some future work.

Academic Writing Skills

Identifying Limitations and Indicating Research Gaps

After reviewing the literature, you advance to your present research to focus on the specific research problem you will be dealing with in the final part of your research paper's introduction. This part is mainly to identify the limitations in a certain field and therefore to indicate a research gap. After the identification of the research gap, a statement of purpose is usually given to state the objectives of your research. In the end of the introduction, as an optional choice, a statement of research value is sometimes provided to indicate possible benefits or applications.

A research gap is a significant research area that has never been touched by other researchers. It is a part that has been absent from present research literature and you figure it out with your research approach. When writing this part, you may indicate that an important part of the research field has been neglected by other researchers in the previous literature. Or you could pose a new question in your research area that has never been noticed by other researchers. There are certain language conventions to obey while writing this part. You can use special signal words to indicate the missing information, such as "*however, but, although, while*", and some modifiers like "*few, little, no*". The following sentences are often used to indicate research gaps.

However, few studies have been done on ...

In spite of this, very little is known about ...

Although many studies have been done on X (a research topic), *little* information is available on Y (another research topic).

Until now *no* field experiments of ... have been reported.

No such finding could be available in ...

Nevertheless, there is *limited* evidence that ...

Unfortunately, there are *few* studies on ...

To our best knowledge, *few* studies/researchers have examined/explored ...

To date, *little* research has investigated/focused on ...

So far, *little* evidence is available on ...

After the research gap is indicated, a statement of purpose follows to state the specific objective(s) of a research paper. Such sentences are usually used here: “The aim of the present paper is to ...” “The purpose of this study was to ...” Then, you may include a statement of value at the end of the introduction to justify your research. It states that you have made pragmatic contributions by putting into use the research outcomes or highlights the theoretical significance of the research in developing a certain research field.

When writing the statement of value, you usually use modal auxiliaries as well as other expressions (weak verbs, for example) to indicate tentativeness. With an attitude of modesty, you try to use words that do not sound too confident of the benefits of your work. Such modal auxiliaries as “may, should, could” are often used here. (see Task 3 below)

Notes: When presenting a gap, try to avoid strong criticism and use respectful and impersonal expressions instead.

Task 1

Identify the sentences concerned with the research gap(s) and the research purpose in Text B.

Sentence(s) to indicate the research gap(s):

Sentence(s) to state the research purpose:

Task 2

Underline the statements which identify the research gap(s) in the following excerpts.

Excerpt 1:

A common ML presumption is that algorithms can learn better with more data and consequently provide more accurate results. However, massive datasets impose a variety of challenges because traditional algorithms were not designed to meet such requirements. For example, several ML algorithms were designed for smaller datasets, with the assumption that the entire dataset can fit in the memory. Another assumption is that the entire dataset is available for processing at the time of training. Big Data breaks these assumptions, rendering traditional algorithms unusable or greatly impeding their performance.

Excerpt 2:

A comprehensive catalogue of MT evaluation techniques and their rich literature is given by Reeder. For the most part, these various human evaluation approaches are quite expensive. Moreover, they can take weeks or months to finish. This is a big problem because developers of machine translation systems need to monitor the effect of daily changes to their systems in order to weed out bad ideas from good ideas. We believe that MT progress stems from evaluation and that there is a logjam of fruitful research ideas waiting to be released from the evaluation bottleneck. Developers would benefit from an inexpensive automatic evaluation that is quick, language-independent, and correlates highly with human evaluation. We propose such an evaluation method in this paper.

Task 3

Complete the following statements to indicate a proper degree of tentativeness.

1. If we want our applications to access similar web APIs with the same flexibility as people browse similar webpages, we will _____ (need to/have to) fundamentally rethink the way in which the interfaces to those web APIs are designed.
2. Our IoT platform provides an immediacy for the health care practitioner that _____ (will improve/improves) clinical decision-making and ensure the right level of support _____ (should/can) be deployed at the earliest point of need to prevent an escalating crisis.
3. These recent industry and standardisation tendencies _____ (show/suggest) that TCP _____ (will/may) gain extensive support in IoT scenarios soon.
4. This tutorial _____ (seems to provide/is intended to provide) an overview of the basic theory of HMMs (as originated by Baum and his colleagues), provide practical

details on methods of implementation of the theory, and describe a couple of selected applications of the theory to distinct problems in speech recognition.

Task 4

The following excerpt is taken from an article in the field of Cloud Computing. Rearrange the following sentences in the correct order. The first two sentences have already been identified.

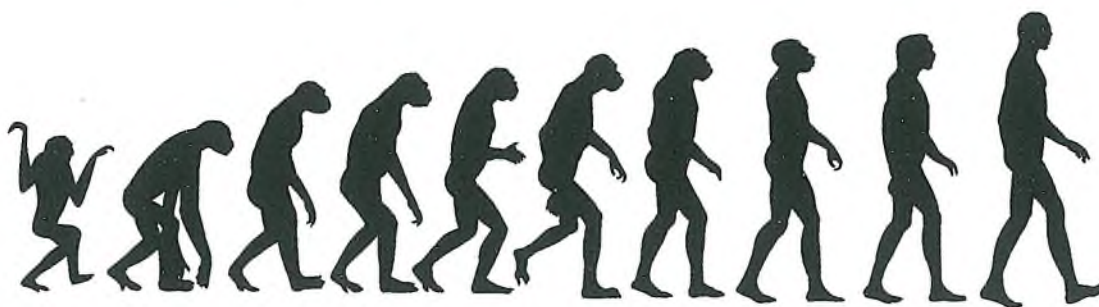
- 1 Cloud Computing is traditionally divided into three market segments: Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS).
- 2 To better understand cloud communications, it is useful to understand the different service models of Cloud Computing.
- ☐ Recent research on Cloud Computing has focused on the implementation of Service Level Agreements (SLA) and operation of large data centres.
- ☐ Rather than centralising Cloud Computing resources in large data centres, Distributed Cloud Computing resources are aggregated from a grid of standard PCs hosted in homes, offices and small data centres.
- ☐ The best known is SaaS where the customer purchases access to an application that is hosted and runs in the cloud.
- ☐ However, in case of Force Majeure such as natural disasters, strikes, terrorism, unpreventable accidents, etc., SLA can no longer be applied.
- ☐ PaaS refers to access to platforms that allow the customers to deploy their own applications in the cloud, and IaaS is at a lower level with access to the systems, storage, network connectivity, and OS management.

Unit 6

Biology

As many more individuals of each species are born than can possibly survive; and as, consequently, there is a frequently recurring struggle for existence, it follows that any being, if it vary however slightly in any manner profitable to itself, under the complex and sometimes varying conditions of life, will have a better chance of surviving, and thus be NATURALLY SELECTED.

— Charles Robert Darwin



1 Academic Reading

Text A: Coincidental Killers

Academic Reading Skills: Creating and Using Mental Images

2 Academic Writing

Text B: Genome-wide Meta-analysis of Cognitive Empathy: Heritability, and Correlates with Sex, Neuropsychiatric Conditions and Cognition

Academic Writing Skills: Describing Processes and Procedures

1

Academic Reading

Lead-in Questions

1. What attitude should we adopt towards microbes? How can humans better coexist with them?
2. When we contract infectious diseases, should we use antibiotics?

Text A

Coincidental Killers

Ed Young

- 1 In the late 17th century, the Dutch scientist Antonie van Leeuwenhoek created a new type of microscope lens and brought an entire world of tiny organisms into focus. Looking at his own dental plaque¹, he wrote: "I then most always saw with great wonder, that in the said matter there were many very little living animalcules², very prettily a-moving." These little creatures were intriguing but seemingly unimportant, and few others picked up the baton from van Leeuwenhoek. That changed in the 19th century, when Louis Pasteur and Robert Koch proved that some of these microbes were behind important diseases.
- 2 That framing has stuck. Microbes are everywhere, but we take their presence on phones, keyboards, and toilet seats as a sign of filth and squalor. They fill our bodies, helping us to digest our food and safeguard our health, but we view them as adversaries to be drugged and conquered.
- 3 This antagonism is understandable. Aside from those of us with access to microscopes, most people will never see microbes with their own eyes. And so we tend to identify

1 plaque: 牙斑菌

2 animalcules: 微生物

microbes with the disease-causing minority among them, the little buggers³ that trigger the tickling mist of a sneeze or the pustule⁴ on otherwise smooth skin. We become aware of their existence when they threaten our lives, and for much of our history, that threat was substantial. Epidemics of smallpox⁵, cholera⁶, tuberculosis⁷, and plague have traumatised humanity, and the fear of these diseases has contaminated our entire culture, from our religious rites to Hollywood films such as *Contagion* (2012)⁸ or *Outbreak* (1995).

- 4 When microbes aren't killing us, we are largely oblivious to them. So, we construct narratives of hosts and pathogens⁹, heroes and villains, us and them. Those that cause disease exist to reproduce at our expense, and we need new ways of resisting them. And so we study how they evolve to outfox our immune system or to spread more easily from one person to another. We identify genes that allow them to cause disease and we label those genes as "virulence factors". We place ourselves at the centre of their world. We make it all about us.
- 5 But a growing number of studies show that our anthropocentric¹⁰ view is sometimes unjustified. The adaptations that allow bacteria, fungi¹¹ and other pathogens to cause us harm can easily evolve outside the context of human disease. They are part of a microbial narrative that affects us, and can even kill us, but that isn't about us. This concept is known as the coincidental evolution hypothesis or, as the Emory University microbiologist Bruce Levin described it in 2008, the "shit happens" hypothesis.
- 6 This hypothesis does not apply to all infections, and is almost certainly irrelevant to viruses, which always need to reproduce in a host. Nor does it apply to the many bacteria and fungi, such as *Staphylococcus aureus*¹² or *Candida albicans*¹³, that are

3 buggers: 混蛋, 难题。

4 pustule: 脓疱

5 smallpox: 天花

6 cholera: 霍乱

7 tuberculosis: 结核病

8 编者注: 作者笔误, 影片上映时间应为 2011 年。

9 pathogens: 病原体

10 anthropocentric: 人类中心论的

11 fungi: 真菌

12 *Staphylococcus aureus*: 金黄色酿脓葡萄球菌

13 *Candida albicans*: 白色念珠菌

long-standing human pathogens and well-adapted to us. But it does explain some weird aspects of many diseases.

- 7 Why, for example, would bacteria harm the hosts that they depend on for survival? In some cases, the answer is obvious: They cause symptoms such as sneezing or coughing that help them to spread. But what about *S.pneumoniae*¹⁴? Strains that sit harmlessly in a host's airways are already capable of spreading to another individual. The virulent forms, which descend deeper into the respiratory tract, are actually less contagious. The same goes for bugs such as *Hemophilus influenzae*¹⁵ and *Neisseria meningitidis*¹⁶, which can inflame the protective membranes¹⁷ around the brain and lead to life-threatening cases of bacterial meningitis. In doing so, they risk capsizing their own ship without any hope of boarding a new one.
- 8 The coincidental evolution hypothesis helps to resolve these paradoxes. It tells us that at least some human diseases have nothing to do with us at all.
- 9 The coincidental evolution hypothesis explains a number of other recent discoveries about microbes. Scientists have found antibiotic resistance genes in bacteria that have been frozen for 30,000 years, or isolated in million-year-old caves. We might think of antibiotics as modern inventions, but they're actually weapons that bacteria have been using against each other for aeons¹⁸, or at least well before Alexander Fleming noticed a funky mould in a Petri dish in 1928. Antibiotic resistance genes evolved as part of this ancient war, but they also help today's microbes to deal with the medicines that we mass-produce.
- 10 Likewise, many of the "virulence genes" that help pathogens to cause disease have counterparts in marine microbes with no track record of infecting humans. And some supposedly pathogenic bacteria were often common parts of the environment. "These organisms become accidental pathogens," says the microbiologist Arturo Casadevall from Yeshiva University in New York. "They'll still be there even if you remove all the animals from the planet. And yet, evolution selected for just the right combination of traits to cause disease in humans."

14 *S.pneumoniae*: 肺炎链球菌

15 *Hemophilus influenzae*: 流感嗜血杆菌

16 *Neisseria meningitidis*: 脑膜炎奈瑟菌

17 membranes: 膜

18 aeons: 极漫长的时期, 亦可拼写为 "eons".

- 11 *Vibrio cholerae*¹⁹, the bacterium that causes cholera, is a good example. Scientists used to regard it as a human pathogen that spreads when the faeces²⁰ of infected people seep into water supplies. We now know that it's mainly a marine species that attaches itself to the shells of small crustaceans²¹, and occasionally makes its way into our water supply. "In the last decade, people have begun to accept that a lot of these opportunistic pathogens that people assumed were only in the environment transiently between human hosts are actually environmental bacteria that occasionally end up in humans," says Diane McDougald from the University of New South Wales, who studies *V.cholerae*.
- 12 Many of the pathogens we fear most are mere tourists on the human body. Their real homes are oceans, caves, or soils. To understand them, we need to understand them within their natural ecology. Soil, for example, is an extreme habitat for a microbe: harsh and constantly changing. It can quickly oscillate from flood to drought, from scalding heat to freezing cold, and total darkness to intense solar radiation. It's rife with other competing microbes, and crawling with hungry predators. We fear lions and tigers and bears; bacteria have to contend with phage²² viruses, nematode²³ worms, and predatory amoebas²⁴.
- 13 All of these conditions can lead to adaptations that make microbes accidentally suited for life in a human host. We are, after all, just another environment. A thick capsule that shields a microbe from dehydration could also shield it from our immune system. A spore²⁵ that is adapted for travelling through the air can be easily inhaled into a respiratory tract.
- 14 Scientists have demonstrated many of these coincidental adaptations by exposing bacteria to a natural threat and showing that they then become better at infecting humans and other mammals. *Escherichia coli*²⁶, for example, is a common gut²⁷ bacterium, and a darling of laboratory scientists. In its natural environments,

19 *Vibrio cholerae*: 霍乱弧菌

20 faeces: 粪便

21 crustaceans: 甲壳纲动物

22 phage: 抗生素, 噬菌体。

23 nematode: 线虫

24 amoebas: 变形虫

25 spore: 孢子

26 *Escherichia coli*: 大肠杆菌

27 gut: 肠

whether the soil or the gut of a mammal, it is menaced by predatory amoebas, which threaten to engulf and digest it. In 2010, the French scientist Frantz Depaulis and colleagues found an *E.coli* strain called 536 that resists these predators, with genes that protect it from the amoebas' digestive enzymes and allow it to scavenge nutrients such as iron. Rather than being disintegrated, it actually grows inside its would-be predator and eventually kills it from within.

- 15 Many of these protective genes also allow strains of the mostly harmless *E.coli* to cause severe illness in humans, mice and other mammals. This makes perfect sense. Many of our immune cells, like macrophages²⁸, engulf and digest microbes just as amoebas do, so an amoeba-proof bacterium is also a macrophage-proof one. By adapting to their natural predators, strains of *E.coli* are coincidentally pre-adapted to foil our immune system.
- 16 The coincidental evolution hypothesis might be irksome to some. What are the odds that an adaptation to one challenge would perfectly predispose an organism to another? The answer, it seems, is: pretty high. Evolution, however, is all about small probabilities manifesting through long timescales and large numbers – and microbes have both. They have been living on the planet for billions of years, and there are countless legions of them.
- 17 Casadevall likes to say that each microbe holds a different hand of cards – adaptations that allow it to cope with its environment. Most of these combinations are meaningless to us. A bacterium might be able to resist being digested by other cells, but it might not be able to grow at 37 degrees Celsius. It might grow at the right temperature, but it might not be able to tolerate our slightly alkaline²⁹ pH levels. But that doesn't matter. There are so many microbes out there that some of them will end up with a hand that lets them muscle their way into our game. "If you take all the microbial species in the world and imagine that they have these traits randomly, you can find pathogenic microbes for practically anything," says Casadevall.

28 macrophages: 巨噬细胞

29 alkaline: 碱性的

New Words and Expressions

coincidental /kəʊ,ɪnsɪ'dentl/ *adj.* happening completely by chance without being planned 巧合的, 碰巧的

intriguing /ɪn'tri:ɡɪŋ/ *adj.* something that is intriguing is very interesting because it is strange (由于奇特、神秘或出人意料而)非常有趣的, 迷人的

adversary /'ædvəsəri/ *n.* a country or person you are fighting or competing against 对手, 敌手

antagonism /æn'tæɡənɪzəm/ *n.* hatred between people or groups of people 对抗, 对立, 敌对

epidemic /epɪ'demɪk/ *n.* a large number of cases of a disease that happen at the same time 流行病, 传染病

traumatise /'trɔ:mətaɪz, 'trau-/ *vt.* to shock someone so badly that they are affected by it for a very long time 使受精神创伤

contaminate /kən'tæmɪneɪt/ *vt.* to influence something in a way that has a bad effect 玷污, 毒害

oblivious /ə'blɪvɪəs/ *adj.* not knowing about or not noticing something that is happening around you 不知道的, 未觉察到的

unjustified /ʌn'dʒʌstɪfaɪd/ *adj.* done without an acceptable reason 没有正当理由的

adaptation /,ædæp'teɪʃən/ *n.* the process of changing something to make it suitable for a new situation 适应

virulent /'vɪrələnt/ *adj.* a poison, disease etc. that is virulent is very dangerous and affects people very quickly (毒物) 剧毒的; (疾病) 迅速致命的

respiratory /rɪ'spɪrətəri, 'respə'rentəri, rɪ'spaɪərə-/ *adj.* relating to breathing or your lungs 与呼吸有关的; 与肺有关的

capsize /kæp'saɪz/ *v.* if a boat capsizes, or if you capsize it, it turns over in the water (使) (船) 倾覆

paradox /'pærədɒks/ *n.* a situation that seems strange because it involves two ideas or qualities that are very different 自相矛盾 (的情况)

oscillate /'ɒsɪleɪt/ *vi.* to keep changing between two extreme amounts or limits (在数量或限度之间剧烈地) 波动, 摇摆, 变化

dehydrate /,di:haɪ'dreɪt/ *v.* to lose too much water from your body, or to make this happen (使) (人体) 脱水

scavenge /'skævəndʒ/ *v.* if an animal scavenges, it eats anything that it can find (动物) 以 (任何找到的东西) 为食

disintegrate /dɪs'ɪntɪɡreɪt/ *v.* to break up, or make something break up, into very small pieces (使) 碎裂, (使) 分裂, (使) 解体

foil /fɔɪl/ *vt.* to prevent something bad that someone is planning to do 挫败, 阻止

predispose /,prɪ'dɪs'pəʊz/ *vt.* to make someone more likely to suffer from a particular health problem 使易患 (某种疾病)

Building Your Vocabulary

Task 1

Match the words in Column A with their appropriate explanations in Column B.

Column A	Column B
1 ____ manifest	A the right or opportunity to use or benefit from something
2 ____ trigger	B settle or find a solution to
3 ____ access	C become visible or obvious
4 ____ resolve	D a supposition or proposed explanation
5 ____ intense	E cause to happen or exist
6 ____ hypothesis	F of extreme force, degree, or strength

Understanding the Text

Task 2

Choose the main idea for Paragraphs 5–15 of Text A.

- A Microbes harm their human hosts to enhance their chances of survival.
- B Microbes' ability to cause disease is an adaptation against their hosts.
- C Microbes' adaptation to the environment may sometimes be detrimental to humans.

Task 3

Summarise the main idea of Paragraphs 14–15 in Text A.

Handwriting practice area with dotted lines for writing.

Academic Reading Skills

Creating and Using Mental Images

You can use notes to keep a record of important information. As you read, highlight key terms, important ideas and noteworthy details. You may then note down the important information either on the page next to the reading or in a notebook. Timelines, flow charts, outlines and spidergrams are commonly used note-taking methods.

Timelines list events in chronological order.

Flow charts are diagrams to represent workflows, processes or procedures. Flow charts, which normally flow from top to bottom and left to right, use different types of boxes to represent the steps, and connect them with arrows to show their order.

Outlines are a straightforward way to organise the main ideas and supporting details. When you lay out your outlines which reflect the structure of your reading, start with the main points, and then add supporting points. Use numbers, letters, or indentation to show the order of importance.

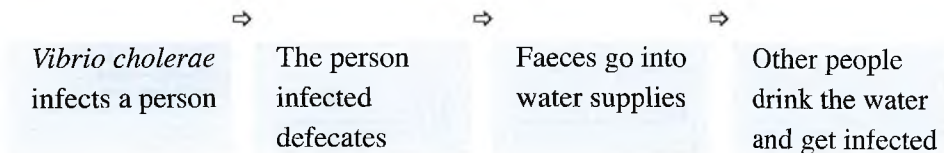
Spidergrams present graphically the relations between important ideas. You normally put the main idea in the centre, and then subdivide it into branches which may each be further divided to provide more details.

You can develop a form of shorthand in note-taking, using symbols and abbreviations that are commonly used or inventing your own. Bear in mind to keep your notes simple.

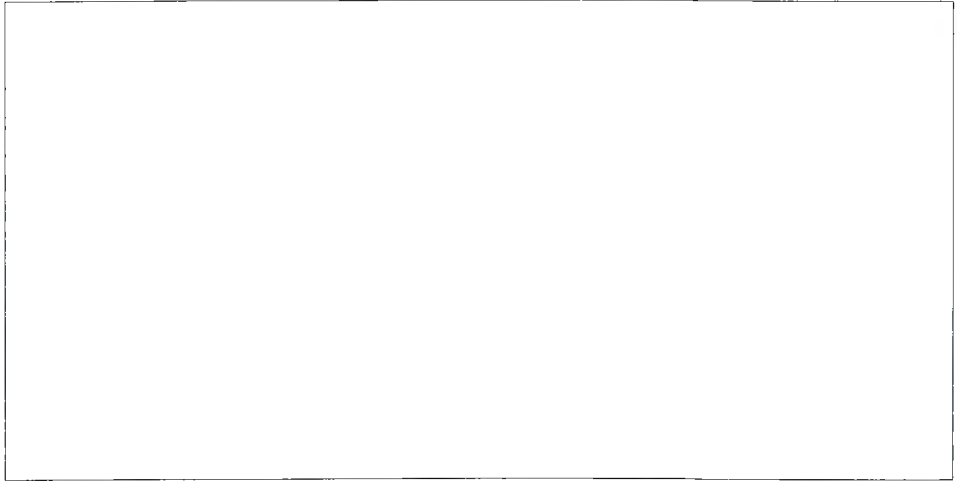
The following flow chart explains how humans are infected by bacteria according to Paragraph 11 of Text A. Examine it carefully and make revisions wherever necessary.

Task 1

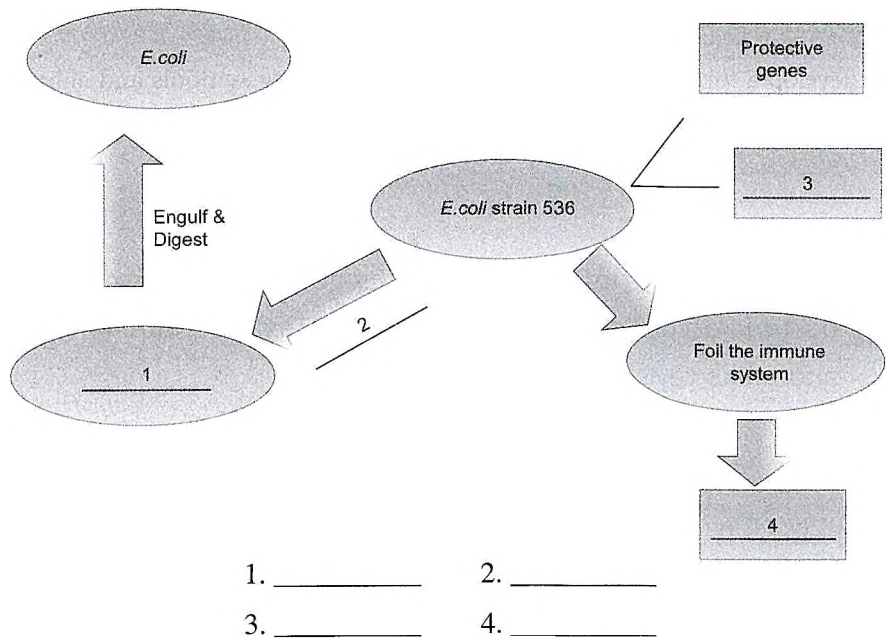
Vibrio cholerae:



Your visual picture:



Task 2 Read Paragraphs 14–15 of Text A. Use the skill of visualisation to complete the mental map below on how *E.coli* develops pathogenicity.



Task 3 Read Paragraphs 5–15 of Text A and draw an outline for this section.



The following excerpt describes the process of intracellular offspring production by some Firmicutes³⁰. Draw a flow chart to illustrate the reproduction process.

In large *Epulopiscium* spp. this unique reproductive strategy begins with asymmetric cell division. Instead of placing the FtsZ ring at the centre of the cell, as in binary fission, Z rings are placed near both cell poles in *Epulopiscium*. Division forms a large mother cell and two small offspring cells. The smaller cells contain DNA and become fully engulfed by the larger mother cell. The internal offspring grow within the cytoplasm of the mother cell. Once offspring development is complete the mother cell dies and releases the offspring.

30 Firmicutes: 厚壁菌群

2

Academic Writing

Text B

Genome-wide Meta-analysis³¹ of Cognitive Empathy³²: Heritability, and Correlates with Sex, Neuropsychiatric³³ Conditions and Cognition

V Warrier et al

Materials and methods

[...]

Brisbane Longitudinal Twin Study

- 1 In addition, 1,497 participants (891 females and 606 males) of Caucasian ancestry with genotype data from the Brisbane Longitudinal Twin Study (BLTS) completed the short version (14 questions) of the Eyes Test online as part of a study on genetic and environmental foundations of political and economic behaviours. Participant ages ranged from 18 to 73 (M=37, s.d.=14). Twin heritability was estimated from 749 twin individuals (including 122 complete monozygotic³⁴ pairs and 176 complete dizygotic³⁵ pairs). All participants provided online consent and the study was approved by the QIMR Berghofer Human Research Ethics Committee.

Measures

- 2 The “Reading the Mind in the Eyes” Test (Eyes Test) is a brief questionnaire of

31 meta-analysis: 整合分析, 也称统合分析。

32 cognitive empathy: 认知共情

33 neuropsychiatric: 神经精神病学的

34 monozygotic: 同卵的

35 dizygotic: 异卵的

cognitive empathy. Participants are shown scaled, black and white photographs of eye regions of actors and they have to choose the cognitive state portrayed from the four options provided. The Eyes Test has good test-retest reliability (for example, reliability of 0.833 in the Italian version, and 0.63 in the Spanish version), and scores are unimodally and near-normally distributed in the general population. In the BLTS data set, there was a modest test-retest correlation of 0.47 in 259 participants who retaken the test after a gap of nearly 2 years. For each correct answer on the Eyes Test, participants score 1 point, so the scores ranged from 0 to 36 on the full version of the Eyes Test and 0 to 14 on the short version of the Eyes Test. Further details are provided in Supplementary Note section 1.

Genotyping³⁶, imputation³⁷ and quality control

[...]

BLTS cohort

- 3 The BLTS participants were genotyped on Illumina HumanCoreExome-12 v1.0 or Human610-Quad v1.0 chips³⁸. These samples were genotyped in the context of a larger genome-wide association project. Genotype data were screened for genotyping quality (GenCall<0.7 from the Human610-Quad v1.0), individual and SNP call rates (<0.95 and <0.99 for exome³⁹ markers on the HumanCoreExome-12 v1.0 chip), Hardy-Weinberg Equilibrium ($P<10^{-6}$) and MAF (<0.01). The data were checked for non-European ancestry, pedigree⁴⁰, sex and Mendelian errors. Data from the two different chips were separately phased using SHAPEIT2 and imputed to the 1,000 Genomes reference panel (Phase 1 v3) using Minimac3. After imputation SNPs with a MAF<0.05% were excluded, leaving 11,133,794 SNPs for analyses. We further excluded SNPs with imputation $r^2<0.6$ for meta-analysis.

36 genotyping: 基因分型

37 imputation: 基因型填补

38 Illumina HumanCoreExome-12 v1.0 or Human610-Quad v1.0 chips: Illumina 公司的两个全基因组分型产品

39 exome: 外显子组; 测序外显子组。

40 pedigree: 门系, 家谱。

Academic Writing Skills

Describing Processes and Procedures

To describe a process or a procedure effectively, one should bear the following tips in mind:

1. “WHW” principle – when you describe a process or a procedure, explain in detail where something happened, how something was done, and why something was done. Usually the information about the study site(s) comes first (*where*), and then follows the information about the specific steps taken in the process (*how*) and reasons for making certain choices (*why*). Explanations are given only when necessary.

Where something happened: the location where the study was carried out. For example:
This study was conducted at Xiayang forest farm (26°48'N, 117°58'E, 229–246m mean elevation), Southeast Wuyi Mountain.

How something was done: the specific steps taken in the processes or procedures. For example:
Then the Glu-C digestion was performed at 30°C for 18h with an enzyme-to-protein ratio of 1/40 (w/w).

Why something was done: the reason why a certain step was taken. For example:
Briefly, PVC tubes were carefully driven into the soil to avoid major disturbances from human and stem flows.

Information related to the “WHW” principle is very important, but that doesn't necessarily mean that you have to mention the “WHW” in each single step you intend to describe. Do remember to include only the most important and relevant information.

2. It is advised to describe a procedure step by step, or in other words, in chronological order. Read the following example and pay attention to the underlined signal words/phrases indicating the time sequence:

*For protein digestion, the protein mixture was reduced by 10mM DTT at 60°C for 1h and alkylated by 20mM IAA in the darkness at room temperature for 30min. **After that**, 100mM NH₄HCO₃ buffer (pH = 8.2) was added until the protein concentration was about 1 mg/mL. **Then** the Glu-C digestion was performed at 30°C for 18h with an enzyme-to-protein ratio of 1/40 (w/w). The digestion solution was **then** stored at -20°C for further analysis.*

Sometimes information may be given in a combined order. For example:

*Data was obtained from three experiments. **In the first**, we trained the five wild type mice to run back and forth on two elevated 96cm straight or “L” shaped linear tracks (Rubin et al, 2015). **Before** beginning with Ca^{2+} imaging, we trained the mice for 8–11 days, **until** the mice ran at least 60 times the entire length of each track in two consecutive days. [...] **In the second experiment**, we imaged the five Thy1-GCaMP6f mice. [...] **In the third experiment**, we imaged the prefrontal cortex of ...*

3. Passive voice is more often used because readers are more interested in the processes or procedures you intend to describe than in the people doing the work. For example:

Upon arrival at the laboratory, some of the samples were stored in a 4°C fridge for soil biochemical analysis and the others were freeze-dried for microbial community analysis.

4. Verb tense: the past simple is more frequently used because the processes or procedures described must be something done before you write your paper. But if you want to write about a general situation or a standard procedure, such as how to operate a machine step by step, you should use the present simple. For example:

*Following an approach set out in Baillie et al (2008), we randomly **selected** 1,500 species from a list of all described reptilian species (Uetz, 2010), using the sample function in R [sample (x, size); R Development Core Team, 2007]. A sample of 1,500 species **is** sufficiently large to report on extinction risk and trends, and **buffers** against falsely detecting improvements in extinction risk (Baillie et al, 2008).*

However, there are also journals which may require consistent use of the present simple in the description of processes and procedures.

Identify the experimental steps described in the following excerpt

Task 1 from Text B. Draw a flow chart of this process. Pay special attention to the verbs and sequential expressions.

The BLTS participants were genotyped on Illumina HumanCoreExome-12 v1.0 or Human610-Quad v1.0 chips. These samples were genotyped in the context of a larger genome-wide association project. Genotype data were screened for genotyping quality (GenCall<0.7 from the Human610-Quad v1.0), individual and SNP call rates (<0.95 and <0.99 for exome markers on the HumanCoreExome-12 v1.0 chip),

Hardy-Weinberg Equilibrium ($P < 10^{-6}$) and MAF (< 0.01). The data were checked for non-European ancestry, pedigree, sex and Mendelian errors. Data from the two different chips were separately phased using SHAPEIT2 and imputed to the 1,000 Genomes reference panel (Phase 1 v3) using Minimac3. After imputation SNPs with a MAF $< 0.05\%$ were excluded, leaving 11,133,794 SNPs for analyses. We further excluded SNPs with imputation $r^2 < 0.6$ for meta-analysis. (Para. 3)

Task 2

The following excerpt is taken from an article in the field of animal studies which tries to explore judgement made by dairy calves under ambiguous situations and the reasons for individual differences in judgement bias. Rearrange the following sentences in the correct order. The 4th, 5th and 6th sentences have already been identified.

- ☐ Positive and negative events occurred alternatively during the 5 daily training trials.
- ☐ Groups of 2 to 3 calves were familiarised with the apparatus for 10min 1d before the training phase began but all animals were trained alone.
- ☐ To facilitate training of the negative event animals that touched the empty bottle received an air-puff in the face as a mild punishment.
- ☒ 4 During this first 3d of training, animals were only trained to associate one side of the apparatus (alternately assigned to calves) with a reward (i.e. milk).
- ☒ 6 Training then introduced negative events: an unrewarded (empty bottle) placed on the opposite side from where the calves had previously received a reward.
- ☒ 5 All animals successfully found the milk holder in less than 30s by the third day of training.
- ☐ We used a spatial learning task adapted from Destrez et al.
- ☐ The last trial consisted of only rewarded events to avoid any negative association with the testing area.
- ☐ Five consecutive training trials took place each day.

Rewrite the following excerpt to accentuate the process rather than the

Task 3

people doing the work. Pay special attention to the usage of proper voices.

Over the last several years, Gazit's team – among others – has developed an alternative strategy for efficiently getting genes into MSCs without viruses. The researchers start by packing the wound with the usual collagen matrix and waiting for a couple of weeks for the stem cells to infiltrate the scaffold. They then create a solution containing numerous copies of their gene of interest alongside gas-filled micron-sized bubbles encased by a thin shell of fat molecules. After injecting this solution into the fracture site, they go over the area with an ultrasound wand, much as it's done by obstetricians to check on the health of a fetus. The wand's ultrasound pulses burst the microbubbles, briefly punching nano-sized holes in any adjacent stem cells, which allows the genes in the solution to enter.

The first sentence has been revised as a sample:

Over the last several years, an alternative strategy has been developed for efficiently getting genes into MSCs without viruses. _____

The following excerpt describes a series of experiments on

Task 4

molecular biology. Fill in the blanks with the appropriate forms (voices and tenses) of the given verbs.

The experiments (1) _____ (perform) on male stroke-resistant SHR/N (SHRSR) and stroke-prone SHR/A3 (SHRSP) rats from a breeding colony maintained

by the investigators. Almost all loci⁴¹ (2) _____ (be) the same in these two rat strains except those relevant to stroke. Age-matched male rats from each strain (12 SHRSP rats and 12 SHRSR rats) (3) _____ (feed) with a standard rat chow and water *ad libitum*⁴² until age 8 weeks. Subsequently, animals from each strain (4) _____ (randomise) to one of 2 dietary regimens: a “stroke-permissive diet” high in sodium (HS) and 1% NaCl drinking solution; a “stroke-protective diet” low in sodium and high in potassium (LS) and regular drinking water. All animals (5) _____ (house) at 23°C on a 12-hour light-dark cycle. The SHRSP rat strain in the HS environment (6) _____ (show) stroke symptoms and (7) _____ (die) at 12 weeks of age. The stroke symptoms (8) _____ (define) as severe lethargy⁴³, loss of balance, poor grooming, convulsive rhythmic movement of the forelimbs, immobility, and kangaroo-like posture. The brain tissues (9) _____ (collect) for RNA extraction and subsequent microarray analysis. The study protocols (10) _____ (approve) by the Animal Care Committee of the University of Texas-Houston.

Task 5

Write a summary report on the experiment process according to the following figure and the list of notes taken during the experiment. You are suggested to use the past simple tense (as the experiment has already been completed) and use conjunctions when necessary.

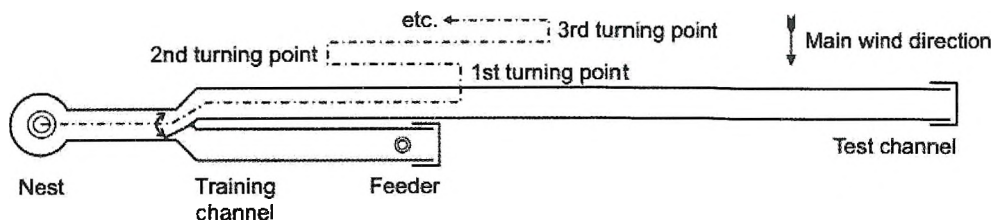


Figure 1 Experimental arrangement. Nest and feeder were connected by a training channel; a test channel was arranged in parallel close to the training channel on the upwind side. This avoided food odours being blown into the test channel and distracting the ants. Experimental animals were guided into either the test or training channel by a swing door. Above the test channel, desert ants' search behaviour around the assumed feeder position is illustrated schematically. The initial three turning points are indicated; the initial six turning points were evaluated.

41 loci: 基因定位点

42 water *ad libitum*: 供水充足 (小白鼠可随意饮水)

43 lethargy: 嗜眠症

1. Conduct an experiment to examine how the distance component of food site vectors is established in desert ants.
2. Manipulate the animals' homebound journey by setting up the test equipment.
3. Train individually marked *C. fortis* ants to visit a feeder (3.2cm Petri dish filled with cookie crumbs about 20m from the nest through a straight channel (Figure 1).
4. If an ant reaches the feeder and takes up a food item, capture it after it takes up the food.
5. Release the ant closer to home, just 10m from the nest, thus halving the distance of the return journey.
6. When the ant begins its next foraging trip, guide it into a test channel running in parallel close to the training channel.
7. Record the animal's search for food until completion of the initial six U-turns (Figure 1).

The first sentence has been written for you:

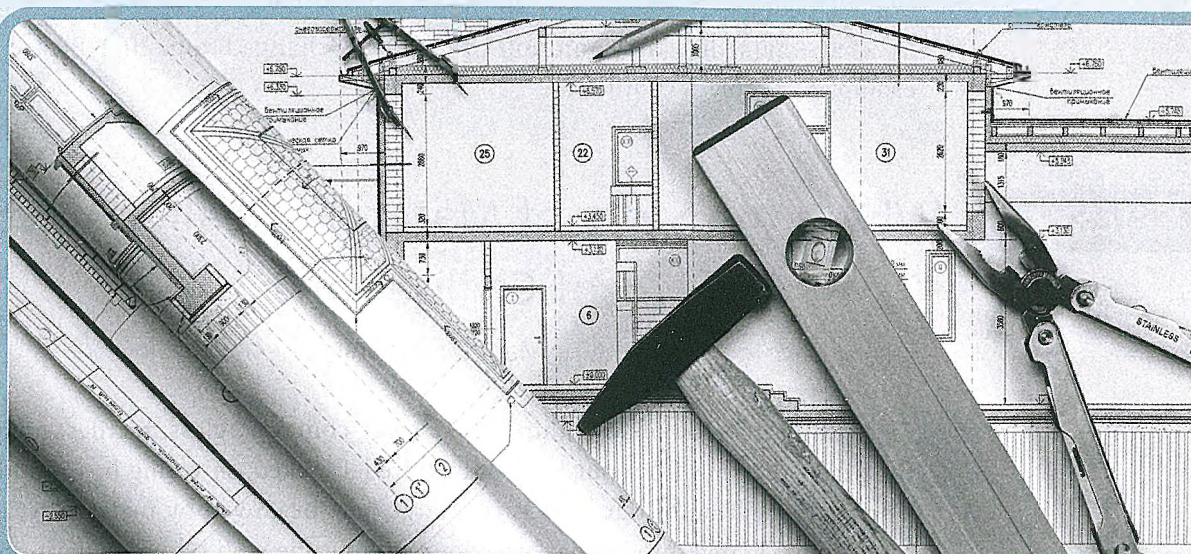
To examine how the distance component of food site vectors is established in desert ants, we manipulated the animals' homebound journey.

Unit 7

Materials Science

Materials scientists are the architects of the future's commonplace miracles.

— Ivan Amato



1 Academic Reading

Text A: Why Is the Story of Materials Really the Story of Civilisation?

Academic Reading Skills: Understanding and Analysing Data

2 Academic Writing

Text B: Preparation of Highly Conductive, Transparent, and Flexible Graphene/Silver Nanowires Substrates Using Non-thermal Laser Photoreduction

Academic Writing Skills: Describing Tables and Figures

1

Academic Reading

Lead-in Questions

1. What different feelings can various materials (e.g. steel, silk, plastic, wood, etc.) bring to people?
2. In what way(s) has the evolution of materials changed your life? Please use examples to explain your idea.

Text A

Why Is the Story of Materials Really the Story of Civilisation?

Mark Miodownik

- 1 Everything is made of something. Take away concrete, glass, textiles, metal, and the other materials from our lives and we are left naked, shivering in a muddy field. The sophistication of our lives is in a large part bestowed by material wealth, we would quickly revert to animal behaviour without the stuff of our civilisation: What makes us human is our clothes, our homes, our cities, our things, which we animate through our customs and language. This becomes very apparent if you ever visit a disaster zone. Thus the material world is not just a display of our technology and culture, it is part of us, we invented it, we made it and it makes us who we are.
- 2 The fundamental importance of materials is made clear from the naming of ages of civilisations – the stone, iron and bronze ages – with each new era being brought about by a new material. Iron and steel were the defining materials of the Victorian era, allowing engineers to give full rein to their dreams of creating suspension bridges, railways, steam engines and passenger liners. Isambard Kingdom Brunel used them as a manifesto to transform the landscape and sow the seeds of modernism. The 20th century is often hailed as the age of silicon, after the breakthrough in materials science that ushered in the silicon chip and the information revolution. Yet

a kaleidoscope of other new materials also revolutionised modern living. Architects took mass-produced sheet glass and combined it with structural steel to produce skyscrapers that invented a new type of city life. Plastics transformed our homes and dress. Polymers were used to produce celluloid and ushered in a new visual culture, the cinema. The development of aluminium alloys¹ and nickel superalloys² enabled us to fly cheaply and accelerated the collision of cultures. Medical and dental ceramics allowed us to rebuild ourselves and redefine disability and ageing – and as the term “plastic surgery” implies, materials are often the key to new treatments used to repair our faculties (hip replacements) or enhance our features (silicone implants³ for breast enlargement).

- 3 My obsession with materials started as a teenager. I was puzzled by their obscurity, despite being all around us. How many people can spot the difference between aluminium and steel? Woods are clearly different from one another, but how many people can say why? Plastics are confusing; who knows the difference between polythene⁴ and polypropylene⁵? Eventually I enrolled in a degree at Oxford University's material science department, went on to do a PhD in jet engine alloys and am now professor of materials and society and director of the Institute of Making at University College London. On my journey I have found a hidden world of makers who create the stuff on which we all rely, from aircraft manufacturers to clothing makers. Materials are at the heart of every company I visit and it is hard not to conclude that although Google and Twitter may dominate technology headlines, and cosmologists may be most popular with the media, materials transformation is still what makes the world go around.
- 4 Starting next week in a new series of columns for *Observer Tech Monthly* I am going to tell the story of stuff. Each month I will pick a different material and uncover the human needs and desires that brought it into being, and decode the materials science and engineering behind it. Along the way, we will find that the real differences between materials are deep below the surface, a world that is shut off from most unless they have access to sophisticated scientific equipment. So to understand materiality is necessarily a journey into the inner space of materials. Pretty much

1 aluminium alloys: 铝合金

2 nickel superalloys: 镍基高温合金

3 silicone implants: 硅胶假体

4 polythene: 聚乙烯

5 polypropylene: 聚丙烯

the whole of materials science is concerned with the microscopic worlds. Doing so explains why some materials smell and others are odourless; why some can last for 1,000 years and others crumble in the sun; how some glass can be bulletproof, while a wine glass shatters at the slightest impact. The journey into this microscopic world reveals the science behind our food, our clothes, our gadgets, our jewellery, and of course our bodies.

- 5 Take for example, a piece of thread, which exists at the same scale as hair. It is a synthetic structure at the limit of our eyesight that has allowed us to make ropes, blankets, carpets, but most importantly, clothes. Textiles are one of the earliest synthetic materials; when we wear a pair of jeans we are wearing a miniature woven structure, the design of which is older than Stonehenge⁶. Clothes have kept us warm and protected for all of recorded history, as well as keeping us fashionable. But they are hi-tech too. In the 20th century we learnt how to make space suits from textiles strong enough to protect astronauts on the moon as well as solid textiles for artificial limbs called carbon fibre composites⁷.
- 6 But there is more to materials than the science. Those who make things all have a different understanding of the practical, emotional and sensual aspect of their materials. For instance, we know the sounds of the doors in our houses, and can distinguish between someone leaving or entering from the subtle differences in keys rattling and hinges creaking. As a child I could always tell whether it was my mother or my father coming up the stairs, from the subtle differences in the sound of the creaky stairs. These acoustic personalities of buildings are often overlooked during the design process. Carpet makes a room feel warmer but also changes the acoustic signature of the room. The clickity-clack of high heels and the party they announce are muted; the squeak of rubber tennis soles and the sport they anticipate is banished; the comforting solid thump of sensible shoes on their way to work is no longer proclaimed. Installing carpet is a kind of auditory gag, which may of course be used intentionally for that purpose such as when designers want to create a sense of intimacy and calm. It is this diversity of material knowledge that I intend to capture in these columns.
- 7 Because materials are built from atoms, we cannot avoid talking about the rules that govern them, which are described by quantum mechanics. This means that as we

6 Stonehenge: (英国) 巨石阵

7 carbon fibre composites: 碳纤维复合材料

enter the atomic world, we must abandon common sense, and talk instead of wave functions and electron states. More materials are being designed from scratch at this scale, and can perform seemingly impossible tasks. Silicon chips designed using quantum mechanics have already brought about the information age. Silicon is now changing the way we light our homes (light emitting diodes⁸) and harvest energy from the sun (solar cells).

- 8 The central idea behind materials science is that changes at invisibly small scales manifest themselves as changes in a material's behaviour at the human scale. It is this process that our ancestors stumbled upon to make bronze and steel, even though they did not have the microscopes to see what they were doing – an amazing achievement. When you hit a piece of metal you are not just changing its shape, you are changing the inner structure of the metal, which is why metals get harder when you hit them. Our ancestors knew this from experience but didn't know why. Nevertheless this gradual accumulation of knowledge got us to the 20th century before any real appreciation of the structure of materials was understood. In these columns I will be championing this skill of making. This is not just because it is the hallmark of human civilisation but because the deindustrialisation of the developed world has devalued making.
- 9 Making is not just an economic activity, it is the equal of literature, performance or mathematics as a form of human expression. By eschewing material knowledge we cease to understand the world around us. We wring our hands about climate change or urban sprawl without any recognition that our ignorance of materiality might be the cause. We feel proud of the technological marvel that is a smartphone, and yet we upgrade – ditch one for a newer model – at the first opportunity. We may assuage our conscience by hoping that they are recycled with some technology equal in sophistication to their fabrication techniques but they are not; most are disposed of in industrial blenders.
- 10 The ages of civilisation are named after materials precisely because they transformed and shaped society. By distancing ourselves from the act of making, by buying and consuming stuff but never having any experience of their manufacture, the developed world finds itself not to be the illiterate society that education ministers fear, but an unmakerly society. In my view this practical ignorance is every bit as dangerous to a modern democracy as a lack of literacy. By swapping

8 light emitting diodes: 发光二极管

a material and industrial understanding of the world for one based on facts and information, we find ourselves uncivilised in a different way.

- 11 This series of columns won't be an exhaustive survey of materials, nor a catalogue of the most important ones. But I will aim to capture the fabric of our lives through materiality. After all, everything is made of something.

New Words and Expressions

bestow /br'stəu/ *vt.* to present; to give as a gift 给予; 授予

revert /rɪ'vɜ:t/ *vi.* to go back to a previous state 恢复; 回到 (常与 to 连用)

animate /'ænɪmeɪt/ *vt.* to give life or energy to something 使有生气; 赋予……生命

rein /reɪn/ *n.* give full/free rein to: to allow an emotion or feeling to be expressed freely 放任; 不加约束

hail /heɪl/ *vt.* to describe someone or something as being very good 把……称赞为, 把……誉为

usher /'ʌʃə/ *vt.* to cause something new to start, or to be at the start of something new 开创, 引领 (常与 in 连用)

kaleidoscope /kə'læɪdəskəʊp/ *n.* a pattern, situation, or scene that is always changing and has many details or bright colours 千变万化的图案

collision /kə'lɪʒən/ *n.* an accident in which two or more bodies hit each other while moving in different directions 碰撞; 冲突

obscurity /əb'skjʊərɪti/ *n.* something that is

difficult to understand, or the quality of being difficult to understand 费解的事物; 晦涩, 费解

cosmologist /kɒz'mɒlədʒɪst/ *n.* an astronomer who studies the evolution and space-time relations of the universe 宇宙学家

materiality /mə'tɪəri'ælɪti/ *n.* the state or quality of being physical or material 物质性

crumble /'krʌmbəl/ *vi.* to break apart into lots of little pieces 碎裂; 崩塌

synthetic /sm'tetɪk/ *adj.* produced by combining different artificial substances 合成的; 人造的

composite /'kɒmpəzɪt/ *n.* something made up of different parts or materials 复合材料

hinge /hɪndʒ/ *n.* a piece of metal fastened to a door that allows it to swing open and shut 铰链, 合页

acoustic /ə'ku:stɪk/ *adj.* relating to sound and the way people hear things 声音的, 听觉的

banish /'bænɪʃ/ *vt.* to not allow someone or something to stay in a particular place 驱逐, 赶走

proclaim /prə'kleɪm/ *vt.* to say publicly that something important is true 宣布, 声明

manifest /'mænɪfɛst/ *vt.* manifest oneself: to appear or to become easy to see 显现, 显露

stumble /'stʌmbəl/ *vi.* to walk in an unsteady way 蹒跚而行

eschew /ɪs'tʃu:/ *vt.* to deliberately avoid doing or using something 回避; 避开

sprawl /sprɔ:l/ *n.* a large area of buildings that are spread out in an untidy and unattractive way (建筑群) 杂乱扩展的大片地区

assuage /ə'sweɪdʒ/ *vt.* to make an unpleasant feeling less painful 缓和; 减轻

illiterate /ɪ'lɪtərɪt/ *adj.* not able to read or write 文盲的

Building Your Vocabulary

Task 1

Match the expressions in Column A with their appropriate explanations in Column B.

Column A	Column B
1 _____ proclaim	A unable to read or write
2 _____ eschew	B avoid
3 _____ usher in	C unclearness
4 _____ obscurity	D expansion
5 _____ illiterate	E announce formally
6 _____ sprawl	F start
7 _____ revert to	G get rid of
8 _____ banish	H relating to sound
9 _____ acoustic	I return to

Understanding the Text

Task 2

Choose the statement that best summarises the theme of Text A.

- A The significance of materials extends beyond scientific scope and into the emotional spectrum.
- B Materials are so fundamentally important to us that we have to reexamine materiality in order to understand the world.
- C Quantum mechanics vastly changed the way people see materials and will continue to hold an important position in materials science.

Task 3

Read the following paragraph from Text A and identify the supporting details/examples for the three aspects mentioned in the topic sentence. Complete the following table by writing down the corresponding number of sentences for each aspect.

But there is more to materials than the science. Those who make things all have a different understanding of the practical, emotional and sensual aspect of their materials.

① For instance, we know the sounds of the doors in our houses, and can distinguish between someone leaving or entering from the subtle differences in keys rattling and hinges creaking. ② As a child I could always tell whether it was my mother or my father coming up the stairs, from the subtle differences in the sound of the creaky stairs. These acoustic personalities of buildings are often overlooked during the design process. ③ Carpet makes a room feel warmer but also changes the acoustic signature of the room. ④ The clickity-clack of high heels and the party they announce are muted; ⑤ the squeak of rubber tennis soles and the sport they anticipate is banished; ⑥ the comforting solid thump of sensible shoes on their way to work is no longer proclaimed. Installing carpet is a kind of auditory gag, which may of course be used intentionally for that purpose such as when designers want to create a sense of intimacy and calm. It is this diversity of material knowledge that I intend to capture in these columns. (Para. 6)

Aspect of materials	The corresponding number of sentences
practical aspect	
emotional aspect	
sensual aspect	

Academic Reading Skills

Understanding and Analysing Data

Tables and figures are frequently used to present data in scientific research. They are clear and effective to demonstrate research results. These visual materials can be used to present a large amount of information or data to an audience in order to facilitate communication and present research results more effectively. Science students and researchers should be able to read and correctly interpret graphic information in published works.

Tables

Lists of numbers or texts that require more than two columns or rows to present are better presented in tables because tabular presentation is clearer and more efficient. Here is a sample to illustrate the structure of a table in academic papers.

Table x Caption of the table

Header	Column 1	Column 2	Column 3	Column 4
Row 1	Values	Values	Values	Values
Row 2	Values	Values	Values	Values
Row 3	Values	Values	Values	Values
Row 4	Values	Values	Values	Values

Notes: This is where the author supplies important additional information to the data, if any, to facilitate understanding.

Here is a table from an academic paper.

Table 1 Element analysis results of complex of lanthanum (III) with N-(2-amino ethyl) maleamic acid radical

Formula	Value of calculation / %				Experimental value / %			
	La	C	H	N	La	C	H	N
LaL ₃	32.44	28.00	2.23	6.53				
LaL ₃ ·H ₂ O	31.13	26.87	2.69	25.08				
LaL ₃ ·2H ₂ O	29.92	21.53	3.01	6.03				
LaL₃·3H₂O	29.17	25.18	3.36	5.88	30.06	27.33	3.21	5.55
LaL ₃ ·4H ₂ O	27.72	23.98	3.60	5.59				

Bold indicates the element analysis result of N-(2-amino ethyl) maleamic acid lanthanum (III) is close to **LaL₃·3H₂O**.

Figures

In research papers, figures, as commonly used visual presentations of data, include graphs, diagrams, drawings, maps, photos, etc. Here, as the most common form of figures, graphs will be discussed. Graphs are often used to demonstrate relationships between values, such as comparison and change. Similar to a table, the basic elements of a graph also include the caption, the values, and the notes. The rows and columns of a table, however, are often transformed into axes in a graph to create clearer visual impressions. The most common forms of graphs are line graphs and bar graphs.

Line graphs plot a series of related values that depict a change in Y as a function of X. Typically, the independent variable is plotted along the X-axis (horizontally), and the dependent variable along the Y-axis (vertically). Line graphs are powerful to demonstrate a trend or a correlation between numerical values. With two or more lines on a graph, it can be used to compare or contrast them (see Figure 1).

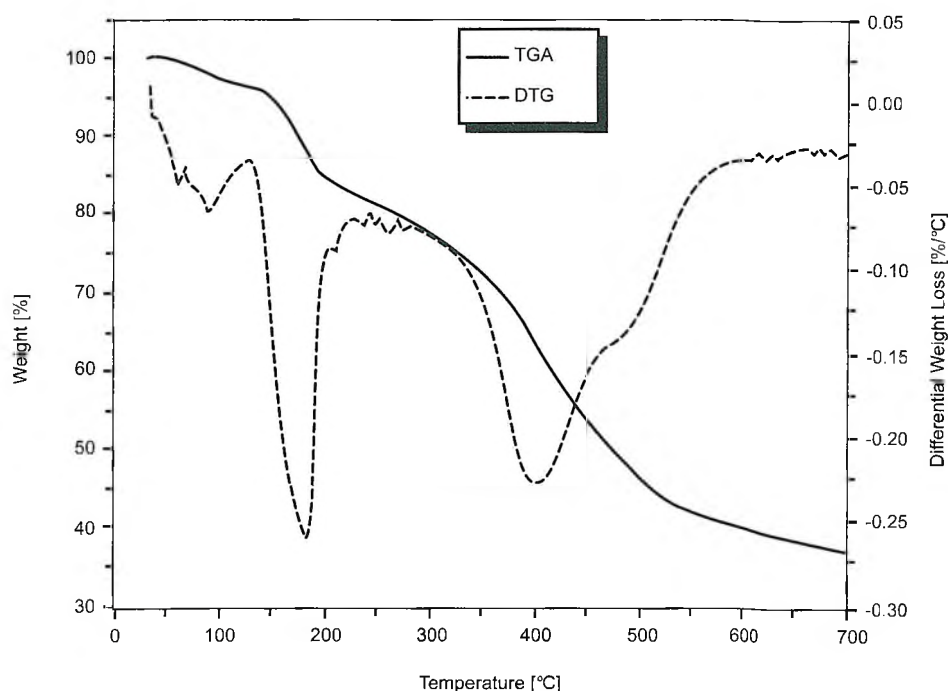


Figure 1 TGA and DTG curves of LaL_3 complex in nitrogen atmosphere

Bar graphs are used when you wish to compare the values of a single variable among several groups or over time.

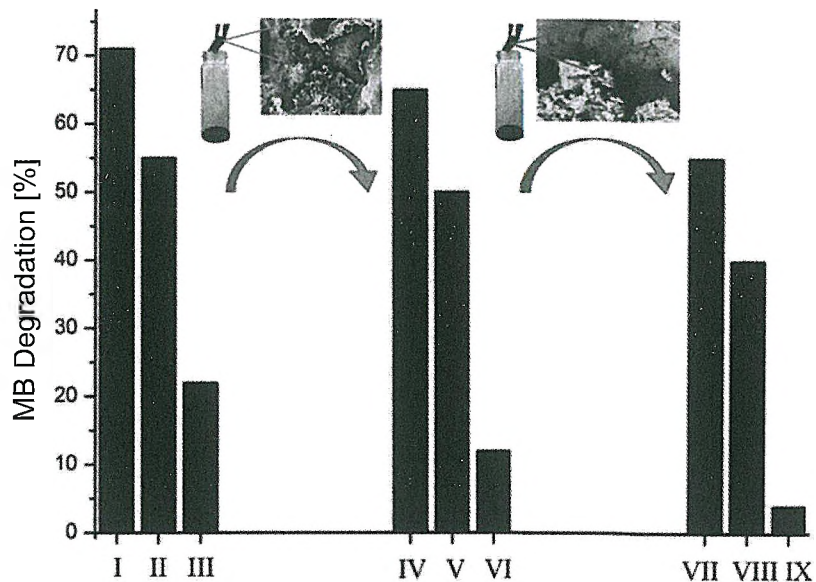


Figure 2 Recyclability after re-deposition of co-PPr on degraded GF/co-PPr

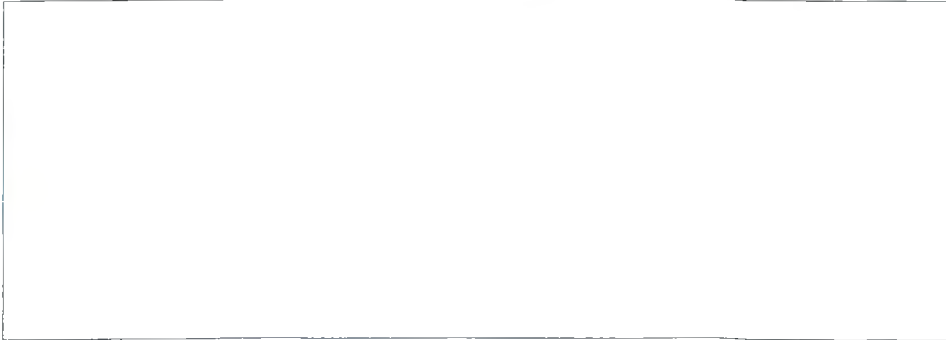
Tips for reading graphic information

When you study tables and figures, you should first look at the caption, the notes, and other words. These will help you understand what the graphic material is about. Next, you will learn to analyse the relationships between rows and columns (in a table), or the relationships between values along X-axis and Y-axis (in a graph). And you can take notes on the graphic material with one or two sentences about what the graphic information mainly shows.

Task 1 Read the following paragraph and draw a figure to show the development trend of material consumption in the 20th century according to the description.

The volume of materials that flow through the national economy grew substantially over the past century. Indeed, the growth of material consumption dramatically outstripped the growth rate of the population. The total consumption of nonfuel materials (construction, mining, and industrial materials) was 17 times greater in 1989 than it was in 1900. The most accelerated growth during this period occurred between 1950 and 1975, when the nation was rapidly building its current infrastructure of

suburbs, highways, airports, and skyscrapers. Since the 1970s, growth in consumption has been more modest and more dramatically influenced by business cycles.



Task 2 Read the table and answer the following questions.

Table 2 US per capita consumption of materials, 1776 and 1996 (pounds)

Materials	1776	1996
Aluminum	0	93
Cement	12	742
Clay	100	326
Coal	40	7,581
Copper	1	23
Iron ore	20	603
Lead	2	13
Natural gas	0	8,164
Petroleum	0	7,520
Phosphate	1	340
Potash	1	48
Salt	4	404
Sand, gravel, stone	1,000	19,061
Sulfur	1	111
Zinc	0.5	12

1. Why does the author present the information in a table rather than in a line chart or bar chart?
2. What is the major trend of per capita consumption of materials from 1776 to 1996?
3. From 1776 to 1996, which material experienced the biggest increase?
4. What is the second most used material in 1776? What about this material's situation in 1996?

The following figure and table are taken from the article

Task 3 “Possibilities of Using Cellulose⁹ Fibres in Building Materials” in the field of materials science. Identify their components and the relationships between the components.

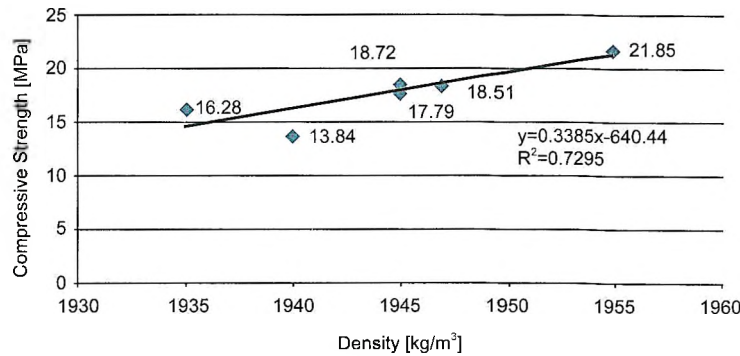


Figure 3 Linear relationship between the compressive strength and density of cellulosic cement-based composites

Table 3 Mixture proportion of fibre-cement composites

Mixture samples		CEM I 42.5 R (wt.%)	Sand (wt.%)	Water (wt.%)	Cellulosic fibres (wt.%)	
					GW-500	G-700T
Alternative 1	WP1	21.05	62.96	15.79	0.2	-
	WP2	21.05	62.16	15.79	1.0	-
	WP3	21.05	58.16	15.79	5.0	-
Alternative 2	RP1	21.05	62.96	15.79	-	0.2
	RP2	21.05	62.16	15.79	-	1.0
	RP3	21.05	58.16	15.79	-	5.0
Reference mixture		21.05	63.16	15.79	-	-
RF						

9 cellulose: 有纤维质的

Task 4

Read the caption of the figure below and try to understand what the figure is about.

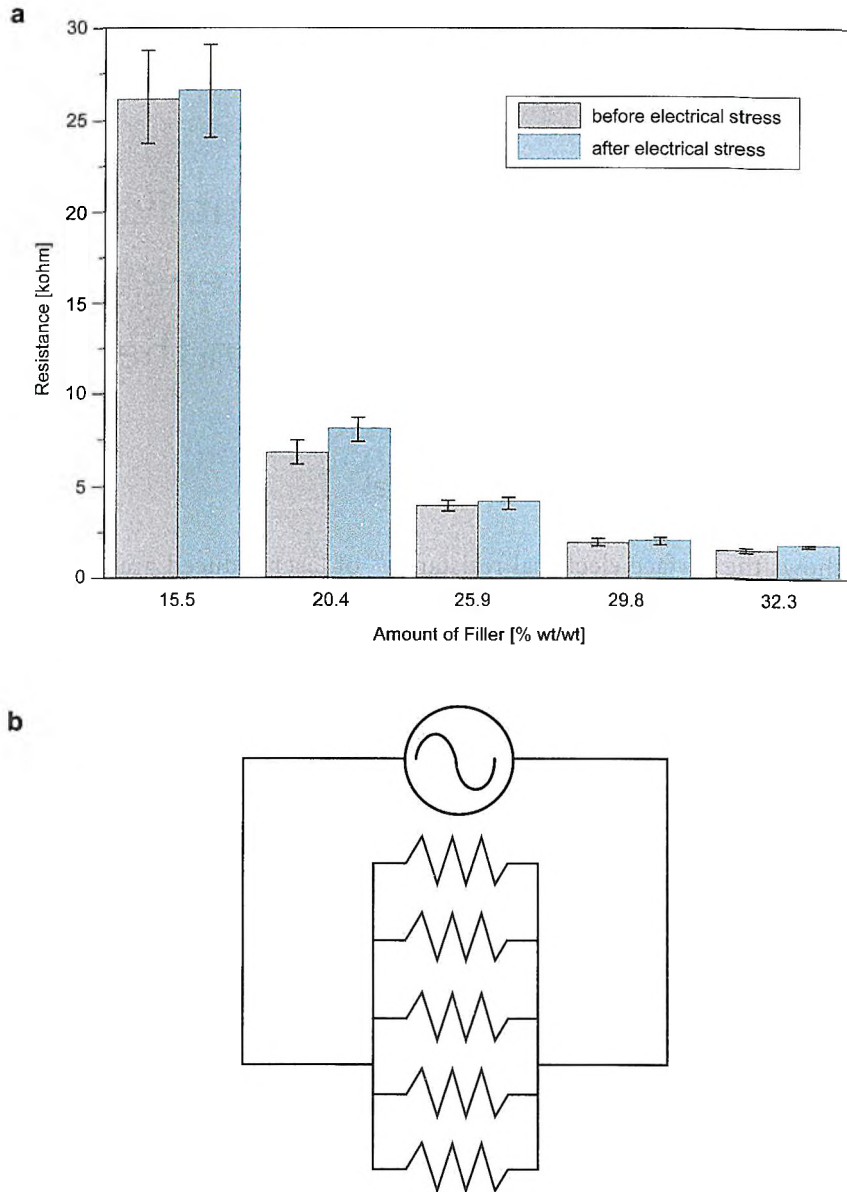


Figure 4 (a)Electrical resistance of 3D-printed wires containing various weight percentages of carbon black filler before and after being subjected to 12 V AC for 7 days. (b) Circuit diagram of the electrical test setup.

2

Academic Writing

Text B

Preparation of Highly Conductive, Transparent, and Flexible Graphene/Silver Nanowires Substrates Using Non-thermal Laser Photoreduction

Badawi Anis et al

- 1 Figure 5 shows the surface electrical resistance¹⁰ of laser reduced graphene oxide¹¹ doped with SNWs¹² (rGO-SNWs) with various dipping times¹³. The performance of the rGO-SNWs varied according to the dipping time. The surface electrical resistance of pristine graphene oxide showed very high resistance ($\geq 20 \text{ M}\Omega$). This could be attributed to the presence of the hydroxyl and epoxy¹⁴ functional groups on the basal plane in addition to carbonyl and carboxyl¹⁵ groups at the edges which in turn reduce the conductivity of graphene sheets.
- 2 From Figure 5 one can observe that the rGO-SNWs sheet resistance¹⁶ decreases with increasing the dipping time in the GO solution. The surface electrical resistance of the rGO-SNWs decreased from $4 \times 10^3 \Omega \square^{-1}$ to $30 \Omega \square^{-1}$ after 60min of dipping. As the time increases the number of graphene oxide layers will increase and a continuous and well-connected sheets will form. [...] Figure 5 shows a comparison of the light

10 electrical resistance: 电阻

11 laser reduced graphene oxide: 激光还原氧化石墨烯

12 SNWs: 银纳米线

13 dipping times: 浸泡时间

14 hydroxyl and epoxy: 羟基和环氧树脂

15 carbonyl and carboxyl: 羰基和羧基

16 sheet resistance: 膜电阻

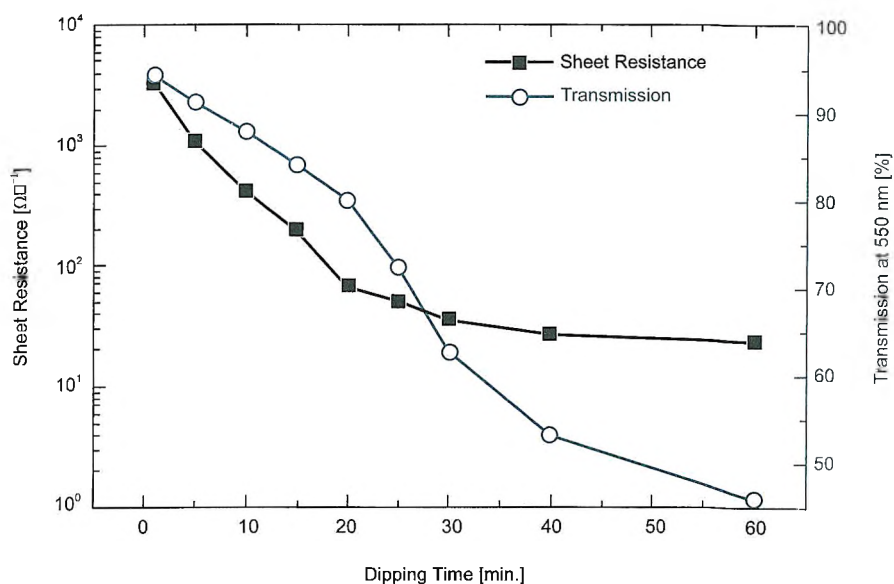


Figure 5 Surface electrical resistance and transmission of laser reduced graphene on PET substrate with various dipping time

transmittance¹⁷ of rGO-SNWs at various dipping times in the GO solution with the sheet resistance. For the optical transmittances of rGO-SNWs, the transparency decreased with the dipping time in the GO solution. This could be attributed to the increase of the GO film. From Figure 5 one can observe that the optimum conditions are 80% transparency with $70\Omega\Box^{-1}$ after 20min of dipping in GO solution.

17 light transmittance: 透光率

Academic Writing Skills

Describing Tables and Figures

Tables and figures are commonly used in academic papers to present research results. Common types of figures include graphs, photos, drawings, maps, etc. Tables and figures have important differences:

1. The title of a table should be put above it while the title of a figure under it;
2. Tables and figures should be numbered separately;
3. Tables are used mainly for presenting specific data or texts while figures are used mainly for presenting a trend of the data;
4. You could use “Fig.,” the abbreviation of “Figure” in your text, but there is no abbreviation for “Table”.

Information conventions

Competent academic writers usually include three information elements when they describe tables or figures. These three elements are: 1) a general statement which tells the readers the location of the figures; 2) statements which present the most important and relevant findings; 3) statements which further explain or comment on the findings. Here are further illustrations and examples for these three elements.

Element 1: location statement. For example:

The logic behind this selection is demonstrated in Fig.3.

Element 2: important and relevant findings. For example:

The addition of graphite can effectively stabilise friction coefficient of the EP-matrix polymer.

In some cases, you can combine Element 1 and 2 in one single sentence. For instance:

The ware rates decrease as foamed copper is introduced to the composites (Fig. 4b).

Element 3: comments and explanations.

In the comment statements, competent academic writers usually: 1) generalise from the results; 2) provide possible reasons; and/or 3) compare the results with those from other research. Comment statements can be organised in one of the following two ways: 1) you could provide a short comment after one important result you describe; or 2) you

could provide your comments after all the important findings are described.

A well-structured description does not always contain each one of the above mentioned three information elements. Remember to keep the most important and relevant information according to your needs.

Language conventions

1. Choosing proper modifiers

Adjectives or adverbs such as *sharp/sharply* or *significant/significantly* could be used to indicate the amount or speed of a change.

The language you use to describe your results has as much power as the tables and graphs themselves, if not more. Results do not speak for themselves. Your descriptions should point to the trend/comparison/similarity/difference you want your readers to notice. For example, the modifiers used in the descriptions (Statement A and B) of the following Figure 6 serve different purposes.

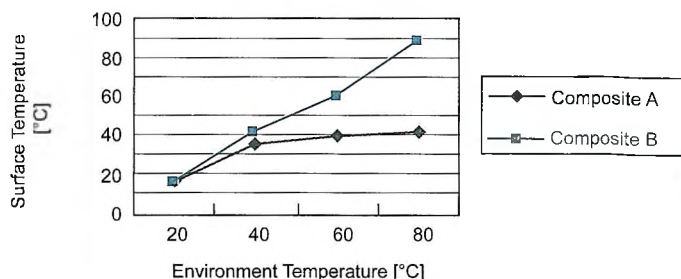


Figure 6 Surface temperature change of Composite A & B under different environment temperatures

- A When the environment temperature increased, the surface temperatures of both composites (A & B) show **similar** trend of rising **simultaneously**. (showing that both composites will heat up when the environment gets hotter)
- B The surface temperature of Composite B rose **sharply** from 20°C to 90°C as the environment temperature went up while, at the same time, that of Composite A only **mildly** increased by 20°C. (showing that Composite B is not as heat-resistant as A)

2. Choosing proper verb tenses

Generally speaking, two tenses are used in describing tables and figures: simple past and simple present. When writing your reports of tables/figures according to the three-element format, you should follow the verb tense conventions below.

Element 1: locating the figure (present tense). For example:

*Figure 6 **shows** the changes in surface temperature of two composites as the environment heats up.*

Element 2: presenting the findings (past tense). For example:

*The surface temperature of Composite B **rose** sharply from 20°C to 90°C as the environment temperature **went up** while, at the same time, that of Composite A only mildly **increased** by 20°C.*

Element 3: commenting on the results (present tense and modal auxiliaries). For example:

*This result **is** consistent with earlier findings suggesting that Composite A **has** better thermostability than B.*

*This result **may** lead to the conclusion that Composite B **is** not as heat-resistant as A.*

Task 1 Read Text B and locate the three basic elements of information.

The following excerpt is from the article "Thermal Treatment Effects on Charge Storage Performance of Graphene-Based¹⁸

Task 2 Materials for Supercapacitors¹⁹" in the field of materials science. Identify the information elements. What results are highlighted? And how are the results and comments organised?

Figure 7 shows the gravimetric²⁰ specific capacitance C_g and surface-normalised specific capacitance C_s as a function of the treatment temperature. The highest specific capacitance was obtained for RGO-400 ($C_g = 96 \text{ F/g}$). The enhancement of the specific capacitance at 400°C can be partially attributed to the development of BET surface area²¹. Because all materials investigated here have relatively low BET surface area, the corresponding gravimetric capacitance is still below the level characteristic for high-surface-area carbons (which can reach up to 150–200 F/g in aqueous²² H_2SO_4 solutions).

18 graphene-based: 基于石墨烯的

19 supercapacitors: 超级电容

20 gravimetric: (测定) 重量的, 重量分析的。

21 BET surface area: BET 比表面积。BET 是 Brunauer、Emmett 和 Teller 三位科学家名字的首字母组合。

22 aqueous: 含水的; 似水的。

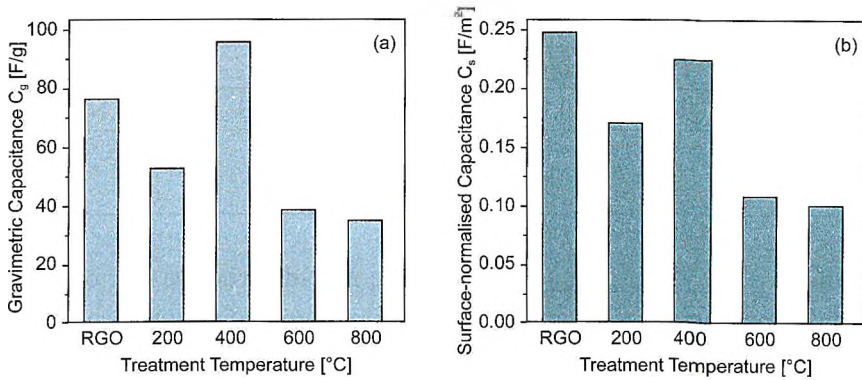


Figure 7 Plot of gravimetric capacitance of the graphene-based supercapacitors and their specific capacitance normalised by BET surface areas

To allow comparison with other material-specific reports and ascertain the effect of treatment temperature, irrespective of surface area, Figure 7 also shows area-specific capacitance values C_s normalised to the BET surface area. Increasing the treatment temperature appeared to have in general a detrimental effect on the weight-normalised capacitance, C_g ; however, the surface-normalised capacitance C_s obviously rebounded at about 400°C and reached 0.22 F/m². This value is higher than the range (0.07–0.13 F/m²) reported for commercial activated carbons and many other high-surface-area carbons (including carbide-derived carbons and carbon nano-tubes) in aqueous H₂SO₄ electrolyte solutions.

Task 3

Read the following two tables and fill in the blanks with the appropriate form of the given verbs.

Table 4 Rebound hammer test report for geopolymer concrete samples

Age of rebound hammer test		Types of geopolymer concrete					
		Control	0.75% nS	3% nS	6% nS	0.02% CNT	1% TiO ₂
7	Rebound hammer NO	23	22	24	<10	20	27
days	Predicted compressive strength	14 ± 5.1 MPa	13 ± 4.95 MPa	16 ± 5.4 MPa	NA	10 ± 4.5 MPa	21 ± 6.0 MPa
28	Rebound hammer NO	30	25	29	23	25	27
Days	Predicted compressive strength	25 ± 6.3 MPa	18 ± 5.7 MPa	24 ± 6.2 MPa	14 ± 5.25 MPa	18 ± 5.7 MPa	21 ± 6.0 MPa

Table 5 Ultrasonic pulse velocity test report for geopolymer concrete samples

Age of ultrasonic pulse velocity test	Types of geopolymer concrete					
	Control	0.75% nS	3% nS	6% nS	0.02% CNT	1% TiO ₂
7 days	7.14	7.21	4.18	4.81	7.26	7.12
28 days	7.16	6.95	4.27	4.67	6.93	6.95

Table 4 and 5 _____ (show) the non-destructive test of geopolymer concrete incorporating nano material²³ and also for control condition. Rebound hammer test²⁴ _____ (show) the predicted compressive strength according to the rebound number and standard curve. It _____ (find) that the predicted compressive strength at 7 days _____ (be) almost similar to the actual compressive strength after crushing whereas at 28 days the result _____ (be) not similar mainly for 3% and 6% nano silica²⁵ and 1% TiO₂ addition. In case of ultrasonic pulse velocity test²⁶, the values _____ (be) greater than 4.5 km/s. So it _____ (say) the quality of concrete for all cases _____ (be) excellent as per the IS code which was prepared basically for the cement-based concrete.

Read Figure 8 and Figure 9 and complete the following paragraph
Task 4 of data description. Consult the writing instructions or guidance in brackets.

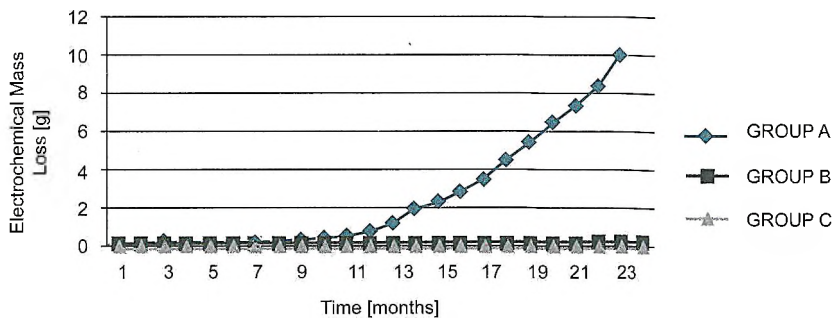


Figure 8 Electrochemical mass loss of Groups A, B and C versus time

- 23 geopolymer concrete incorporating nano material: 聚合物混凝土结合纳米材料
 24 rebound hammer test: 反弹锤击试验
 25 nano silica: 纳米二氧化硅
 26 ultrasonic pulse velocity test: 超声波脉冲速度测试

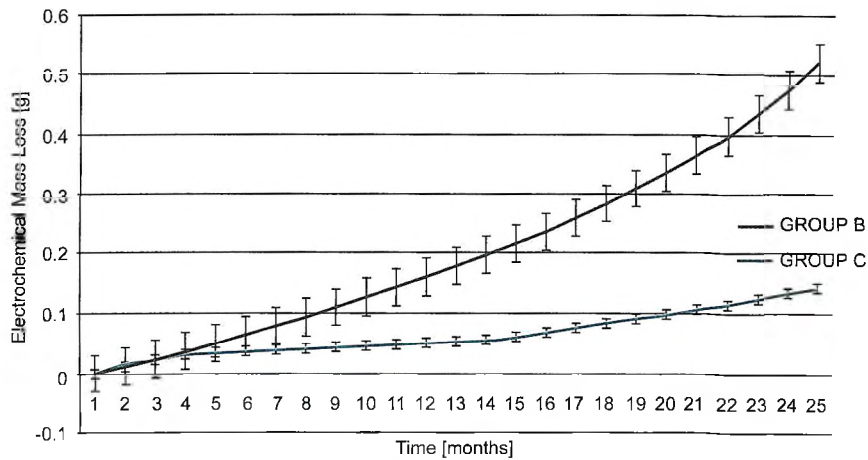


Figure 9 Electrochemical mass loss of Groups B and C versus time

This research studied the comparison of degrees of corrosion in three groups of metal composites: Groups A, B and C. Group B and Group C contained corrosion inhibitors, while A contained no corrosion inhibitors. Also, Group C had the corrosion inhibitor as an admixture, while Group B had the corrosion inhibitor as sprayed. Figure 8 shows the comparison of electrochemical mass loss of Groups A, B and C. It is noticed that the two groups B and C, containing corrosion inhibitors, had better behaviour in the corrosion than Group A, containing no corrosion inhibitor. Figure 9 shows _____

(to summarise Figure 9, particularly based on your understanding of its caption. Element 1)

Group C _____

(to compare the results between Group B and Group C. Element 2)

Finally, it is obvious that _____

(to draw up a brief summary of comparison among all the three groups: Group A, Group B, and Group C. Element 3)

As is shown/illustrated in Figure 9, in Month 20, _____

(to point out another important result from Figure 9 in Month 20. Element 2)

It clearly demonstrates the fact that _____

(to emphasise the difference between Group B and Group C. Element 3)

In conclusion, _____

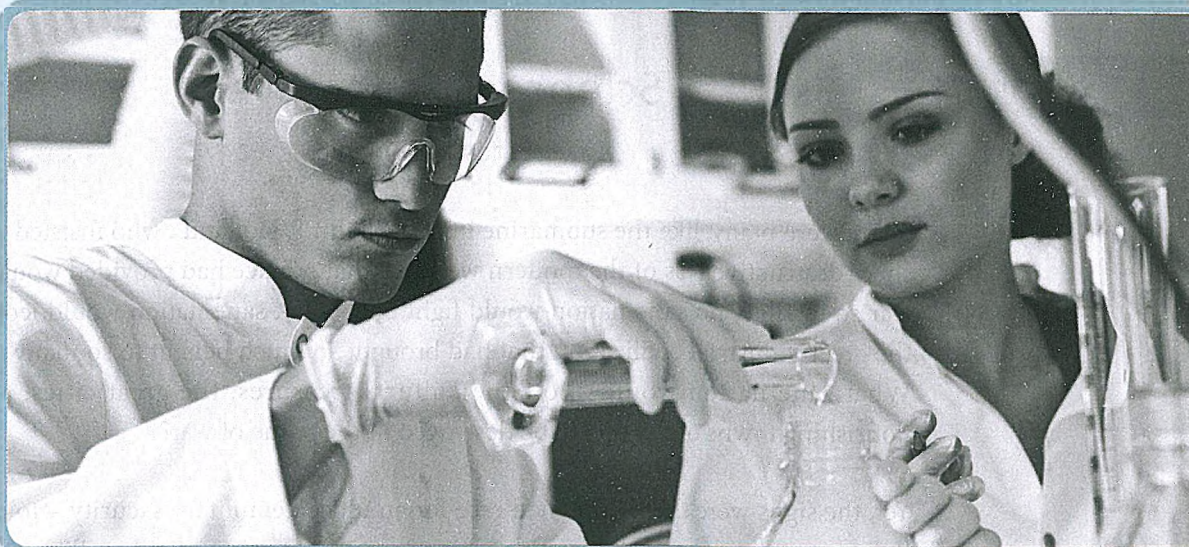
(to conclude and imply the most important finding shown in Figure 8 and Figure 9. Element 3)

Unit 8

Chemistry

Chlorine is a deadly poison gas employed on European battlefields in World War I. Sodium is a corrosive metal which burns upon contact with water. Together they make a placid and nonpoisonous material, table salt. Why each of these substances has the properties it does is a subject called chemistry.

— Carl Sagan



1 Academic Reading

Text A: Science Goes to War

Academic Reading Skills: Analysing Unknown Words Through Context

2 Academic Writing

Text B: Direct Conversion of CO₂ into Liquid Fuels with High Selectivity over a Bifunctional Catalyst

Academic Writing Skills: Expressing with Nominalisation

1

Academic Reading

Lead-in Questions

1. Is science a blessing or a curse?
2. How can you make your future career contribute to the development of society?
3. Which is more important in scientific research, to procure national benefit or to better serve humanity?

Text A

Science Goes to War

Ernest Volkman

- 1 There were many, like the submarine inventor John P. Holland¹, who insisted that the sheer destructiveness of the modern weapons that science had provided would make war so terrible, no sane nation would fight one. What sane nation would jeopardize the great industrial civilization that had brought so much benefit to so many people? What sane nation would allow the cultural treasures, the great cities, and the flourishing towns of Europe to be destroyed in the name of war?
- 2 Yet, the signs were there. The mad race for an ultimate military security – to develop the most advanced and destructive armaments, to construct mighty military edifices, to build huge fleets, to conscript millions into vast armies, to overlay the Continent with a network of railroads to move troops rapidly, to stockpile mammoth amounts of ammunition and guns, to prepare war plans that called for armies of millions to fall upon each other in mass battles – amounted to a gargantuan powder keg, needing only a single political spark to set off Armageddon². For nearly a hundred years, Europe had enlarged the Napoleonic concept of total war with scientific

1 John P. Holland: 约翰·菲利普·霍兰, 爱尔兰潜艇科学家, 他研制了世界上第一艘可以实战的潜艇“霍兰-6”号, 被公认为“现代潜艇的鼻祖”。

2 Armageddon: 基督教《圣经》所述世界末日之时善恶对决的最终战场, 引申出“伤亡惨重的战役”“毁灭世界的大灾难”“世界末日”等含义。

and technical advances to make it even more total. Science finally had been mated indissolubly to war, creating a juggernaut that no one was able to stop. But politicians and generals thought that although modern total war would be incredibly violent, like nothing ever seen before, it would be short. European generals dreamed and planned for a short war in which the outcome would be decided in one great Napoleonic campaign, the decisive blow, using the full might of modern armaments, to destroy an enemy totally. No one who directed any of these mighty military machines seemed to doubt that the fruits of modern science – metallurgy³, chemistry, industrial machinery, radio, turbines, diesel engines, hydraulics⁴, fire-control mathematics, gun ballistics⁵, optics⁶ – provided an irresistible force that guaranteed victory.

- 3 But when the great cataclysm of World War I finally arrived, events soon proved this belief a tragic chimera. By the end of 1914, the mass armies were locked in stalemate and, inevitably, they turned to the one weapon they believed would end that stalemate and bring victory: science. And that would cause the gravest crisis of conscience in the history of science, one that remains unresolved to this day.
- 4 It was a crisis summarized in the career of one of the century's greatest scientific minds, a mind that simultaneously could provide the science to improve mankind's existence and the science to destroy it.
- 5 In 1909, the German chemist Fritz Haber electrified his scientific discipline with an amazing discovery: recovering nitrogen⁷ from the atmosphere by "fixing" it in a process that involved synthesizing ammonia⁸. The implications of that discovery were enormous: With the process, agricultural fertilizer could be mass-produced at low cost, the development that set off the agricultural revolution in which per-acre yields increased a hundredfold, allowing nations for the first time to produce enough food to feed their growing populations.

3 metallurgy: 冶金学, 是一门研究如何经济地从矿石或其他原料中提取金属或金属化合物, 并制成具有一定性能的金属材料的科学。

4 hydraulics: 水力学

5 ballistics: 弹道学; 发射学。

6 optics: 光学

7 nitrogen: 氮气, 氮元素。氮元素以单质(氮气)形式存在时无毒, 以化合物形式存在时常常有毒。

8 ammonia: 氨, 或称氨气, 是一种无色气体, 有强烈的刺激气味, 极易溶于水。液氨是一种制冷剂。氨也是制造硝酸、化肥、炸药的重要原料。

- 6 Fertilizer had been relatively expensive up to that point, since its key ingredient, sodium nitrate⁹, had to be imported, mostly from large deposits in Chile. Haber's process eliminated the need for that chemical in fertilizer production. The feat would earn him the Nobel Prize in Chemistry, and in 1911 he was named head of one of the scientific institutes Kaiser Wilhelm¹⁰ had set up to establish German preeminence in science.
- 7 Haber was busily at work at the institute in 1914 when an alarmed German High Command approached him for help in solving a serious logistical problem, one that involved a subject Haber knew well: sodium nitrate. The problem, Haber was told, stemmed from the fact that sodium nitrate was a key ingredient in explosives. Germany's stock of the chemical would be exhausted by 1916, given the vast expenditure of artillery ammunition the war required. And since a British naval blockade had cut off Chilean supplies, Germany faced the prospect of running out of ammunition. Could Herr Doktor Haber find a solution?
- 8 In short order, Haber found the solution, which amounted, basically, to a refinement of the process that had revolutionized world agriculture, this time involving a new process of making gun cotton that eliminated the need for sodium nitrate. Haber had taken a fateful step, for he was now using his scientific genius not for the betterment of mankind – as he once claimed – but for its destruction. And there was worse to come. In 1915 the German High Command approached him again, this time with a problem involving battlefield tactics. Artillery, even great barrages of fire, was proving insufficient in driving enemy troops out of their trenches; they simply dug deeper. Was there some kind of chemical Haber could devise that would be fired into enemy trenches, forcing the troops out? Haber's solution was a relatively mild form of chlorine gas, highly irritating (though not fatal) to the human respiratory system. But the gas, launched toward enemy trenches by large blowers, proved subject to wind conditions, and was easily defeated by soldiers holding wet handkerchiefs over their noses and mouths. The trench deadlock continued, and desperate for something to end it, the High Command turned to Haber again. This time he had a drastic solution in mind: poison gas, the greatest evil ever spawned by the union of science and war until the advent of thermonuclear weapons.

9 sodium nitrate: 硝酸钠, 易溶于水和液氨, 可助燃, 有氧化性, 与有机物摩擦或撞击能引起燃烧或爆炸。

10 Kaiser Wilhelm: 威廉二世, 末代德意志皇帝和普鲁士国王, 也是世界大战闪电战计划创始人。

- 9 In 1916 the German army created the Chemical Warfare Service and named Haber as its chief, with a mandate to come up with poison gases that could be packed inside the warheads of artillery shells, then exploded in enemy lines and spread deadly fumes capable of wiping out entire armies. Working nearly around the clock, Haber finally came up with phosgene¹¹, which could kill a man in seconds, and mustard gas¹², which even if not fatal, almost invariably caused blindness and severe lung damage. That year, these terrible new weapons were unleashed against a section of French-held front; several thousand soldiers were killed outright and many thousands of others ran in panic. The Germans didn't have the mobility to exploit the success, and the Allies quickly came up with countermeasures, especially the invention of the gas mask.
- 10 But poison gas would now forever be linked to the name of Fritz Haber, an odiousness that did not appear to bother him. He was puzzled when he heard about the controversy that broke out in Britain when a prominent British chemist, Frederick Soddy¹³, refused to do any research connected with poison gas, and was threatened with incarceration in the Tower of London as a "traitor." To Haber, the Soddy case made no sense; he assumed that Soddy was as fervid a British patriot as Haber was a German one. And in a time of war, a scientist's first duty was to his country, not to mankind or pure science. If a scientist was called upon to invent new and greater ways of killing other human beings, that was an unfortunate but necessary by-product of war.
- 11 For that reason, Haber felt no discomfort when his closest friend, Albert Einstein, berated him for using his great scientific talents to slaughter so many of his fellow human beings. In Haber's view, Einstein was a hopeless dreamer when it came to politics, a brilliant scientific mind who did not understand that Germany's very existence hung in the balance. He insisted that in a total war, where science was the major determinant of national survival, a scientist was just as much a soldier as the man with a rifle on the front line. Despite the great chasm between them on this subject, Haber and the lonely pacifist remained good friends. The same, however, cannot be said of relations between Haber and his wife. Appalled that her husband was personally responsible for so many deaths, she begged him to stop and follow the example of his close friend Einstein, who refused to have anything to

11 phosgene: 光气, 又称碳酰氯, 高毒, 不燃, 化学反应活性较高, 遇水后有强烈腐蚀性。

12 mustard gas: 芥子气, 曾在一战的战场上给交战双方造成了惨重损失, 有“毒气之王”之称。

13 Frederick Soddy: 弗雷德里克·索迪, 英国物理学家、化学家, 获 1921 年诺贝尔化学奖。

do with any science that involved war. She and her husband had increasingly bitter arguments on the subject; finally, when she realized she could not sway him, she committed suicide.

- 12 Even this tragic event in his life failed to change Haber's mind. At the end of the war, he was shocked to learn that the victorious Allies were considering bringing him up on war crimes charges. The plan was dropped after a number of scientists rushed to Haber's defense, notably the distinguished British biologist J.B.S. Haldane¹⁴, who argued that the war was intrinsically evil before the invention of poison gas, and would have remained evil because of high-explosive shells, the tank, the submarine, and barbed wire. So how could Haber take something that was already evil beyond comprehension and make it even more evil?
- 13 An interesting question. The answer would come sooner and more terribly than Haldane could have realized.

New Words and Expressions

jeopardize /'dʒepədɑɪz/ *vt.* to risk losing or spoiling something important 危及; 危害; 损害

armament /'ɑ:məmənt/ *n.* the weapons and military equipment used by an army 军备; 武器

edifice /'edɪfɪs/ *n.* a building, especially a large one (尤指宏伟的) 建筑

conscript /kən'skrɪpt/ *vt.* to make someone join the army, navy etc. 征召 (入伍)

stockpile /'stɒkpɑɪl/ *vt.* to keep adding to a supply of goods, weapons etc. that you are keeping ready to use if you need them in the future 贮存, 积聚 (物资、武器等)

mammoth /'mæməθ/ *adj.* extremely large 庞大的, 巨大的

gargantuan /gɑ:'gæntʃuən/ *adj.* extremely large 巨

大的, 庞大的

juggernaut /'dʒʌgənɔ:t/ *n.* a very powerful force, organization etc. whose effect or influence cannot be stopped 不可抗拒的强大力量; 无法控制的强大机构

guarantee /,gærən'ti:/ *vt.* to make it certain that something will happen 使必然发生, 确保

cataclysm /'kætəklɪzəm/ *n.* a violent or sudden event or change, such as a serious flood or earthquake (突发的) 剧变, 灾难 (如洪水、地震)

chimera /kaɪ'mɪərə/ *n.* something, especially an idea or hope, that is not really possible and can never exist 幻想; 妄想

stalemate /'steɪlmət/ *n.* a situation in which it seems impossible to settle an argument or

14 Haldane: 霍尔丹, 生于英国, 后入印度籍, 生物学家。

disagreement, and neither side can get an advantage 僵局; 僵持

conscience /'kɒŋʃəns/ *n.* the part of your mind that tells you whether what you are doing is morally right or wrong 良知, 良心

eliminate /ɪ'lɪmɪneɪt/ *vt.* to completely get rid of something that is unnecessary or unwanted 消除; 根除

feat /fi:t/ *n.* something that is an impressive achievement, because it needs a lot of skill, strength etc. to do 业绩, 功绩, 壮举

preeminence /pri:'emɪnəns/ *n.* the quality of being much more important, more powerful, or better than any others of its kind 卓越, 杰出

deadlock /'dedlək/ *n.* a situation in which a disagreement cannot be settled 僵局

spawn /spɔ:n/ *vt.* to make a series of things happen or start to exist 使大量生产, 使大量出现; 酿成

advent /'ædvent/ *n.* the advent of something: the time when something first begins to be widely used 某事物的出现 (到来)

invariably /ɪn'veəriəbli/ *adv.* if something

invariably happens or is invariably true, it always happens or is true 始终不变地; 总是

unleash /ʌn'li:ʃ/ *vt.* to suddenly let a strong force, feeling etc. have its full effect 突然释放, 发泄 (力量, 感情等)

outright /aʊt'reɪt/ *adv.* immediately and without any delay 立刻, 立即

countermeasure /'kaʊntəmeɪʒə/ *n.* an action taken to prevent another action from having a harmful effect 对策, 对付措施

incarceration /ɪn,kɑ:sə'reɪʃən/ *n.* putting or keeping someone in prison 监禁, 关押

chasm /'kæzəm/ *n.* a big difference between two people, groups, or things (两个人、团体或事物间的) 巨大差距; 显著分歧

appall /ə'pɔ:l/ *vt.* to make someone feel very shocked and upset 使惊骇, 使震惊

sway /sweɪ/ *vt.* to influence someone so that they change their opinion 影响 (某人); 使改变看法

intrinsically /ɪn'trɪnsɪkli/ *adv.* in an essential or natural way 从本质上 (讲)

Building Your Vocabulary

Task 1

Complete the following sentences using the expressions in the box below. Change the form where necessary.

ultimate	electrifying	synthesize	unleash	advent
subject to	unresolved	spark	yield	in panic

1. Lithium-oxygen batteries have been touted as the “_____” battery due to their theoretical energy density, which is ten times that of a lithium-ion battery.
2. In _____ advance, researchers create circuit within living plants.
3. An ability to _____ molecules remains essential training for the next generation of chemists; it is simply part of the indispensable core of the subject.
4. These long and expensive procedures may produce tiny _____ of the target molecule.
5. This technology could _____ the self-driving revolution for older vehicles.
6. With the _____ of self-driving cars and digital personal assistants, AI is beginning to change people’s lives for the better arrival.
7. Governments should ensure that all data they collect is available to the public for analysis, _____ privacy and national security considerations.
8. It points out several gaps and _____ issues which make it difficult to assess Cu requirements.
9. This work sheds new light on the biological mechanisms of individual vulnerability to alcohol-heightened aggression and is sure to _____ new research aimed at reducing the burden of violence on our society.
10. The Indian IT industry is _____ and rightly should be because technology is overtaking them and they haven’t invested in it.

Understanding the Text

Task 2 Choose the statement that best summarises the theme of Text A.

- A Scientists are greatly tortured when they exploit war to promote science.
- B Science pushed the war to an end more destructively and more quickly.
- C Science has empowered modern weapons to bring about destruction whose magnitude is beyond imagination.

Task 3 Complete the following statements based on Text A.

1. According to Paragraph 2, Napoleon's concept of war is _____ by science.
2. Fritz Haber's career is mentioned in order to _____.
3. Albert Einstein is compared with Fritz Haber, since _____.
4. In the following paragraph after Paragraph 13, the author would most probably write about _____.

Academic Reading Skills

Analysing Unknown Words Through Context

Unknown words tend to pose a challenge in reading comprehension. However, readers can tackle it by resorting to the context for proper analysis and accurate understanding of unknown words. Context clues are hints found within a sentence, a paragraph, or a passage. They can be classified as follows:

CONTEXT CLUE 1: Definition/Description

The new term may be formally defined, or sufficient explanation may be given within the sentence or in the following sentence. For example:

*The coating includes substances known as **photocatalysts**, which trigger chemical reactions in light.*

The word “photocatalysts” in the sentence means “substances which trigger chemical reactions in light”.

CONTEXT CLUE 2: Examples

Sometimes when a reader finds a new word, an example might be found nearby that helps to explain its meaning. Expressions like *including*, *such as*, and *for example*, point out example clues. For example:

*Total synthesis of a natural product can give chemists access to non-natural derivatives that might have **pharmacological** effects – as, for example, in the discovery of new antibiotics.*

The word “pharmacological” in the sentence means “of or relating to medicine”. The example given thereafter is a familiar subject in the field of medicine – the discovery of new antibiotics.

CONTEXT CLUE 3: Restatement

The reader may discover the meaning of an unknown word by identifying it with some synonym nearby because it repeats an idea expressed in familiar words nearby. For example:

*The vintners’ trick of adding **litharge** (lead monoxide) to wines was still common in the 18th and 19th centuries.*

It can be inferred from the context that “lead monoxide” in brackets is a restatement of “litharge”, hence further explaining the meaning of “litharge”.

CONTEXT CLUE 4: Tone

The author sets a tone, and the meaning of the unknown word must be in agreement with the tone. For example:

*The most alarming of all man's assaults upon the environment is the contamination of air, earth, rivers, and sea with dangerous and even **lethal** materials.*

The word “dangerous” in the parallel structure sets the negative tone of the sentence; therefore, “lethal” means “deadly”.

CONTEXT CLUE 5: Common-sense knowledge

Common-sense knowledge provides an effective clue to a word's meaning. For example:

*Hence the arrival of the newest weapon in the corporate team-building **arsenal**: bouncy castles.*

The word “arsenal” in the sentence means “a place where weapons are stored”.

CONTEXT CLUE 6: Word formation

Knowledge of prefixes, roots, and suffixes can be a direct clue to meaning. For example:

*The destinies of the two races in this country are **indissolubly** linked together.*

The root of “indissolubly” is “solve”, then by adding both prefixes (dis-, in-) and suffixes (-ble, -ly) it becomes an adverb meaning “that cannot be dissolved or separated”.

CONTEXT CLUE 7: Structure

Sentence structures can be constructive clues to the meaning of the unknown word.

Cause and effect

Causal relationships may imply the meaning of the unknown word. Words like “because”, “since”, “therefore”, “thus”, “so”, etc. may signal context clues. For example:

*These pesticides in water body can impair the structure and function of the aquatic ecosystem. Thus, pesticides are a main source of **contaminants** in aquatic bodies of agro-ecosystem.*

From the cause-and-effect relationship indicated by the words “thus” and “source”, it can be inferred that the word “contaminants” means “pollutants”.

Contrast

Contrastive patterns can help the reader find the meaning of an unfamiliar word by identifying the antonym of a familiar term. Words like “although”, “however”, and “but” may signal contrast clues. For example:

*On a global scale, the electricity we generate in thousands of power stations is enormous. Even so, it **pales** in comparison to the amount of energy the earth receives*

every day from the sun.

From the comparison and contrast indicated by “even so” and “in comparison to”, it can be inferred that the word “pales” is opposite in meaning to “enormous”.

Task 1 Use the context clues to determine the meaning of the underlined words in the following sentences.

1. This pollution is for the most part irrecoverable; the chain of evil it initiates not only in the world that must support life but in living tissues is for the most part irreversible.
2. Vanilla and sugar were added to make chocolate sweeter and more palatable for European tastes.
3. Unfortunately, the accompanied water/solvent electrolysis process aggravates the expansion and delamination of graphitic materials, which lead to ineffective current supply or broken circuit.
4. It took hundreds of millions of years to produce the life that now inhabits the earth – eons of time in which that developing and evolving and diversifying life reached a state of adjustment and balance with its surroundings.
5. Industrial applications of the developed nanostructures have been explored as well for solar light absorber and photocatalysis to split water and decompose organic material using visible light.
6. The apple trees were coming into bloom but no bees droned among the blossoms, so there was no pollination and there would be no fruit.
7. Chemical stability diagrams describe how solution chemistry changes upon the thermodynamically driven dissolution of a species into solution as the system progresses toward equilibrium.
8. For the first time in the history of the world, every human being is now subjected to contact with dangerous chemicals, from the moment of conception until death.

Task 2

Use the context clues to determine the meaning of the underlined words in the following paragraphs.

Paragraph 1:

Researchers have designed a new kind of sensor using modified, fluorescent gold nanoparticles that can detect extremely low concentrations of melamine, a nitrogen-rich organic compound used to adulterate milk, infant formula and various other foodstuffs. The sensor will be useful in identifying food products contaminated with melamine. Melamine-contaminated milk is known to have deleterious effects on children's kidneys. Animal studies suggest that melamine-containing foods could trigger bladder cancer. Existing techniques to detect melamine in foods are selective and fail to track minute traces. To devise a better melamine sensor, the researchers synthesized fluorescent gold nanoparticles using amino-mercapto-triazole, an organic compound. The nanoparticles were capped with the organic compound. They then used these nanoparticles to identify traces of melamine in real samples such as cow milk and infant formula.

Paragraph 2:

In fact, it is difficult to detect the decline of productivity that results from erosion directly, because the productivity reduction caused by erosion often occurs so slowly that it may not be recognized until crop production is no longer economically viable. Moreover, improved technology often masks productivity decline caused by erosion, leading to increased rather than decreased yields. Various indirect methods (e.g. the comparative-plot method, transect method, and desurfacing experiments) have been carried out extensively in the study of erosion-productivity relationships.

Paragraph 3:

Smart windows rely on liquid crystals which respond to the effect of an electric field. Under its influence such molecules will then align themselves all in the same direction and so not impede the passage of light. Remove that field and they relax to a random arrangement and the glass becomes opaque. To make smart windows requires two panes of glass, suitably coated with indium tin oxide to make the surfaces conducting. Between them is sandwiched a solution of the liquid crystals as a film only 20 microns thick.

Complete the following table to identify the meaning (in English or Chinese) of the underlined words and the context clue(s) you've used.

One roadblock to chemical weapons disposal is that heat and humidity quickly break down enzymes that can disable the deadly chemicals. Now, researchers have developed a highly stable compound that can inactivate nerve agents like sarin in a matter of minutes.

To create the compound, chemists Omar Farha and Joseph Hupp of Northwestern University in Evanston, Illinois, turned to nature for inspiration. Bacteria produce enzymes called phosphotriesterases¹⁵, which deactivate certain pesticides and chemically related nerve gases at lightning speed, wiping out weapons in milliseconds. But such enzymes are fragile and easily degraded. The chemists set out to reproduce the mechanism by which phosphotriesterase breaks down these chemicals to create a human made catalyst that would survive the most inhospitable conditions.

They started with metal-organic frameworks (MOFs), a recently developed class of porous compounds composed of metals arranged in a crystalline network linked by carbon-based molecules. MOFs are highly adaptable materials: Scientists can switch out the metals or the linkers to optimize the material for a variety of applications, such as carbon dioxide capture and hydrogen or methane storage. And because MOFs are porous, they have large surface areas that can rapidly create chemical bonds, making them good candidates for catalysts.

In the natural enzyme, phosphotriesterase, two zinc atoms act as so-called Lewis acids, which accept electrons to bind with the nerve agent. Once the agent has bonded, hydrolysis occurs – a water molecule attacks the agent, slicing and dicing essential chemical bonds, thereby deactivating it. The scientists designed MOFs with a similar structure, but they replaced the zinc with zirconium, which likewise behaves as a Lewis acid and makes for an ultrastable MOF.

Words	Meaning	Context clue(s) used
disable		
inspiration		
degraded		
inhospitable		
optimize		
hydrolysis		

15 phosphotriesterases: 磷酸三酯酶

Academic Writing

Direct Conversion of CO₂ into Liquid Fuels with High Selectivity over a Bifunctional Catalyst

Peng Gao et al

- 1 Owing to the growing energy demand, dwindling fossil fuel reserves and increasing atmospheric CO₂ concentration, renewable energy sources, such as solar, wind and biomass, foresee increasing usage. However, the widespread utilization of renewable energy sources is currently limited by their intermittent and fluctuating nature. In this context, the chemical conversion of carbon dioxide (CO₂) into value-added products with the assistance of hydrogen (H₂) would represent a promising solution to the storage of renewable energy.
- 2 Thus, much attention has been paid to CO₂ hydrogenation to various C₁ feedstocks (for example, methane¹⁶ [CH₄], methanol¹⁷ [CH₃OH], carbon monoxide [CO] and formic acid¹⁸) and considerable progress has been achieved. However, the extreme inertness¹⁹ of CO₂ and the high kinetic barriers for the formation of C–C bonds mean it is still a great challenge to synthesize C₂₊ (hydrocarbons with two or more carbons) products directly from CO₂, such as gasoline (C₅–C₁₁ hydrocarbons), which is a very important transportation fuel widely used around the world. CO₂ is well-known to be a very stable molecule ($\Delta_f G^\circ = -396 \text{ kJ mol}^{-1}$), the end product of any combustion process, either biological or chemical, along with water. Another key bottleneck problem is the assembly of the atoms and formation of chemical bonds

16 methane: 甲烷

17 methanol: 甲醇

18 formic acid: 甲酸

19 inertness: 惰性

to convert the relatively simple CO_2 molecules into the much more complex and energetic $\text{C}_5\text{--C}_{11}$ hydrocarbons. Obviously, significant catalytic advances are required for the large-scale production of liquid fuels directly from CO_2 hydrogenation. Currently, the Fischer–Tropsch synthesis (FTS) route using modified Fe-based catalysts can be utilized to produce hydrocarbons directly from CO_2 . However, the maximum $\text{C}_5\text{--C}_{11}$ hydrocarbon fraction was limited by the Anderson–Schulz–Flory distribution to $\sim 48\%$ with an undesirable CH_4 fraction of $\sim 6\%$. Furthermore, the heat of adsorption of CO_2 is lower than that of CO because of the thermodynamic stability of CO_2 , which leads to a much lower coverage of CO_2 over the catalyst, and thus a low CO_2 reactivity and high CH_4 selectivity.

- 3 In the present work, a bifunctional catalyst that contains partially reducible metal oxides (In_2O_3) and H-form Zeolite Socony Mobil-5 (HZSM-5) zeolites²⁰ exhibits an excellent performance for the direct production of gasoline-range hydrocarbons from CO_2 hydrogenation with a high selectivity. The C_{5+} selectivity in hydrocarbon distribution reached up to 78.6% with only 1% for CH_4 selectivity at a CO_2 conversion of 13.1%. There was no obvious catalyst deactivation over 150 hours, and a much better performance for CO_2 hydrogenation to C_{5+} hydrocarbons was observed using a pellet catalyst with internal gas recycling. Such results thus suggest a promising potential for its industrial application.

20 H-form Zeolite Socony Mobil-5 (HZSM-5) zeolites: 美国公司开发的由 Zeolite Socony Mobil 缩写命名的 ZSM 系列高硅铝比沸石分子筛催化剂

Academic Writing Skills

Expressing with Nominalisation

Understanding nominalisation

Nominalisation is an essential feature of academic writing. Nominalisations are formed by adding affixes to verbs or adjectives or by conversion of the parts of speech of such words as verbs or adjectives. For example:

- A *The **development** of chemical strategies to couple O₂ **reduction** directly to substrate **oxidation** would enable sustainable synthetic methods.*
- B *In large part, the increased use of energy sources which emit carbon dioxide is driven by improved domestic **availability** of natural gas.*

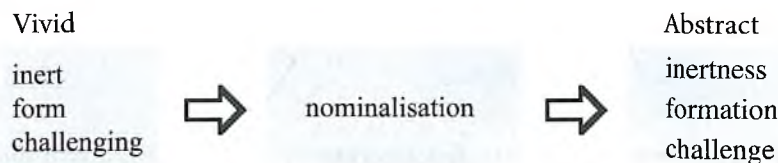
In these two examples, nominalisations (such as *development*, *reduction*, *oxidation*, and *availability*) are used. It can be shown clearly in the following figure.



Features of nominalisation

Content: Abstract

- A *CO₂ is extremely **inert**; hence it seems that there are a lot of barriers preventing it from becoming active enough to react to other elements so as to **form** C-C bonds. As a result, it is still very **challenging** for researchers to synthesize C₂₊ (hydrocarbons with two or more carbons) products directly from CO₂.*
- B *However, the extreme **inertness** of CO₂ and the high kinetic barriers for the **formation** of C-C bonds mean it is still a great **challenge** to synthesize C₂₊ (hydrocarbons with two or more carbons) products directly from CO₂.*



Compare these two sentences. They express the same information, but nominalisations are used in the second sentence. The difference lies in that adjectives or verbs are used to describe a situation or action vividly in non-academic writing whereas nominalised words are more abstract and frequently used in academic writing.

Tone: Objective

- A *In this context, if we can add hydrogen (H₂) to help turn carbon dioxide (CO₂) into valuable products, e.g. gasoline, we believe it would probably help to solve the difficult problem we face. We mean it is still very difficult to find a good way to store the global-warming gas, carbon dioxide.*

- nominalisation  everyday speech/writing

- B *In this context, the chemical conversion of carbon dioxide (CO₂) into value-added products with the assistance of hydrogen (H₂) would represent a promising solution to the storage of renewable energy.*

+ nominalisation  academic writing

By using nominalisation, the second version becomes much more objective in tone than the first version. Objectivity is the major element dictated by academic writing.

Form: Concise

- A *The chemical process lasted for 150 hours. In such a long process, the catalyst did not fail to function at a significant level. The process of converting carbon dioxide by adding hydrogen to it performed much better than using other approaches before. In our work, a pellet catalyst with internal gas recycling was used. Therefore, C₅₊ hydrocarbons were efficiently obtained by adding hydrogen to CO₂. (65 words)*

a longer version
unpacking



packing
a shorter version



- B *There was no obvious catalyst deactivation over 150 hours, and a much better performance for CO₂ hydrogenation to C₅₊ hydrocarbons was observed using a pellet catalyst with internal gas recycling. (30 words)*

Nominalisation enables the text to be more concise by packing a great deal of information into a few words while readers can get a lot of information by unpacking the compact academic text.

However, it is suggested that nominalisation should not be overused although it is an essential feature of academic writing. What counts most in academic writing is to convey information as clearly as possible.

Task 1

Read the following excerpt from Text B, underline the nominalised words and discuss the effect of using them.

Another key bottleneck problem is the assembly of the atoms and formation of chemical bonds to convert the relatively simple CO_2 molecules into the much more complex and energetic $\text{C}_5\text{--C}_{11}$ hydrocarbons. Obviously, significant catalytic advances are required for the large-scale production of liquid fuels directly from CO_2 hydrogenation. Currently, the Fischer–Tropsch synthesis (FTS) route using modified Fe-based catalysts can be utilized to produce hydrocarbons directly from CO_2 . However, the maximum $\text{C}_5\text{--C}_{11}$ hydrocarbon fraction was limited by the Anderson–Schulz–Flory distribution to ~48% with an undesirable CH_4 fraction of ~6%. Furthermore, the heat of adsorption of CO_2 is lower than that of CO because of the thermodynamic stability of CO_2 , which leads to a much lower coverage of CO_2 over the catalyst, and thus a low CO_2 reactivity and high CH_4 selectivity. (Para. 2)

Task 2

Find and discuss the usage of nominalised words in the following sentences.

1. The thick silicate layer also places serious mass transfer limitations on substrate diffusion that can reduce the apparent activity of enzyme catalysis.
2. The consumption of biofuels and biomass-derived chemicals can diversify energy supply, reduce the dependence on fossil fuels and result in less adverse environmental impacts.
3. To enhance the economic feasibility of large-scale production of biofuels, extensive research has been undertaken to improve the hydrolysis²¹ efficiency and reduce the cost.
4. The safety of chemical processes should be supported by the harmonization of

21 hydrolysis: 水解, 是利用水将物质分解形成新的物质的过程。

incident and accident investigation across national borders.

5. Hazards are also arising from a lack of consultation, poor supervision and willful violations from people in response to overspecification of procedures or generally poor procedures.
6. In chemical industrial parks that specialize in petrochemical production, efficient use of both materials and energy can be improved by the integration of materials, products, and energy.
7. Rather than having to design a process to fit an enzyme's requirements, directed evolution has made it possible to fit the catalyst to optimal manufacturing conditions.
8. Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances.

Task 3 Find the nominalised words in the following sentences and then unpack them.

1. The release of carbon dioxide as well as light hydrocarbon gases is understood to contribute significantly to the increase in global average temperature through the greenhouse effect.

Unpacking: _____

2. By post-deposition heat treatment, an improvement in the transport properties of SnS film can be achieved.

Unpacking: _____

3. Organic solar cells have been a subject of growing research interest over the past quarter century, and are now developed to the point where they are on the verge of introduction into the market.

Unpacking: _____

-
-
4. Despite its synthetic versatility, the use of diazomethane²² in the laboratory is often considered to be problematic or risky because of its explosiveness and toxicity.

Unpacking: _____

5. Inadequate waste disposal leads to higher increase of public exposure to toxic compounds having negative effects on environment and human health. It was proved that the REE's²³ bioaccumulation through the food chain can cause ailments due to the exposure of humans to low concentration of REEs.

Unpacking: _____

22 diazomethane: 重氮甲烷

23 REE: the rare earth element, 稀土元素。

Task 4 Pack the given information in the following paragraphs and rewrite them into academic texts.

1. Many people are worried about their health and public health problems. So products, like baby bottles, have attracted much attention. First, the chemical BPA used in products, like baby bottles, is shown to be harmful to health. Then, scientists find the related compound BPS can be used instead.

Packing: _____

2. Some researchers do additional experiments. They suspect that iron-reducing bacteria might be the culprit. The iron-reducing bacteria grow dramatically if they are under the oxygen-free conditions. Actually, the oxygen-free conditions are favorable to arsenic. They mobilize As (III).

Packing: _____

3. A lot of research is done on e-cigarette aerosol emission. When scientists do research, they typically research on benchmarking toxicant levels and how toxic cigarettes are compared with benchmarking toxicant levels. However, e-liquids have distinct chemical makeup and their unique properties. And the studies do not fully explore and explain that.

Packing: _____

4. If a larger number of pesticides are used in most sections of the world, it would make more food produced, make people not or less hungry, and make people healthier.

Packing: _____

5. To keep the battery safe, we must make separators stable in terms of thermal energy, that is to say, they shouldn't be too hot, or overheated. So researchers have developed new polymers. The new polymers have high melting points, and then, they help to make them more stable, that is, more integral. However, their mechanical properties are still not very good. That makes it difficult to process. It is not easy to process esp. when the battery is assembled.

Packing: _____

6. Chemically synthesizing proteins helps to promote biochemical and biophysical studies in a unique way. By *unique*, we mean it is a molecular tool. Even, it is able to control protein structures as precisely as up to the atomic level. When preparing diverse water-soluble proteins, people probably use the mainstream methods. That is to say, they have peptide segments produced. And they have them chemoselectively condensed. Nowadays, integral membrane proteins become the target of biotechnological research and drug development. However, chemically synthesizing them has been hampered. Why? Because it is difficult to prepare and handle poorly soluble transmembrane peptides.

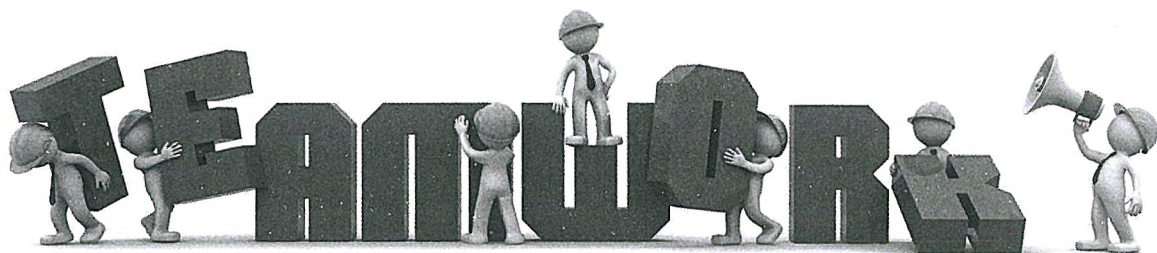
Packing: _____

Unit 9

Management

Servant leadership is easy for people with high self-esteem. Such people have no problem giving credit to others. They have no problem listening to other people for ideas. They have no problem in building other people up.

— Ken Blanchard



1 Academic Reading

Text A: If Humble People Make the Best Leaders, Why Do We Fall for Charismatic Narcissists?

Academic Reading Skills: Understanding Collocation

2 Academic Writing

Text B: Looking Too Old? How an Older Age Appearance Reduces Chances of Being Hired

Academic Writing Skills: Using Hedging Expressions Properly

1

Academic Reading

Lead-in Questions

1. What are the most important qualities of a great leader?
2. What should a leader do to keep his/her team members keen and motivated?

Text A

If Humble People Make the Best Leaders, Why Do We Fall for Charismatic Narcissists?

Margarita Mayo

- 1 The research is clear: When we choose humble, unassuming people as our leaders, the world around us becomes a better place.
- 2 Humble leaders improve the performance of a company in the long run because they create more collaborative environments. They have a balanced view of themselves – both their virtues and shortcomings – and a strong appreciation of others' strengths and contributions, while being open to new ideas and feedback. These “unsung heroes” help their believers to build their self-esteem, go beyond their expectations, and create a community that channels individual efforts into an organized group that works for the good of the collective.
- 3 For example, one study examined 105 small-to-medium-sized companies in the computer software and hardware industry in the United States. The findings revealed that when a humble CEO is at the helm of a firm, its top management team is more likely to collaborate and share information, making the most of the firm's talent.
- 4 Another study showed that a leader's humility can be contagious: When leaders behave humbly, followers emulate their modest attitude and behavior. A study of 161

teams found that employees following humble leaders were themselves more likely to admit their mistakes and limitations, share the spotlight by deflecting praise to others, and be open to new ideas, advice, and feedback.

- 5 Yet instead of following the lead of these unsung heroes, we appear hardwired to search for superheroes: over-glorifying leaders who exude charisma.
- 6 The Greek word *Kharisma* means “divine gift,” and charisma is the quality of extraordinary charm, magnetism, and presence that makes a person capable of inspiring others with enthusiasm and devotion. German sociologist Max Weber defined charisma as “of divine origin or as exemplary, and on the basis of it, the individual concerned is treated as a leader.” Research evidence on charismatic leadership reveals that charismatic people are more likely to become endorsed as leaders because of their high energy, unconventional behavior, and heroic deeds.
- 7 While charisma is conducive to orchestrating positive large-scale transformations, there can be a “dark side” to charismatic leadership. Jay Conger and Rabindra Kanungo describe it this way in their seminal book: “Charismatic leaders can be prone to extreme narcissism that leads them to promote highly self-serving and grandiose aims.” A clinical study illustrates that when charisma overlaps with narcissism, leaders tend to abuse their power and take advantage of their followers. Another study indicates that narcissistic leaders tend to present a bold vision of the future, and this makes them more charismatic in the eyes of others.
- 8 Why are such leaders more likely to rise to the top? One study suggests that despite being perceived as arrogant, narcissistic individuals radiate “an image of a prototypically effective leader.” Narcissistic leaders know how to draw attention toward themselves. They enjoy the visibility. It takes time for people to see that these early signals of competence are not later realized, and that a leader’s narcissism reduces the exchange of information among team members and often negatively affects group performance.
- 9 It’s not that charismatic and narcissistic people can’t ever make good leaders. In some circumstances, they can. For example, one study found that narcissistic CEOs “favor bold actions that attract attention, resulting in big wins or big losses.” A narcissistic leader thus can represent a high-risk, high-reward proposition.

- 10 And it's not that humble leaders can't ever be charismatic. Researchers agree that we could classify charismatic leaders as "negative" or "positive" by their orientation toward pursuing their self-interested goals versus those of their groups. These two sides of charismatic leadership have also been called personalized and socialized charisma. Although the socialized charismatic leader has the aura of a hero, it is counteracted with low authoritarianism and a genuine interest in the collective welfare. In contrast, the personalized charismatic leader's perceived heroism is coupled with high authoritarianism and high narcissism. It is when followers are confused and disoriented that they are more likely to form personalized relationships with a charismatic leader. Socialized relationships, on the other hand, are established by followers with a clear set of values who view the charismatic leader as a means to achieve collective action.
- 11 The problem is that we select negative charismatic leaders much more frequently than in the limited situations where the risk they represent might pay off. Despite their grandiose view of themselves, low empathy, dominant orientation toward others, and strong sense of entitlement, their charisma proves irresistible. Followers of superheroes are enthralled by their showmanship: Through their sheer magnetism, narcissistic leaders transform their environments into a competitive game in which their followers also become more self-centered, giving rise to organizational narcissism, as one study shows.
- 12 If humble leaders are more effective than narcissistic leaders, why do we so often choose narcissistic individuals to lead us?
- 13 The "romance of leadership" hypothesis suggests that we generally have a biased tendency to understand social events in terms of leadership and people tend to romanticize the figure of the leader.
- 14 My own research shows that our psychological states can also bias our perceptions of charismatic leaders. High levels of anxiety make us hungry for charisma. As a result, crises increase not only the search for charismatic leaders, but also our tendency to perceive charisma in the leaders we already follow.
- 15 Economic and social crises thus become a unique testing ground for charismatic leaders. They create conditions of distress and uncertainty that appear to be ideal for the ascent of charismatic figures. Yet at the same time, they also make us more vulnerable to choosing the wrong leader. Crises and other emotionally laden events

increase our propensity to romanticize the grandiose view of narcissistic leaders. The paradox is that we may then choose to support the very leaders who are less likely to bring us success. In a time of crisis, it's easy to be seduced by superheroes who could come and "rescue" us, but who possibly then plunge us into greater peril.

- 16 While this may sound hopeless, there is another way of looking at it. Essentially, we have the leaders we deserve. As we collectively select and construct our leaders to satisfy our own needs and desires, we can choose humility or socialized charisma over narcissism.

New Words and Expressions

charismatic /ˌkærɪzˈmætɪk/ *adj.* having a natural ability to attract or interest other people and make them admire you 有超凡魅力的, 有感召力的

unassuming /ˌʌnəˈsjuːmɪŋ, -ˈsuː-/ *adj.* showing no desire to be noticed or given special treatment 不张扬的; 谦逊的, 不摆架子的

helm /helm/ *n.* the wheel or control which guides a ship or boat (船的) 舵轮

at the helm: in charge of something 掌权, 领导

humility /hjuːˈmɪlɪti/ *n.* the quality of not being too proud about yourself (used to show approval) 谦逊, 谦恭 (含褒义)

contagious /kənˈteɪdʒəs/ *adj.* if a feeling, attitude, or action is contagious, other people are quickly affected by it and begin to have it or do it (感情、态度、行动) 有感染力的; 会蔓延的

emulate /ˈemjʊleɪt/ *vt.* to do something or behave in the same way as someone else, especially because you admire them (尤因为仰慕而) 效仿, 模仿

deflect /dɪˈflekt/ *vt.* to do something to stop people paying attention to you, criticizing you etc.

转移, 引开 (注意、批评等)

exude /ɪɡˈzjuːd/ *vt.* if you exude a particular quality, it is easy to see that you have a lot of it 充分显露 (某一品质)

endorse /ɪnˈdɔːs/ *vt.* to express formal support or approval for someone or something (正式) 赞同, 支持, 认可

orchestrate /ˈɔːkɪstreɪt/ *vt.* to organize an important event or a complicated plan, especially secretly 精心策划

narcissism /ˈnɑːsɪsɪzəm/ *n.* when someone is too concerned about their appearance or abilities or spends too much time admiring them (used to show disapproval) 自我陶醉, 自恋 (含贬义)

grandiose /ˈgrændiəs/ *adj.* grandiose plans sound very important or impressive, but are not practical (计划等) 浮夸的; 不切实际的

counteract /ˌkaʊntərˈækt/ *vt.* to reduce or prevent the bad effect of something by doing something that has the opposite effect 对抗; 抵消

authoritarianism /ɔːˌθɒrɪˈteəriənɪzəm/ *n.* the state of being authoritarian or the belief that people

with power, especially the State, have the right to control other people's actions 权力主义, 独裁主义

enthrall /ɪn'tʁɔ:l/ *vt.* to make someone very interested and excited, so that they listen or watch something very carefully 迷住(某人); 使(某人)着迷

hypothesis /haɪ'pɒθəsɪs/ *n.* an idea that is

suggested as an explanation for something, but that has not yet been proven to be true 假设; 假说

propensity /prə'pensɪti/ *n.* a natural tendency to behave a particular way 倾向; 习性

plunge /plʌndʒ/ *v.* to move, fall, or be thrown suddenly forwards or downwards (使)突然向前倒下(或跌落)

Building Your Vocabulary

Task 1 Complete the following sentences using the expressions in the box below. Change the form where necessary.

contagious	endorse	collaborative	at the helm of
balanced	prototypically	hardwire	

1. The program presented a(n) _____ view of the two sides of the conflict.
2. What actions can be taken to engage new players in a(n) _____ endeavor?
3. Others think that the rules for what is “musical” are _____ in our brains to some degree.
4. Therefore, the hypotheses concerned the use of _____ healthy food, the use of ready-to-serve meals, as well as the use of certain shopping places, of local shops and of public nutrition counseling.
5. A disease is _____ if it is caused by some type of infecting organism that is spread human to human. Not all diseases involve an organism and not all organism involved diseases are transmitted person to person.
6. _____ ABC company are business partners who bring a strong combination of skills, experience, and a commitment in managing and growing this company locally and beyond the country.
7. Although groups of infants who will be gifted have higher BSID scores, the BSID cannot be _____ as a method of identifying individual infants who will later demonstrate superior cognitive function.

Understanding the Text

Task 2 Text A has 16 paragraphs which can be roughly separated into 4 parts. Write some notes about the focus of each part.

Section 1 (Paras. 1–4): Humble leaders make a better world.

Section 2 (Paras. 5–7): _____

Section 3 (Paras. 8–15): _____

Section 4 (Para. 16): _____

Task 3 Paraphrase the following paragraph from Text A.

One study suggests that despite being perceived as arrogant, narcissistic individuals radiate “an image of a prototypically effective leader.” Narcissistic leaders know how to draw attention toward themselves. They enjoy the visibility. It takes time for people to see that these early signals of competence are not later realized, and that a leader’s narcissism reduces the exchange of information among team members and often negatively affects group performance. (Para. 8)

Your paraphrase: _____

Academic Reading Skills

Understanding Collocation

Collocation is a group of two or more words that are usually used together. It is a pair, a group, or a sequence of words that habitually co-occur with a frequency greater than chance. For example, “heavy drinker,” “to take a risk,” and “to save time” are typical collocations in English.

Linguistic researchers study large amounts of data to analyze with statistics on how often people use certain word combinations or collocations. In terms of frequency, there are strong collocations and weak collocations. Strong collocations refer to words that are almost always used together, for example “to make noise.”

Learning collocations is important because it helps us understand words that are used in larger groups or in a chunk of language. Using these chunks can lead to more fluent English.

In academic texts, the word “research” is often used together with nouns such as “finding,” “methodology,” “purpose,” and “evidence” and with verbs such as “show,” “suggest,” and “demonstrate.” If you say “research approach” or “research proof,” people may still understand you but it will sound very unnatural. To native speakers, it may even sound funny if you fail to use the strong collocations such as “research method” or “research evidence.” It takes considerable efforts to improve the competence with the language to combine these words correctly.

Types of collocations

There are several different types of collocations made from combinations of verbs, nouns, adjectives etc. Some of the most common types are:

adverb + adjective: remarkably similar

adjective + noun: limited information

noun + noun: gender stereotype

noun + verb: research suggests

verb + noun: report findings

verb + adverb: vary significantly

verb + adjective: remain unchanged

Task 1 Identify the different types of collocations in the following text.

Flamholtz's (1986) leadership model provides a framework within which to consider the effects of cultural differences, especially in terms of understanding people's conception of an organization, coordination and control of organizational activities, and the roles and relations of employees within the organization. The model consists of a continuum of six leadership styles ranging from very directive to nondirective. At the directive end of the continuum, autocratic leaders believe they have the authority to make decisions or set goals and do not feel the need to provide explanations concerning their decisions or goals. Benevolent autocratic leaders tend to rely on authority as the main source for decision-making, but provide explanations concerning their decisions and goals. Consultative leaders are less directive and seek feedback from employees on ideas and goals before making a final decision. At the nondirective end of the continuum, participative leaders work with employees to develop and set goals, but retain the final decision-making authority. Consensus leaders seek employee input and all votes are considered nominally equal. Finally, laissez-faire leaders permit employees to make their own decisions.

Task 2 Fill in the blanks with the words in the box.

highly	challenge	certainly
critical	constantly	slightly

One popular notion is that entrepreneurs and people who enjoy 1. _____ changing, innovative environments are more creative than others. But there are many types of creativity in business. Some managers, for instance, are 2. _____ creative at fixing things that are broken and enjoy the 3. _____ of returning a system to a previous state of optimal functioning. While it's 4. _____ true that entrepreneurs excel at original thinking, so do many non-entrepreneurs. In reality, what sets entrepreneurial individuals apart is something 5. _____ different – something both broader and deeper than what is typically evoked by the word “creativity.” A 6. _____ aspect to this dimension is openness to new experiences.

Task 3 Correct the collocation errors in the following sentences.

1. In the business books, social capital has been identified as social resources that firms utilize to achieve sustainable success and competitive advantage.
2. Aside from building a thriving enterprise, a small-business owner's most important responsibility is making an environment that inspires and motivates employees.
3. Economists reformed a new theory of inflation in the late 1980s.
4. The article does not sustain its conclusions with enough convincing evidence.
5. In other words, an effectual argument is one where the conclusion necessarily follows from the premises. It is impossible for the conclusion to be false if the premises are true.
6. Secondly, preferences of the same social or occupational group may vary very much from one country to another due to the cultural, social, and economic conditions.
7. Using more explicit study evidence in policymaking could also offer alternatives and options for study evidence in more transparent ways.

Task 4 Complete the following sentences based on the Chinese hints in brackets. Pay special attention to collocations.

1. E-commerce has brought _____ (巨大的商机) to small companies. However, they may not be able to invest enough capital in the Research and Development and the application of technology, which may affect their online business.
2. Scholars _____ (提出了各种各样的理论) to explain the failure of market leaders in the face of technological change.
3. _____ (学界已有了相当的共识) on the nature of strategic thinking. It is about thinking creatively about strategic options and alternative ways to compete.
4. Which style will be favored will likely be dependent on the cultural values prevalent in a given country. For example, previous _____ (研究表明) that Chinese managers value power distance.
5. Thus, particularly for managers in multinational corporations, our findings indicate that an understanding of _____ (国家与国家之间的差异) is imperative.

2

Academic Writing

Text B

Looking Too Old? How an Older Age Appearance Reduces Chances of Being Hired

Michèle Céline Kaufmann, Franciska Krings, and Sabine Sczesny

- 1 This research examined the impact of FAA (facial age appearance) on hiring, demonstrating that older-appearing candidates were less likely to be hired than younger-appearing candidates because older age-appearance triggered impressions of lower health and fitness. Building on models of face-based impression formation (Freeman & Amady, 2011; Zebrowitz, 2006), we predicted and found that candidates with older-appearing faces – but not chronologically older candidates – triggered impressions of low health and fitness, compared to younger-appearing candidates. These impressions reduced perceptions of person-job fit, which lowered hiring probabilities for older-appearing candidates.
- 2 Past research has largely focused on age stereotypes triggered by chronological age as the driving force of discrimination against older workers. Our study points to an additional path: trait impressions derived from faces, particularly with respect to health and fitness. We demonstrated that these impressions play a crucial role in recruitment and that they have behavioral consequences, ultimately reducing older-appearing candidates' chances of being hired. The results of our study showed that impressions of low health only emerged with older-appearing candidates but not with chronologically older candidates. Thus, knowing the candidate's age did not activate possible associations between old age and low health in decision makers. Indeed, physical and psychological health problems are neither more prevalent in older than in younger workers (Ng & Feldman, 2012), nor do they seem to be a consistent part of the older worker stereotype (Posthuma & Campion, 2009). Our results indicated that impressions of poor health and fitness were primarily triggered

by age-related facial features, suggesting that this path is specific to appearance-based age bias.

- 3 The findings of our study have several implications for theory and practice: First, they call for an extension of current models of age discrimination to include the effects of FAA on personnel decisions, through health and fitness impressions. This path is not considered in current models of age discrimination (e.g. Shore & Goldberg, 2005). It is crucial, however, because facial age information and face-based trait impressions, are omnipresent in the context of work. Second, our findings imply that applicants and employees who do not fit the ideal image of health and fitness, i.e. whose facial appearance signals low health or fitness, may experience the discrimination that is typically directed toward older workers. Frequent consequences of discrimination against older workers include lower chances of being hired, less access to training, and greater risk of being laid-off in times of downsizing (Gordon & Arvey, 2004; Weiss & Maurer, 2004). Moreover, age discrimination is particularly prevalent in organizations characterized by rapid change, i.e. organizations that have grown and expect to grow rapidly (Dieckman & Hirnisey, 2007). The results of our study suggest that not only people with older FAA but also people who appear less healthy may be treated similarly, independently of their actual age and health status, and this may be particularly true in organizations that underscore dynamism and change.
- 4 Another implication of this study is that some organizational measures typically used to combat age discrimination may be ineffective. For example awareness trainings that increase employers' awareness of common stereotypes and the risk of bias, are one of the most widely used measures (Kulik & Roberson, 2008). Since people are generally unaware of the impact that face-based impressions have on their judgments (Maddox & Gray, 2002; Sczesny & Kühnen, 2004), however, awareness trainings will only be effective if they specifically target facial appearance (i.e. if they raise people's awareness of the impact that an older facial appearance may have on impressions of health and fitness and, ultimately, personnel decisions). A more promising approach, at least in the first phase of a recruitment procedure, would be the use of anonymous application procedures, because they conceal candidates' membership in social categories. In our study, use of anonymous applications indeed eliminated discrimination. But the effectiveness of anonymization depends on various additional factors, including how it is implemented (Åslund & Skans, 2012). In our study, all dates, demographic information and the picture were removed. This rather strict approach may be necessary to eliminate age discrimination, certain information in the résumé (e.g. listings of time periods with different employers) may still point to the candidate's age.

Academic Writing Skills

Using Hedging Expressions Properly

In academic writing, authors often have to be very cautious about the way they present their hypotheses, results, and conclusions. To indicate less than one hundred percent certainty they hedge, i.e. avoid a categorical statement. They do so for two reasons:

- A to tone down their statements in order to reduce the threat of opposition from other researchers;
- B to avoid overstating their results as they are aware that theirs may not be the final word on the issue.

Since the certainty of any research is hardly ever one hundred percent, the use of hedging (unless overdone) is part of academic writing. In scientific articles, hedging expressions are commonly used in literature review and discussion.

The following words and phrases can be used as hedges.

modal verbs: will, must, would, may, might, could. For example:

*Second, our findings imply that applicants and employees who do not fit the ideal image of health and fitness, i.e. whose facial appearance signals low health or fitness, **may** experience the discrimination that is typically directed toward older workers.*

verbs/phrasal verbs: appear, seem, suggest, indicate, assume, estimate, argue, tend to. For example:

*A clinical study illustrates that when charisma overlaps with narcissism, leaders **tend** to abuse their power and take advantage of their followers.*

nouns: assumption, suggestion, claim, possibility, estimation, most. For example:

*These differences also raise the **possibility** that models derived from FACE studies on high fertility sites could be overestimating any positive “silver-lining” effect of climate change on food production.*

Another example:

*Finally, **most** of this research has focused on the individual level of analysis and has not explored how leadership factors relate to organizational level processes and outcomes.*

adjectives: likely, probable, possible, many, some, several. For example:

*A study of 161 teams found that employees following humble leaders were themselves more **likely** to admit their mistakes and limitations, share the spotlight by deflecting praise to others, and be open to new ideas, advice, and feedback.*

adverbs: perhaps, probably, possibly, apparently, certainly, presumably, approximately, roughly, reasonably, somehow, somewhat, usually, generally. For example:

*However, it is **probably** fair to say that reverse bias is not as common. Moreover, measurement errors and inefficient use of data are **probably** becoming less frequent problems, since measurement error has decreased with technological advances in the molecular era and investigators are becoming increasingly sophisticated about their data.*

Another example:

*The “romance of leadership” hypothesis suggests that we **generally** have a biased tendency to understand social events in terms of leadership and people tend to romanticize the figure of the leader.*

introductory phrases: in our opinion/view, we feel/believe that, to our knowledge, it is our view that, one would expect that. For example:

***One would expect that** the demand for automobiles in China over the next 15 years will be influenced, not just by the price of automobiles and household incomes, but also by demographic changes and the rise of aspirations for a middle-class lifestyle.*

Task 1 Read the following excerpt from Text B and underline the hedging expressions.

Past research has largely focused on age stereotypes triggered by chronological age as the driving force of discrimination against older workers. Our study points to an additional path: trait impressions derived from faces, particularly with respect to health and fitness. We demonstrated that these impressions play a crucial role in recruitment and that they have behavioral consequences, ultimately reducing older-appearing candidates' chances of being hired. The results of our study showed that impressions of low health only emerged with older-appearing candidates but not with chronologically older candidates. Thus, knowing the candidate's age did not activate possible associations between old age and low health in decision makers. Indeed, physical and psychological health problems are neither more prevalent in older than in younger workers (Ng & Feldman, 2012), nor do they seem to be a consistent part of the older worker stereotype (Posthuma &

Campion, 2009). Our results indicated that impressions of poor health and fitness were primarily triggered by age-related facial features, suggesting that this path is specific to appearance-based age bias. (Para. 2)

Task 2

Read the following sentences. Discuss with your partners to see how strong the claims sound. Rank the sentences in order of strength.

1. Ignoring strategic behavior may therefore result in inefficient allocation of public resources for improving school quality.
2. Prevalence of such behavior can result in misestimating the attractiveness of certain schools if stated ranks are interpreted at face value.
3. Simultaneously identifying preferences and heterogeneity in sophistication will therefore require restricting behavioral rules and parametric assumptions.
4. A small change in students' reports could potentially have large effects on the assignment probabilities.
5. In addition, the results for the operation conditions, which were carefully stated, might be used as a reference for modern spas, which nowadays tend to include also Hammams.

Task 3

Improve the sentences by adding hedging expressions in places where necessary, so that the sentences may sound more cautious.

1. Trust is essential in virtual teams, because team members cannot easily monitor one another's behaviors, and such trust emerges among pro-socially motivated team members.
2. Because a transactive memory develops when group members interact or are trained together, it is disrupted when members are replaced.
3. But even when such information is noticed, it is not integrated into the impression that is formed. It is considered irrelevant or attributed to external factors or circumstances, rendering it not useful for forming impressions of an individual.
4. Although women are more emotional than men, there are some emotions that are reserved for men.
5. On a practical note, our findings indicate the importance of maintaining a balance between forces of globalization, which will encourage uniformity of practices and

a broad range of behaviors across cultures, and forces of cultural determination and adaptation.

6. A reason for that is that VCFs invest only in NTBFs with a positive evaluation. The range in our samples was between 3 and 5, where 5 is the best rating and 0 the worst. Another reason is that the founders perceive the advancement of the technology differently than the evaluating institution.
7. This result is influenced by the behavior of the NTBF. If it already plans to go international, it will try to raise a higher investment sum.
8. Consequently, the investors are more willing to invest more money to support the internationalization of the company. They will know that a higher amount of capital is needed.

Task 4

Insert appropriate words or phrases from the box into the spaces. One word or phrase may fit two blanks.

will likely	indicate	suggested
tended to	infer	may

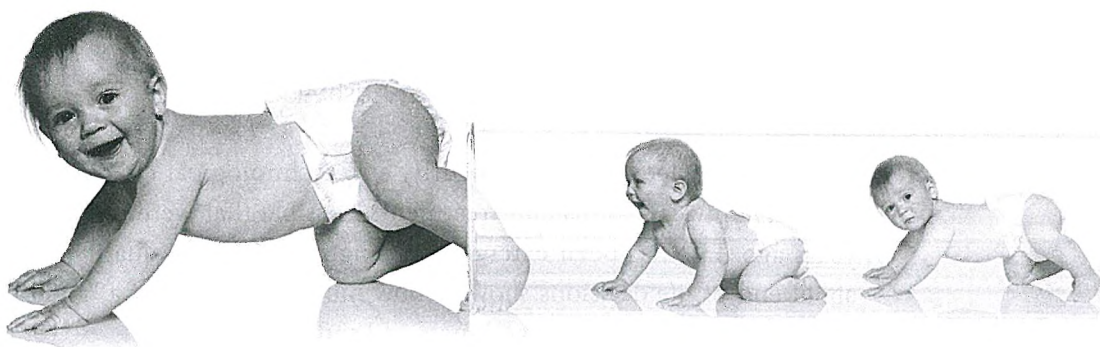
These findings 1. _____ that reconciliation between the rationalist and culturalist approach is long overdue. It does appear that managers in both countries are familiar with a range of styles and use these styles frequently and interchangeably; this 2. _____ be due to the impact of globalization of businesses. When a manager in Turkey is exposed to a Western counterpart, he or she develops and incorporates a new range of potential styles. The same 3. _____ also be true for managers from the US who are exposed to Turkish management styles. However, managers in our study 4. _____ favor particular styles over and above others. Which style will be favored 5. _____ be dependent on the cultural values prevalent in a given country. For example, previous research 6. _____ that Turkish managers value power distance. In our study, consistent with this value orientation, Turkish managers emphasized the Autocratic style. We should emphasize, however, that we did not test the link between culture and managerial behavior directly. We 7. _____ this relationship based on previous research that indicates the cultural characteristics of Turkey and the United States.

Unit 10

Medicine

I will remember that I do not treat a fever chart, a cancerous growth, but a sick human being, whose illness may affect the person's family and economic stability. My responsibility includes these related problems, if I am to care adequately for the sick.

— Hippocratic Oath



1 Academic Reading

Text A: The Doctor's Dilemma: Is It Ever Good to Do Harm?

Academic Reading Skills: Understanding Ellipsis and Substitution

2 Academic Writing

Text B: Delivering Precision Medicine in Oncology Today and in Future

Academic Writing Skills: Creating Cohesion

1

Academic Reading

Lead-in Questions

1. Do you think doctors can make a decision for the patient without his or her consent?
2. Do you think doctors are morally superior?

Text A

The Doctor's Dilemma: Is It Ever Good to Do Harm?

Gwen Adshead

- 1 Medical knowledge changes swiftly, and technological changes make new and expensive investigations and treatments possible that were only theoretical a few years ago. Life has been extended in length, but not in quality, and the debates about end-of-life decisions show us how much the notion of a “good life” is bound up with the absence of disease, illness and suffering.
- 2 The practice of medicine is not purely technical. It involves a relationship between a person who is seeking help and who may be vulnerable, and a person who has the skills and knowledge to help. Relationships that involve disparities of power, knowledge and vulnerability require some degree of external oversight and regulation. Traditionally, in medicine, this oversight has taken the form of codes of ethics, starting with the *Hippocratic Corpus*¹. Today, bodies such as the General Medical Council² and the Royal Colleges³ define the standards of good medical practice.

1 *Hippocratic Corpus*: 《希波克拉底文集》，是公元前 3 世纪的亚历山大学者们的医学著作集。

2 General Medical Council (GMC): 英国医学总会，英国一个收费的医生注册登记慈善机构。

3 Royal Colleges: 在一些英联邦国家中经过《皇家宪章》允许，可以在大学名称前加入“皇家”二字的大学。

- 3 There has been much discussion of how we make moral choices, but what do we mean by a “moral” decision in medicine? Conventionally, we are distinguishing what is clinically and technically possible from whether it is “right” to intervene at all. For example, if a person’s heart stops, we know we can resuscitate them, but should we do so?
- 4 To answer that question, we do not expect to rely solely on numerical data and we do not anticipate getting an obvious and single answer. We are aware that there may be more than one answer to the question, and those answers may conflict with each other. We will want to get clinical information about the situation: Why did the heart stop? Will restarting the heart make things better or worse for that person in medical terms? We will also want to know what the patient thinks about the situation: Did they anticipate this? Do they want to be resuscitated? And if we don’t know these things, we will want to ask some questions about how best to make a complex decision if we have not heard the wishes of the person concerned.
- 5 Moral reasoning differs from those types of reasoning that are purely computational, logical or algorithmic⁴. To answer ethical questions, we engage in a process of reflection and discussion: We begin a discourse that uses the words “ought” and “should,” as opposed to “can” and “must.” If the patient’s heart has stopped because they are losing blood, then a doctor may say: “We must give the patient more blood or his heart will stop, and we can do so because the blood is here and we know it will work.” However, that statement does not answer the question: “Should we resuscitate the patient if his heart stops?” The doctor’s statement about what can be done is not irrelevant, but it is only a part of the reasoning process involved in deciding whether it is right to resuscitate. If the patient had left instructions that they did not want to be resuscitated if their heart stopped during surgery, then the facts of successful resuscitation practice would be irrelevant to what the doctors should do.
- 6 What we are distinguishing here are facts and values – a distinction developed by David Hume⁵ in the 18th century. Hume says that it is a fallacy to think that because things are a certain way (facts), then they should be that way (values). We cannot derive values from facts, but we do evaluate facts and make moral judgments about them, and this reasoning and reflection process is crucial to medical ethical decision-making.

4 algorithmic: 演算法的，计算程序的。

5 David Hume: 大卫·休谟，苏格兰的不可知论哲学家、经济学家和历史学家，他被视为苏格兰启蒙运动以及西方哲学历史中最重要的物之一。

- 7 For centuries, it was assumed that a good decision ethically in medicine was the same as a good clinical decision. If the doctor did what was medically indicated to benefit the patient, then this was the ethically right thing to do. Although sometimes crudely summarized as “doctor knows best,” this approach to ethical dilemmas in medicine is (arguably) less about the doctor’s status, and more about the tensions between facts and values.
- 8 Medicine as a science utilizes a method of study that focuses on consequences of actions, on causes and effects in nature. These facts about how bodies heal, or how drugs work, are sometimes confused with medicine’s ethical imperative to bring about good consequences for the patient, or at least reduce harmful consequences. Concerns tend to arise when there is friction between the facts and values.
- 9 Modern medical ethics developed out of an examination of medical authority after the Second World War, partly in response to the Nuremberg trials⁶ of doctors who had used medicine to torment and kill citizens, but also in sympathy with a general increase of attention to the human rights of ordinary people which had previously been denied – people of color, women and those made vulnerable by illness.
- 10 Legal cases reflected this change: In one famous instance (Murray v McMurchy, in 1949), while operating on a woman for another purpose, a surgeon tied her fallopian tubes⁷ without her consent, because he foresaw that becoming pregnant would be clinically dangerous for her, and that it would also be dangerous for her to undergo two surgical procedures. She sued for negligence and won: It was not disputed that the surgeon was factually correct, in clinical terms, but he had not considered that the patient’s own view of herself and her body were essential to the decision-making process. He had focused on facts, and assigned no value to the patient’s view, even though it was her body that was being operated on.
- 11 This case brings us to an important issue in moral reasoning generally, which is how we think about words like “good” or “right” or “best,” in relation to a human decision. It is not a question of whether we want doctors to make ethical decisions on a daily basis – it is a fact that this will happen in the world of medical practice. What we want is for doctors to make “good” ethical decisions, or at least the “best possible.” We want to know that they have engaged in the type of thinking that

6 Nuremberg trials: 纽伦堡审判, 指 1945 年至 1946 年在德国纽伦堡进行的由第二次世界大战战胜国对欧洲轴心国的军事、政治和经济领袖的军事审判。

7 fallopian tubes: 输卵管

takes account of values and personal lived experience.

- 12 The doctor is empowered to do harm to the patient in pursuit of doing good, and there is a social acceptance that treatment may entail a deliberately imposed suffering that is not the primary intention of the doctor. This acceptance requires a great deal of trust in the medical profession – and doctors are still the most trusted professional group. The trust that makes these interactions possible assumes that doctors will not be the kind of people who exploit vulnerability and exercise influence for their own ends. There is a question here about how society expects doctors not just to be good technically, but to be good personally.
- 13 There are other accounts of ethical reasoning that may be helpful when thinking about doctors as good people. In his book, *Justice: What's the Right Thing to Do?*, Michael Sandel⁸ has argued that moral decision-makers need to follow an ethical reasoning process that pays attention to justice and the ways that people weigh the value of their decisions. He argues that impartiality is not always the keystone of justice, but rather that justice processes need to pay attention to what people value.
- 14 There remains a question about whether it is just and fair to expect a group of people who are chosen for cognitive intelligence and skills in exam-passing to become morally superior individuals. It is often said that doctors are held to a higher moral standard than other people, but how are they trained to that higher moral standard? After the Harold Shipman⁹ Inquiry, it was recommended that doctors undergo revalidation every five years, but there is no evidence that the revalidation process addresses moral reasoning or the moral identity of doctors. Doctors still do “bad” things, even when they are good people in other ways, and technically good at what they do.
- 15 Medicine needs a way of thinking about ethics that addresses different moral values and intuitions. What remains unclear is how we train doctors to be good people, not just to do good work and make good choices.

8 Michael Sandel: 迈克尔·桑德尔，政治哲学家，哈佛大学教授。

9 Harold Shipman: 哈罗德·希普曼，英国家庭医生、连环杀手。英国医师协会称有足够证据证明希普曼杀了 215 个人，其中 80% 为女性。

New Words and Expressions

dilemma /dɪˈlemə, daɪ-/ *n.* a situation in which it is very difficult to decide what to do, because all the choices seem equally good or equally bad 进退两难的境地, 困境

theoretical /θɪəˈretɪkəl/ *adj.* relating to the study of ideas, especially scientific ideas, rather than with practical uses of the ideas or practical experience 理论的

disparity /dɪˈspærɪti/ *n.* a difference between two or more things, especially an unfair one 不同, 差异

vulnerability /ˌvʌlnərəˈbɪləti/ *n.* the state of being exposed 脆弱性

clinical /ˈklɪnɪkəl/ *adj.* relating to treating or testing people who are sick 临床的

resuscitate /rɪˈsʌsɪteɪt/ *vt.* to make someone breathe again or become conscious after they have almost died 使恢复呼吸; 使苏醒

discourse /ˈdɪskɔːs/ *n.* the language used in particular types of speech or writing 话语; 语篇

irrelevant /ɪˈrelɪvənt/ *adj.* not useful or not relating to a particular situation, and therefore not important 无关紧要的, 不相干的

fallacy /ˈfæləsi/ *n.* a false idea or belief, especially one that a lot of people believe is true 误解, 谬见

derive /dɪˈraɪv/ *vt.* to get something from something else 得到, 获得

imperative /ɪmˈperətɪv/ *n.* something that must be done urgently 紧急的事

instance /ˈɪnstəns/ *n.* an example of a particular kind of situation 例子, 实例

negligence /ˈneglɪdʒəns/ *n.* failure to take enough care over something that you are responsible for 疏忽; 玩忽职守

empower /ɪmˈpaʊə/ *vt.* to give someone more control over their own life or situation 使自主; 使控制局面

entail /ɪnˈteɪl/ *vt.* to involve something as a necessary part or result 使必要; 需要

exploit /ɪkˈsplɔɪt/ *vt.* to try to get as much as you can out of a situation, sometimes unfairly 利用

impartiality /ɪmˌpɑːfɪˈælɪti/ *n.* an inclination to weigh both views or opinions equally 公平, 公正, 不偏不倚

cognitive /ˈkɒɡnɪtɪv/ *adj.* related to the process of knowing, understanding, and learning something 认知的, 认知过程的

revalidation /rɪˌvælɪˈdeɪʃən/ *n.* the cognitive process of re-establishing a valid proof 再次批准

Building Your Vocabulary

Task 1

Complete the following sentences using the expressions in the box below. Change the form where necessary.

empower	entail	exploitation	cognitive	resuscitate
algorithmic	discourse	fallacy	dilemma	imperative

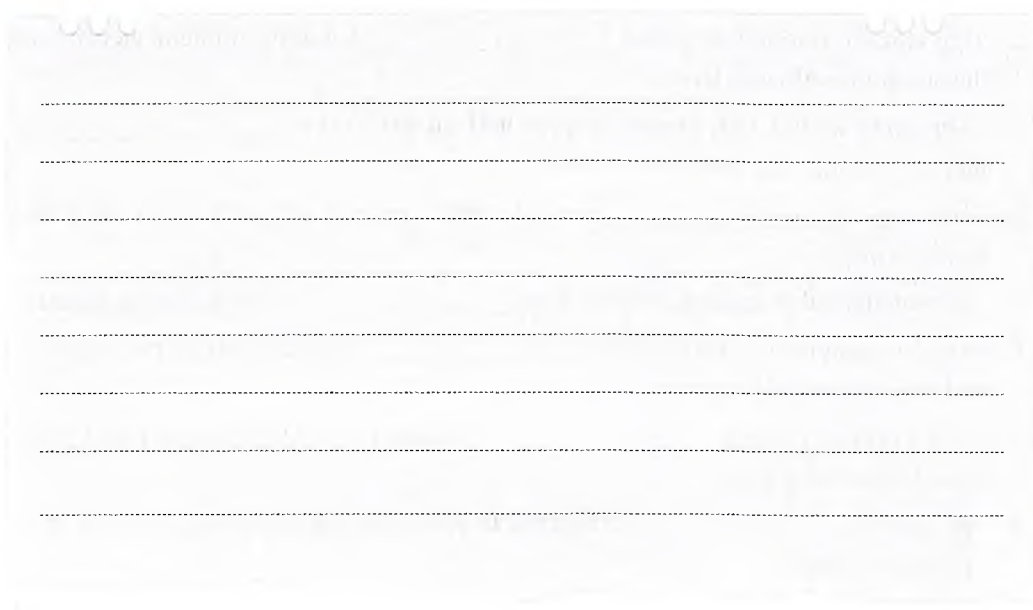
1. It said neither nurse tried to _____ the patient when they found he was not responding.
2. That team developed an initial _____ solution, implemented it, and the solution is already live.
3. Everybody knows that husbands give half an ear to the _____ of their wives and vice versa.
4. This same _____ is what causes some people to be afraid of flying on an airplane.
5. A broad and balanced education is a(n) _____ for raising standards.
6. Brazil is caught up in its own _____ between accelerated growth and environmental preservation.
7. Such a decision would _____ a huge political risk in the midst of the presidential campaign.
8. We must _____ children to speak up and then be sure that their voices are heard.
9. Practices like meditation, exercise, and _____ therapy have shown comparable short-term efficacy to medication.
10. Extra payments should be made to protect the interests of the staff and prevent _____.

Understanding the Text

Task 2 Choose the statement that best summarises the theme of Text A.

- A Doctors are in a better place to make decisions for their patients.
- B Moral reasoning is important for patients to make better decisions themselves.
- C A new way of reasoning is needed to keep pace with medical advancement.

Task 3 Identify two or more supporting details of the article's theme.

A large rectangular area with horizontal dotted lines for writing, intended for the student to provide supporting details for the article's theme.

Academic Reading Skills

Understanding Ellipsis and Substitution

Cohesion is the grammatical and lexical linking within a text or sentence that holds a text together and gives it meaning. It is related to the broader concept of coherence. There are two main types of cohesion: grammatical cohesion, which is based on structural content, and lexical cohesion, which is based on lexical content and background knowledge.

Ellipsis, or omission, refers to the deletion of some elements (nouns, verbs, and even clauses) of a sentence on the condition that the sentence after omission could be understood in the context of the remaining elements. For example:

- A *Periodontal disease is a term generally used to describe specific diseases that affect the supporting tissues and is divided into gum diseases and periodontitis, which affect the deep layer of the periodontium (periodontal ligament, alveolar bone, and dental cementum).* (In this sentence, the nominal phrase “periodontal disease,” used as the subject here, has been omitted before “is divided into.”)
- B *Nine out of the aforementioned 37 reported cases were diagnosed as respiratory complications; four of these cases with a mental illness.* (Here, the predicate verb “were diagnosed” is omitted after the phrase “four of these cases.”)
- C *These include: Is this treatment recommended by the National Institute for Health and Care Excellence (NICE) and, if not, why not?* (In this sentence, “if not, why not?” means “if this treatment is not recommended by NICE, why is it not recommended?” This sentence is an example of clausal ellipsis.)

Substitution is used when a word is not omitted, as in ellipsis, but is substituted for another more general word. Generally speaking, nouns, verbs, or clauses can be substituted. For example:

- A *It found that patients who successfully completed the program had no increase in their oral diabetes medicine and were only half as likely to progress to insulin as **those** who did not complete the program.* (The word “those” is the nominal substitution of “patients” mentioned before.)

- B *The widespread notion that only very rare or novel cases meet the criteria for publication is mistaken. On the contrary, more frequently encountered scenarios prove useful: A common case that presents in an unusual way, or a common management strategy that faces new obstacles or unexpected outcomes, provides useful insights as much as do descriptions of exceedingly rare conditions. (Here, “do” is the verbal substitution of “provide useful insights.”)*
- C *The patients will be too weak to sit up or work the scooter controls safely. If so, they may qualify for a power wheelchair. (In this sentence, the word “so” substitutes the clause “The patients will be too weak to sit up or work the scooter controls safely.”)*

Task 1

Read the following excerpts from Text A and identify the part of ellipsis to understand them better.

Excerpt 1:

Medical knowledge changes swiftly, and technological changes make new and expensive investigations and treatments possible that were only theoretical a few years ago. Life has been extended in length, but not in quality, and the debates about end-of-life decisions show us how much the notion of a “good life” is bound up with the absence of disease, illness and suffering. (Para. 1)

Excerpt 2:

Modern medical ethics developed out of an examination of medical authority after the Second World War, partly in response to the Nuremberg trials of doctors who had used medicine to torment and kill citizens, but also in sympathy with a general increase of attention to the human rights of ordinary people which had previously been denied – people of color, women and those made vulnerable by illness. (Para. 9)

Task 2

Indicate which elements are omitted in each of the following sentences.

1. The sex ratio will be favored which maximizes the number of descendants an individual will have and hence the number of gene copies transmitted.
2. The system makes cells put their ordinary activities on hold and instead switch on their defense systems, in reaction to high levels in the bloodstream of chemicals called purines.
3. People who've never tried e-cigarettes will focus on the potential risks from using them, and in particular whether they're likely to reintroduce smoking to a young generation who have been steadily shunning it in larger and larger numbers over recent decades.
4. On physical examination, he was nourished, conscious, and oriented. His blood pressure (BP) was 100/70 mmHg, pulse rate (PR) 90 beats per minute, respiratory rate (RR) 20 breaths per minute, and temperature (T°) 37.5 °C.
5. Whereas more hepatic NKTs were found in germ-free mice when compared with matched SPF control mice, no change was seen in TLR4-knockout mice.
6. At present, induction of specific pathological changes in animals after screening, or altering the animal's general condition is feasible but not an ideal pattern model.

Task 3

Identify substitution in the following sentences.

1. We found that patients with a cigarette-smoking history of more than 15 pack-years or those who quit smoking less than 15 years ago have a lower likelihood of harbouring EGFR mutations compared with those who have never smoked cigarettes.
2. We analysed the relationship between the amount of alcohol consumption and heart disease as some previous studies did.
3. A good case report is considered to be one that has a clear message, can be generalised, and is relevant to many other clinicians.
4. He rarely prescribed these drugs for further treatment to any patient, as some doctors did.

2

Academic Writing

Text B

Delivering Precision Medicine in Oncology Today and in Future

F. Ciardiello et al

Defining personalized cancer medicine

- 1 In its broadest sense, “personalized medicine” is the tailoring of medical treatment to the characteristics of an individual patient and moves beyond the current approach of stratifying patients into treatment groups based on phenotypic biomarkers. Nowhere in medicine has the impact of personalized medicine been greater than in oncology. For scientists and oncologists, the term “personalized medicine” is often used interchangeably with terms such as “genomic medicine,” “precision medicine” and “precision oncology.” These terms are used to describe the use of an individual patient’s molecular information (including genomics and proteomics) to inform diagnosis, prognosis, treatment and prevention of cancer for that patient. As the transition from stratified cancer medicine to truly personalized cancer medicine intensifies, it is this definition that the ESMO¹⁰ Personalized Medicine Task Force prefers to use when describing personalized cancer medicine. But irrespective of the term used, the direction of travel is clear – precision diagnosis and treatment of cancer at the molecular level – and this change in paradigm has profound implications, from preclinical definition of mechanism of action to the development of molecular taxonomies of cancer, and from genome diagnostics to trial design.

From genomics to clinics – the context and history of personalized medicine

- 2 Although much of cancer biology is based on the central tenet that it is a genetic disease, caused by a clone of cells that expands in an unregulated fashion because

10 ESMO: European Society for Medical Oncology, 欧洲临床肿瘤学会。

of somatically acquired mutations, this view contributed little to cancer treatment until the 21st century. The targeting of *HER2*¹¹ overexpression with the monoclonal antibody, trastuzumab¹², to improve outcome in metastatic breast cancer was the first example of targeted treatment; but the paradigm of targeted interference of an oncogene with a specifically designed small molecular inhibitor is best exemplified by imatinib. The tyrosine kinase inhibitor imatinib, developed to target the *BCR-ABL* fusion gene, a consequence of the Philadelphia chromosome and pathognomonic of chronic myeloid leukemia, transformed the care of patients, changing this aggressive, life-threatening disease to a manageable chronic disease. Around the same time, the initiation of the Cancer Genome Project at the Wellcome Trust's Sanger Institute¹³ using exon Sanger sequencing quickly identified somatic mutations in the *BRAF* gene in the majority of malignant melanoma¹⁴. This opened a window into the biology of these tumors and provided the starting point for successful clinical translation, with the development of vemurafenib¹⁵ that specifically targets the underlying molecular lesion.

- 3 With the launch of large-scale cancer whole genome sequencing (WGS) projects such as the International Cancer Genome Consortium (ICGC) and The Cancer Genome Atlas (TCGA) expected to deliver a complete catalogue of genomic alterations in primary cancers and begin to elucidate the mutational patterns and influences across the natural history of cancers, connecting recurrent genomic alterations to altered pathways and acquired cellular vulnerabilities will open the door to targeted therapies. At the same time, elucidation of the mechanisms underlying the processes generating somatic mutations will lead to new insights into cancer causation and, potentially, new approaches to prevention.

11 *HER2*: 人类表皮生长因子受体-2, 是乳腺癌研究较为透彻的基因之一, 于 20 世纪 80 年代分别由三个研究小组独立发现。*HER2* 基因的过表达不仅与肿瘤的发生和发展相关, 还是一个重要的临床治疗监测及预后指标, 并且是肿瘤靶向治疗药物选择的一个重要靶点。

12 *trastuzumab*: 曲妥珠单抗, 是瑞士制药巨头罗氏开发的重组 DNA 人源化单克隆抗体, 商品名为赫赛汀, 主要用于治疗 *HER2* 阳性乳腺癌。

13 the Wellcome Trust's Sanger Institute: 英国韦尔科姆基金会桑格研究所, 是遗传学和进化学的世界级研究所。

14 *melanoma*: 黑色素瘤

15 *vemurafenib*: 威罗菲尼片, 一种激酶抑制剂, 用于治疗晚期黑色素瘤。

Academic Writing Skills

Creating Cohesion

To achieve cohesion, the link of one sentence to the next, consider the following devices:

1. **Reference:** using the pronouns to refer to either a piece of information that lies in the same text or a piece of information that is not contained in the text. For example:

- A *In 1995, mortality among those covered by Medicaid was 46.9 per 100 person-years; among those with private insurance, it was 24.4 per 100 person-years.* (In this sentence, the pronoun “it” refers to “mortality.”)
- B *These findings are consistent with the reported immunologic benefits of protease inhibitors in previously reported data.* (Here, “These findings” refers to the research results discussed in the previous sentence.)
- C *These declines occurred during an era in which antiretroviral therapies became more numerous and more potent.* (“more” is used as comparative reference)

2. **Substitution:** refer to Academic Reading Skills of this unit.

3. **Ellipsis:** refer to Academic Reading Skills of this unit.

4. **Lexical cohesion:**

- A **Repetition:** repeating a word already used in the previous sentence(s). Repeating key words in a paragraph is an important technique for achieving cohesion. Of course, careless or excessive repetition is boring – and a source of clutter. But used skillfully and selectively, this technique can hold sentences together and focus the reader’s attention on the central idea.
- B **Synonym:** if direct repetition is too obvious, use a synonym of the word you wish to repeat. This strategy is called “elegant variation.”
- C **Antonym:** using the “opposite” word. An antonym can also create sentence cohesion, since in language antonyms actually share more elements of meaning than you might imagine.

- D Collocation: using a commonly paired or expected or highly probable word to connect one sentence to another. (see Academic Reading Skills in Unit 9)

5. **Conjunction:** using a conjunction or conjunctive adverb or phrase to link sentences with particular logical relationships.

- A Identity: to indicate sameness.
that is, that is to say, in other words, ...
- B Opposition: to indicate a contrast.
but, yet, however, nevertheless, still, though, although, whereas, in contrast, rather, ...
- C Addition: to indicate continuation.
and, too, also, furthermore, moreover, in addition, besides, in the same way, again, another, similarly, ...
- D Cause and effect: to indicate a reason or result.
therefore, so, consequently, as a consequence, thus, as a result, hence, it follows that, because, since, for, ...
- E Exemplification: to indicate a shift from a more general or abstract idea to a more specific or concrete idea.
for example, for instance, after all, an illustration of, even, indeed, in fact, specifically, to be specific, that is, to illustrate, ...
- F Concession: to indicate a willingness to consider the other side.
admittedly, I admit, I grant, of course, naturally, some (people) believe, it has been claimed that, once it was believed, there are those who would say, ...
- G Indefinites: to indicate a logical connection of an unspecified type.
in fact, indeed, now, ...

Task 1

Read the two excerpts from Text B and underline the cohesive devices used.

Excerpt 1:

For scientists and oncologists, the term “personalized medicine” is often used interchangeably with terms such as “genomic medicine,” “precision medicine” and “precision oncology.” These terms are used to describe the use of an individual patient’s molecular information (including genomics and proteomics) to inform diagnosis, prognosis, treatment and prevention of cancer for that patient. As the transition from stratified cancer medicine to truly personalized cancer medicine intensifies, it is this definition that the ESMO Personalized Medicine Task Force prefers to use when describing personalized cancer medicine. But irrespective of the term used, the direction of travel is clear – precision diagnosis and treatment of cancer at the molecular level – and this change in paradigm has profound implications, from preclinical definition of mechanism of action to the development of molecular taxonomies of cancer, and from genome diagnostics to trial design. (Para. 1)

Excerpt 2:

With the launch of large-scale cancer whole genome sequencing (WGS) projects such as the International Cancer Genome Consortium (ICGC) and The Cancer Genome Atlas (TCGA) expected to deliver a complete catalogue of genomic alterations in primary cancers and begin to elucidate the mutational patterns and influences across the natural history of cancers, connecting recurrent genomic alterations to altered pathways and acquired cellular vulnerabilities will open the door to targeted therapies. At the same time, elucidation of the mechanisms underlying the processes generating somatic mutations will lead to new insights into cancer causation and, potentially, new approaches to prevention. (Para. 3)

Task 2

Complete the following excerpt on oral diseases by choosing from the sentences provided.

Oral diseases, especially periodontal disease and dental caries, are still major health problems in the world today. 1. _____ Among 687 respondents, only 2 subjects had completely healthy teeth (0.3%), whereas the prevalence of gingival bleeding, dental caries, and moderate or severe periodontitis were 30.9%, 42.4%, and 19.8%, respectively. With an increase in age, the main disease type changed from gingivitis and dental caries to severe periodontitis, which led to severe oral disease consistent with our expectations. 2. _____ All of these factors resulted in a vicious cycle,

causing severe periodontitis. Similarly, as income increased, we found a gradual decrease in the prevalence of moderate and severe periodontitis, suggesting a close relationship between individual income and oral health. 3. _____ For example, we found that most elderly people thought loose teeth was normal and not worthy of attention! Therefore, this kind of unadvisable concept of oral health could also be related to the high prevalence of periodontitis among these subjects.

- A Because the dental plaques of the elderly (>65 years old) could not be well controlled, dental caries and dentition defects were more common in these people, coupled with a worse systemic health condition, and decreased disease resistance or self-repair capacity.
- B Because of insufficient income, the elderly tended to put general health as a first priority, paying more attention to the health of respiratory, circulatory, digestive, and neural systems, while ignoring the importance of oral health.
- C Our study showed that the oral health of the elderly in Southwest China was worrying.

Task 3

Rewrite the underlined parts in the following sentences using cohesive devices.

1. Several reports have described reductions in mortality and in the rate of hospitalization of HIV-infected patients; however, reductions in mortality and in the rate of hospitalization of HIV-infected patients have not been clearly related to specific therapeutic regimens.
2. While enrollment continued during the study period, data from some new patients were included in each quarterly analysis.
3. Nevertheless, the course of cognitive function deterioration in hemodialysis patients is essentially unknown, which makes estimating how rapidly a patient will decline challenging for clinicians. Also, the factors correlated with the cognitive function deterioration in such patients have not yet been clearly defined.
4. A total of 194 elderly patients were screened for study participation. Of the 194 elderly patients, 24 were not included in the study in spite of the presence of exclusion criteria.

5. Weight loss is associated with short-term amelioration and prevention of metabolic and cardiovascular risk, but whether short-term amelioration and prevention of metabolic and cardiovascular risk persist over time is unknown.

Task 4

Use proper cohesive devices to proofread the following two passages.

Passage 1:

Quality of life (QOL) refers to the feelings of individuals in a certain cultural environment and QOL is associated with personal goals, expectations, standards and concerns, and QOL includes physiological health, the mental state, independency, social relationship, personal beliefs, and the environment. QOL provides clinical doctors and researchers with more information about the disease. QOL quantifies the influence of a certain disease on physical, psychological, and social aspects for patients. The influence is beneficial for health risk factor tracing, treatment selection, and prognosis monitoring. The influence supports the concept of modern health and the transition of medical models.

Passage 2:

When males were compared with females, periodontitis was likely to develop into the severe stage in male subjects (35.5% vs 30.9%). Thus, we found that with an increase in age, the severity of the disease still increased gradually: Subjects aged 60 to 64 years were mainly diagnosed with gingivitis or subgingival calculus (45.5%), subjects aged 65 to 70 years and 71 to 75 years were diagnosed with moderate periodontitis (48.4% and 58.3%), and the oldest subjects (76 to 80 years) were likely to be diagnosed with severe periodontitis and moderate periodontitis (respectively 32.6% and 37.0%).